Exposé I. Presheaves

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In numbers 0 through 5 of this exposé we present the elementary, and most often well-known, properties of categories of presheaves.[1] These properties are used constantly throughout the rest of the seminar, and knowing them is essential for understanding the subsequent exposés. The proofs are immediate; for the most part they are omitted. In numbers 6 through 9 we take up a few themes that are used several times later on. The reader in a hurry may omit them on a first reading. Number 10 fixes the terminology used. Appendix 11 is due to N. Bourbaki.

1 Universes

A *universe* is a nonempty set[2] $\mathcal{U}$ satisfying the following properties:

($\mathcal{U}$1) If $x \in \mathcal{U}$ and $y \in x$, then $y \in \mathcal{U}$.

($\mathcal{U}$2) If $x, y \in \mathcal{U}$, then $\{x, y\} \in \mathcal{U}$.

($\mathcal{U}$3) If $x \in \mathcal{U}$, then $\mathcal{P}(x) \in \mathcal{U}$.

($\mathcal{U}$4) If $(x_i)_{i\in I}$ is a family of elements of $\mathcal{U}$ and $I \in \mathcal{U}$, then $\bigcup_{i\in I} x_i \in \mathcal{U}$.

From the preceding axioms one easily deduces the following properties:

- if $x \in \mathcal{U}$, then the singleton $\{x\}$ belongs to $\mathcal{U}$;
- if x is a subset of some $y \in \mathcal{U}$, then $x \in \mathcal{U}$;
- if $x, y \in \mathcal{U}$, the ordered pair $(x, y) = \{\{x, y\}, x\}$, in Kuratowski's definition, is an element of $\mathcal{U}$;
- if $x, y \in \mathcal{U}$, the union $x \cup y$ and the product $x \times y$ are elements of $\mathcal{U}$;
- if $(x_i)_{i\in I}$ is a family of elements of $\mathcal{U}$ indexed by an element $I \in \mathcal{U}$, then the product $\prod_{i\in I} x_i$ is an element of $\mathcal{U}$;
- if $x \in \mathcal{U}$, then $\mathrm{card}(x) < \mathrm{card}(\mathcal{U})$ strictly. In particular, the relation $\mathcal{U} \in \mathcal{U}$ is not satisfied.

Thus one may perform all the usual operations of set theory starting from elements of a universe without the final result ceasing to be an element of that universe.

The first use of the notion of universe is that it supplies a definition of the usual categories: the category of sets belonging to the universe $\mathcal{U}$ ($\mathcal{U}$-**Set**), the category of topological spaces belonging to $\mathcal{U}$, the category of commutative groups belonging to $\mathcal{U}$ ($\mathcal{U}$-**Ab**), the category of categories belonging to $\mathcal{U}$, and so on.

However, the only known universe is the set of symbols of the type $\{\{\varnothing\}, \{\{\varnothing\}, \varnothing\}\}$, etc. All the elements of this universe are finite sets, and this universe is countable. In particular, no

[1] The reader may also consult SGA 3, I, §§1–3.

[2] This definition contradicts Bourbaki's appendix below. However, the interest of empty universes seems debatable.

universe is known that contains an element of infinite cardinality. We are therefore led to add to the axioms of set theory the axiom

$(\mathcal{U}A)$ *For every set x there exists a universe $\mathcal{U}$ such that $x \in \mathcal{U}$.*

Since the intersection of a family of universes is a universe, it follows immediately that every set is an element of a smallest universe. One can show that axiom $(\mathcal{U}A)$ is independent of the axioms of set theory.

We shall also add the axiom:

$(\mathcal{U}B)$ *Let $R\{x\}$ be a relation and let $\mathcal{U}$ be a universe. If there exists an element $y \in \mathcal{U}$ such that $R\{y\}$, then*

The consistency of axioms $(\mathcal{U}A)$ and $(\mathcal{U}B)$ relative to the other axioms of set theory is not proved and, it seems, not provable.

Let $\mathcal{U}$ be a universe and let $c(\mathcal{U})$ be the least upper bound of the cardinals of elements of $\mathcal{U}$; thus $c(\mathcal{U}) \leq \operatorname{card}(\mathcal{U})$. The cardinal $c(\mathcal{U})$ has the following properties:

(FI) If $a < c(\mathcal{U})$, then $2^a < c(\mathcal{U})$.

(FII) If $(a_i)_{i\in I}$ is a family of cardinals strictly smaller than $c(\mathcal{U})$, and if $\operatorname{card}(I)$ is strictly smaller than $c(\mathcal{U})$, then $\sum_{i\in I} a_i < c(\mathcal{U})$.

Cardinals satisfying properties (FI) and (FII) are called *strongly inaccessible cardinals.*

The cardinal 0 and the countably infinite cardinal are strongly inaccessible.

Axiom $(\mathcal{U}A)$ implies:

$(\mathcal{U}A')$ *Every cardinal is strictly bounded above by a strongly inaccessible cardinal.*

One can show (11) that, conversely, the consistency of $(\mathcal{U}A')$ implies the consistency of $(\mathcal{U}A)$, and that the consistency of these axioms implies the consistency of axiom $(\mathcal{U}B)$.

Call a set E *Artinian* if there is no infinite family $(x_n)_{n\in\mathbb{N}}$ such that $x_0 \in E$ and $x_{n+1} \in x_n$. One can then show [*loc. cit.*] that there is a one-to-one correspondence between strongly inaccessible cardinals and Artinian universes, defined as follows: to every strongly inaccessible cardinal c one associates the unique Artinian universe $\mathcal{U}_c$ such that

$$\operatorname{card}(\mathcal{U}_c) = c.$$

If $c < c'$ are two strongly inaccessible cardinals, then

$$\mathcal{U}_c \in \mathcal{U}_{c'}.$$

In particular, the Artinian universes of cardinal smaller than a given cardinal form a well-ordered set for the membership relation. Axiom $(\mathcal{U}A')$ is equivalent to the axiom:

$(\mathcal{U}A'')$ *Every Artinian set is an element of an Artinian universe.*

We note that all the usual sets $(\mathbb{Z}, \mathbb{Q}, \ldots)$ are Artinian sets.

2 $\mathcal{U}$-categories. Presheaves of sets

1.0

In the rest of the seminar, unless the contrary is expressly stated, the universes considered will possess an element of infinite cardinality. Let $\mathcal{U}$ be a universe. A set is said to be *$\mathcal{U}$-small*, or simply *small* when no confusion can result, if it is *isomorphic* to an element of $\mathcal{U}$. We also use the terminology small group, small ring, small category,[3] etc. We shall often suppose, without explicit mention, that the schemes, topological spaces, indexing sets, etc. with which we work are in $\mathcal{U}$, or at least have cardinality in $\mathcal{U}$; however, many of the categories with which we shall work will not themselves be in $\mathcal{U}$.

Definition (1.1)**.** Let $\mathcal{U}$ be a universe and let C be a category. We say that C is a $\mathcal{U}$-category if, for every pair (x, y) of objects of C, the set $\mathrm{Hom}_C(x, y)$ is $\mathcal{U}$-small.

1.1.1

Let C and D be two categories, and let $\mathrm{Fun}(C, D)$ denote the category of functors from C to D. The following assertions are immediate:

(a) If C and D are elements of a universe $\mathcal{U}$ (respectively are $\mathcal{U}$-small), then the category $\mathrm{Fun}(C, D)$ is an element of $\mathcal{U}$ (respectively is $\mathcal{U}$-small).

(b) If C is $\mathcal{U}$-small and D is a $\mathcal{U}$-category, then $\mathrm{Fun}(C, D)$ is a $\mathcal{U}$-category.

Remark (1.1.2)**.** Let D be a category satisfying the following properties:

(C1) the set $\mathrm{Ob}(D)$ is contained in the universe $\mathcal{U}$;

(C2) for every pair (x, y) of objects of D, the set $\mathrm{Hom}_D(x, y)$ is an element of $\mathcal{U}$.

The usual categories constructed from a universe $\mathcal{U}$ have these two properties: $\mathcal{U}$-**Set**, $\mathcal{U}$-**Ab**, and so on. Let C be a category belonging to $\mathcal{U}$. Then the category $\mathrm{Fun}(C, D)$ does not in general have properties (C1) and (C2). For example, the category $\mathrm{Fun}(C, \mathcal{U}\text{-}\mathbf{Set})$ has neither property (C1) nor property (C2). This justifies the definition adopted for a $\mathcal{U}$-category, rather than the more restrictive notion given by conditions (C1) and (C2) above.

Definition (1.2)**.** Let C be a category. The category of presheaves of sets on C relative to the universe $\mathcal{U}$—or, when no confusion can result, the category of presheaves on C—is the category of contravariant functors on C with values in the category of $\mathcal{U}$-sets.

We denote by $\widehat{C}_{\mathcal{U}}$, or more simply by $\widehat{C}$ when no confusion can result, the category of presheaves of sets on C relative to the universe $\mathcal{U}$. The objects of $\widehat{C}_{\mathcal{U}}$ are called $\mathcal{U}$-presheaves, or simply presheaves, on C. When C is $\mathcal{U}$-small, the category $\widehat{C}_{\mathcal{U}}$ is a $\mathcal{U}$-category. When C is a $\mathcal{U}$-category, $\widehat{C}_{\mathcal{U}}$ is not necessarily a $\mathcal{U}$-category.

Construction–Definition (1.3)**.** Let x be an object of a $\mathcal{U}$-category C. The *$\mathcal{U}$-functor represented by x* is the functor

$$h_{\mathcal{U}}(x) : C^{\mathrm{op}} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

constructed as follows.[4] Let y be an object of C.

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[3] A category C is viewed as a set of arrows, equipped with the subset of objects, viewed as the set of identity arrows, and with the source and target maps. Thus the expressions 'C is an element of $\mathcal{U}$' and 'C is $\mathcal{U}$-small' make sense.

[4] C^{op} will always denote the category opposite to C.

(a) If $\mathrm{Hom}_C(y,x)$ is an element of $\mathcal{U}$, then

$$h_{\mathcal{U}}(x)(y) = \mathrm{Hom}_C(y,x).$$

(b) Suppose that $\mathrm{Hom}_C(y,x)$ is not an element of $\mathcal{U}$, and let $R(Z,x,y)$ be the relation: 'The set Z is the target of an isomorphism $\mathrm{Hom}_C(y,x) \xrightarrow{\sim} Z$.' We then put

$$h_{\mathcal{U}}(x)(y) = \tau_Z R(Z).$$

By axiom $(\mathcal{U}B)$, $h_{\mathcal{U}}(x)(y)$ is an element of $\mathcal{U}$. Let $R'(u,x,y)$ be the relation: 'u is a bijection

$$u : \mathrm{Hom}_C(y,x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y)'.$$

We put

$$\varphi(y,x) = \tau_u R'(u).$$

Notice that in both cases (a) and (b) we have a canonical isomorphism

$$\varphi(y,x) : \mathrm{Hom}_C(y,x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y).$$

In case (a), $\varphi(y,x)$ is the identity. Now let $u : y \to y'$ be an arrow of C. Composition of morphisms defines a map

$$\mathrm{Hom}_C(u,x) : \mathrm{Hom}_C(y',x) \longrightarrow \mathrm{Hom}_C(y,x).$$

We set

$$h_{\mathcal{U}}(x)(u) = \varphi(y,x)\ \mathrm{Hom}_C(u,x)\,\varphi(y',x)^{-1}.$$

One verifies immediately that $h_{\mathcal{U}}(x)$, so defined, is a functor $C^{\mathrm{op}} \to \mathcal{U}$-**Set**.

1.3.1

Similarly, using the isomorphisms φ, for every morphism $v : x \to x'$ one defines a morphism of functors

$$h_{\mathcal{U}}(v) : h_{\mathcal{U}}(x) \longrightarrow h_{\mathcal{U}}(x'),$$

and one verifies immediately that this defines a functor

$$h_{\mathcal{U}} : C \longrightarrow \widehat{C}_{\mathcal{U}}.$$

1.3.2

Let $\mathcal{V}$ now be a universe such that $\mathcal{U} \subset \mathcal{V}$. We then have a canonical fully faithful functor

$$\mathcal{U}\text{-}\mathbf{Set} \hookrightarrow \mathcal{V}\text{-}\mathbf{Set},$$

and hence a fully faithful functor

$$\widehat{C}_{\mathcal{U}} \hookrightarrow \widehat{C}_{\mathcal{V}}.$$

The diagram

$$\begin{tikzcd} C \arrow[r, "h_{\mathcal{U}}"] \arrow[dr, "h_{\mathcal{V}}"'] & \widehat{C}_{\mathcal{U}} \arrow[d] \\ & \widehat{C}_{\mathcal{V}} \end{tikzcd}$$

commutes up to canonical isomorphism. When C is an element of $\mathcal{U}$, this canonical isomorphism is the identity.

1.3.3

In practice, the universe $\mathcal{U}$ is fixed once and for all and is not mentioned. We then use the notation $\widehat{C}$ for the category of $\mathcal{U}$-presheaves of sets and

$$h : C \longrightarrow \widehat{C}.$$

For every object x of C, the presheaf $h(x)$ is called the *presheaf represented by* x, and we shall always identify its value at an object $y \in \mathrm{Ob}(C)$ with $\mathrm{Hom}_C(y, x)$.

Proposition (1.4)**.** Let C be a $\mathcal{U}$-category, let F be a presheaf on C, and let X be an object of C. There is an isomorphism, functorial in X and in F,

$$i : \mathrm{Hom}_{\widehat{C}}(h(X), F) \xleftarrow{\sim} F(X),$$

where, when F is of the form $h(Y)$, the map i is just the map

$$h : \mathrm{Hom}(h(X), h(Y)) \longleftarrow \mathrm{Hom}(X, Y).$$

In particular, the functor h is fully faithful.

1.4.1

This proposition justifies the usual abuse of language that identifies an object of C with the corresponding contravariant functor. A presheaf isomorphic to an object in the image of h, or, using the above abuse of language, isomorphic to an object of C, is called a *representable presheaf.*

1.4.2

Let $\mathcal{V}$ be a universe containing a universe $\mathcal{U}$. The category of $\mathcal{U}$-presheaves is a *full* subcategory of the category of $\mathcal{V}$-presheaves; consequently, a $\mathcal{U}$-presheaf is representable if and only if its image in the category of $\mathcal{V}$-presheaves is a representable $\mathcal{V}$-presheaf.

Continuation anchor. The next untranslated source section is Exposé I, Section 2, ‘Projective and inductive limits’.

Exposé I. Presheaves

Sections 2–5

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 2–5, continuing the preceding translated fragment through Section 1.4. Obvious typographical slips in the working French TEX source have been corrected where the mathematics fixes the reading.

2 Projective and inductive limits

Let C be a $\mathcal{U}$-category and let I be a small category. Write $\mathrm{Fun}(I, C)$ for the category of functors from I to C. The category $\mathrm{Fun}(I, C)$ is a $\mathcal{U}$-category. To every object X of C associate the subcategory $\mathcal{X}$ of C having X as its only object and the identity of X as its only arrow. Let $i_X : \mathcal{X} \to C$ be the inclusion functor. There is a unique functor $e_X : I \to \mathcal{X}$, and we shall denote by $k_X : I \to C$ the functor $i_X \circ e_X$. We shall say that k_X is the constant functor with value X. The correspondence $X \mapsto k_X$ is visibly functorial in X, which enables us to define, for every functor $G : I \to C$, the presheaf

$$X \longmapsto \mathrm{Hom}_{\mathrm{Fun}(I,C)}(k_X, G).$$

Definition 2.1. The *projective limit* of G, denoted $\varprojlim_I G$, is the presheaf

$$X \longmapsto \mathrm{Hom}_{\mathrm{Fun}(I,C)}(k_X, G).$$

When the presheaf $\varprojlim_I G$ is representable, we again denote by $\varprojlim_I G$ an object of C representing it. The object $\varprojlim_I G$ is therefore defined only up to isomorphism. When no confusion can result, we use the abbreviated notation $\varprojlim G$.

As G varies, $\varprojlim G$ is a functor from $\mathrm{Fun}(I, C)$ to $\widehat{C}$.

2.1.1

By symmetry, reversing the direction of the arrows in C, one defines similarly the inductive limit of a functor: it is a covariant functor on C with values in the category of $\mathcal{U}$-sets. We shall use the notations $\varinjlim_I G$ and $\varinjlim G$.

Products, fiber products, and kernels are projective limits. Similarly, sums, amalgamated sums, and cokernels are inductive limits.

Definition 2.2. Let I and C be two categories, and let $G : I \to C$ be a functor. We say that the projective limit of G is *representable* if there exists a universe $\mathcal{U}$ such that:

1) the category I is $\mathcal{U}$-small;

2) the category C is a $\mathcal{U}$-category;

3) the presheaf $\varprojlim G$, with values in the category of $\mathcal{U}$-sets, is representable.

It follows from no. 1 that the object $\varprojlim G$ representing the presheaf $\varprojlim G$ does not depend, up to isomorphism, on the universe $\mathcal{U}$. Note also that there is a smallest universe $\mathcal{U}'$ having properties (1) and (2), and that the presheaf $\varprojlim G$ necessarily takes values in the category of $\mathcal{U}'$-sets.

2.2.1

Let I and C be two categories. We say that *I-projective limits in C are representable* if, for every functor $G : I \to C$, the projective limit of G is representable. Finally, let C be a category and let $\mathcal{U}$ be a universe. We say that *$\mathcal{U}$-projective limits in C are representable* if, for every $\mathcal{U}$-small category I and every functor $G : I \to C$, the projective limit of G is representable.

Proposition 2.3. Let C be a category and let $\mathcal{U}$ be a universe. The following assertions are equivalent:

i) $\mathcal{U}$-projective limits in C are representable.

ii) Products indexed by a small set are representable, and kernels of pairs of arrows are representable.

iii) Products indexed by a small set are representable, and fiber products are representable.

Proof. It suffices to observe that there is an isomorphism, functorial in G,

$$\varprojlim G = \operatorname{Ker}\left(\prod_{i \in \operatorname{Ob}(I)} G(i) \rightrightarrows \prod_{u \in \operatorname{Fl}(I)} G(\operatorname{target}(u)) \right),$$

where the pair of arrows is defined by the morphisms

$$\prod_{i \in \operatorname{Ob}(I)} G(i) \xrightarrow{\operatorname{pr}_{\operatorname{target}(u)}} G(\operatorname{target}(u)),$$

$$\prod_{i \in \operatorname{Ob}(I)} G(i) \xrightarrow{\operatorname{pr}_{\operatorname{source}(u)}} G(\operatorname{source}(u)) \xrightarrow{G(u)} G(\operatorname{target}(u)).$$

Moreover, it is clear that

$$\operatorname{Ker}\big(X \underset{v}{\overset{u}{\rightrightarrows}} Y\big) = X \times_{X \times Y} X,$$

where the two morphisms from X to $X \times Y$ are $\operatorname{id}_X \times u$ and $\operatorname{id}_X \times v$.

2.3.1

There are, of course, analogous definitions and assertions for inductive limits, which we shall not spell out. The same applies to finite projective and inductive limits, that is, those relative to a finite category I.

Corollary 2.3.2. Let $\mathcal{U}$-**Set** denote the category of $\mathcal{U}$-sets, and let $\mathcal{U}$-**Ab** denote the category of abelian-group objects of $\mathcal{U}$-**Set**. The $\mathcal{U}$-projective and $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Set** and in $\mathcal{U}$-**Ab** are representable.

Proposition 2.3.3. Let I be a small category, let $F : I \to \mathcal{U}$-**Set** be a functor, and let $G \subset \operatorname{Ob}(I)$ be a small set of objects of I such that, for every object X of I, there exists an object $Y \in G$ and a morphism $Y \to X$ (respectively $X \to Y$). Then $\varprojlim_I F$ (respectively $\varinjlim_I F$) is representable, and

$$\operatorname{card}(\varprojlim_I F) \le \prod_{Y \in G} \operatorname{card}(F(Y))$$

respectively

$$\operatorname{card}(\varinjlim_{I} F) \leq \sum_{Y \in G} \operatorname{card}(F(Y)).$$

Proof. One checks immediately that $\varprojlim_I F$ (respectively $\varinjlim_I F$) is isomorphic to a subobject of $\prod_{Y\in G} F(Y)$ (respectively to a quotient of $\coprod_{Y\in G} F(Y)$), whence the proposition.

Definition 2.4.1. Let C be a category in which finite projective (respectively inductive) limits are representable, and let $u : C \to C'$ be a functor. We say that u is *left exact* (respectively *right exact*) if it "commutes" with finite projective (respectively finite inductive) limits. A functor that is both left and right exact is called *exact*.

2.4.2

It follows from 2.3 that, for a functor to be left exact, it is necessary and sufficient that it carry the final object, i.e. the empty product, to the final object, the product of two objects to the product of their images, and the kernel of a pair of arrows to the kernel of the image pair; equivalently, that it carry the final object to the final object and fiber products to fiber products. Here it is assumed that finite projective limits in C are representable.

2.5.0

Let I, J, and C be three categories, and let $G : I \times J \to C$ be a functor, i.e. a functor from I with values in the category of functors from J to C. Suppose that the projective limits of the functors

$$\begin{aligned} G_i : J &\longrightarrow C \qquad (i \in \operatorname{Ob}(I)) \\ j &\longmapsto G(i,j) \end{aligned}$$

are representable, and that the functor

$$i \longmapsto \varprojlim_{J} G_i$$

admits a representable projective limit. Then it is clear that the functor G admits a representable projective limit and that there is a canonical isomorphism

$$\varprojlim_{I\times J} G \xrightarrow{\sim} \varprojlim_{I} \varprojlim_{J} G_i,$$

and hence a canonical isomorphism

$$\varprojlim_{I} \varprojlim_{J} G_i \xrightarrow{\sim} \varprojlim_{J} \varprojlim_{I} G_j.$$

We shall say more briefly that projective limits commute with projective limits. One sees in the same way that inductive limits commute with inductive limits. But it is not true in general that inductive limits commute with projective limits.

Definition 2.5. Let C be a category with representable fiber products, let I be a category, let $G : I \to C$ be a functor, let $g : G \to X$ be a morphism from G to an object of C, i.e. a morphism from G to the constant functor associated with X, and let $m : Y \to X$ be a morphism of C. Let

$$G \times_X Y : I \longrightarrow C$$

be the functor

$$i \longmapsto G(i) \times_X Y.$$

We say that the inductive limit of G is *universal* if, for every object X, every morphism $g : G \to X$, and every morphism $m : Y \to X$,

a) the inductive limit of the functor $G \times_X Y$ is representable;

b) the canonical morphism

$$\varinjlim(G \times_X Y) \longrightarrow (\varinjlim G) \times_X Y$$

is an isomorphism.

Proposition 2.6. Let $\mathcal{U}$ be a universe. Inductive limits in $\mathcal{U}$-**Set** that are representable, in particular $\mathcal{U}$-inductive limits (2.2.1) in $\mathcal{U}$-**Set**, are universal.

We shall also use another commutation result between projective and inductive limits, which we now present.

Definition 2.7. A category I is *pseudo-filtered* if it has the following properties:

PS 1) Every diagram of the form

$$\begin{tikzcd} & j \\ i \arrow[ur] \arrow[dr] & \\ & j' \end{tikzcd}$$

can be inserted into a commutative diagram

$$\begin{tikzcd} & j \arrow[dr] & \\ i \arrow[ur] \arrow[dr] & & k \\ & j' \arrow[ur] & \end{tikzcd} .$$

PS 2) Every diagram of the form

$$i \overset{u}{\underset{v}{\rightrightarrows}} j$$

can be inserted into a diagram

$$i \overset{u}{\underset{v}{\rightrightarrows}} j \xrightarrow{w} k$$

such that $w \circ u = w \circ v$.

A category I is called *filtered* if it is pseudo-filtered, nonempty, and connected, i.e. if any two objects of I can be joined by a finite chain of arrows, with no condition imposed on the directions of the arrows. Equivalently, in the presence of PS 2, this means that $I \neq \varnothing$ and that, for any two objects a, b of I, there is always an object c of I and arrows $a \to c$ and $b \to c$. A category I is called *cofiltered* if I^{op} is filtered.

Example 2.7.1. If, in I, amalgamated sums (respectively sums of two objects) and cokernels of double arrows are representable, then I is pseudo-filtered (respectively filtered).

Proposition 2.8. Let $\mathcal{U}$ be a universe. Filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Set** commute with finite projective limits.

One is immediately reduced to proving that filtered $\mathcal{U}$-inductive limits commute with fiber products. The proof is left to the reader. One may use the following description of the limit.

Lemma 2.8.1. Let I be a small filtered category, and let $i \mapsto X_i$ be a functor from I to $\mathcal{U}$-**Set**. On the sum set $\coprod_{i \in \mathrm{Ob}\, I} X_i$, let R be the following relation:

(R) Two elements $x_i \in X_i$ and $x_j \in X_j$ are related if there exist an object $k \in \mathrm{Ob}\, I$ and two morphisms $u : i \to k$ and $v : j \to k$ such that the images in X_k of x_i and x_j under the transition maps u and v, respectively, are equal.

Then:

1) R is an equivalence relation.

2) The quotient $\coprod_{i \in \mathrm{Ob}\, I} X_i / R$ is canonically isomorphic to $\varinjlim_I X_i$.

3) Two elements $x_i \in X_i$ and $x_j \in X_j$ are respectively equivalent, modulo R, to two elements of one and the same X_k.

4) For every $i \in \mathrm{Ob}\, I$, two elements α and β of X_i are equivalent modulo R if and only if there exists a morphism $u : i \to j$ such that $u(\alpha) = u(\beta)$.

Assertion 1) follows from PS 1 (2.7). To prove 2), one checks that $\coprod_{i \in \mathrm{Ob}\, I} X_i / R$ has the universal property of the inductive limit; here only PS 1 is used. Assertion 3) follows from the fact that I is connected. Assertion 4) follows from PS 2.

Corollary 2.9. Let γ be a species of algebraic structure "defined by finite projective limits." The reader is asked to give a mathematical meaning to the preceding phrase; let us merely note that the structures of groups, abelian groups, rings, modules, etc. are structures of this kind. Let $\mathcal{U}$-γ denote the category of γ-objects of $\mathcal{U}$-**Set**. The functor that associates with every object of $\mathcal{U}$-γ its underlying set commutes with filtered $\mathcal{U}$-inductive limits. Consequently filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-γ commute with finite projective limits.

Corollary 2.10. Pseudo-filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Ab** commute with finite projective limits.

One is reduced to filtered limits by decomposing the indexing category into its connected components.

We now record a result that we shall use constantly. Let C and C' be two $\mathcal{U}$-categories, and let

$$C \underset{v}{\overset{u}{\rightleftarrows}} C'$$

be two functors, with u *left adjoint* to v. Recall that this means that there is an isomorphism of bifunctors with values in $\mathcal{U}$-**Set**,

$$\mathrm{Hom}_{C'}(u(X), X') \xrightarrow{\sim} \mathrm{Hom}_C(X, v(X')).$$

Proposition 2.11. The functor u commutes with representable inductive limits; the functor v commutes with representable projective limits.

This assertion means that, for every category I and every functor $G : I \to C'$ such that the projective limit of G is representable, the functor $v \circ G$ admits a representable projective limit and there is a canonical isomorphism

$$v(\varprojlim G) \longrightarrow \varprojlim(v \circ G).$$

The reader will write out for himself the assertion concerning the functor u.

Finally, let us point out a computation of projective limits in the case where the indexing category admits products.

Proposition 2.12. Let I and C be two categories, and let $\mathcal{V}$ be a universe. Suppose that $\mathcal{V}$-projective limits in C are representable and that there exists in I a family of objects $(i_\alpha)_{\alpha \in A}$, with $A \in \mathcal{V}$, such that:

1) the products $i_\alpha \times i_\beta$ are representable for every pair $(\alpha, \beta) \in A \times A$;

2) every object of I maps to at least one of the i_α.

Then, for every contravariant functor

$$F : I^{\mathrm{op}} \longrightarrow C,$$

the projective limit of F exists and there is an isomorphism, functorial in F,

$$\varprojlim F \xrightarrow{\sim} \mathrm{Ker}\left(\prod_{\alpha \in A} F(i_\alpha) \rightrightarrows \prod_{(\alpha,\beta) \in A \times A} F(i_\alpha \times i_\beta) \right),$$

where the two arrows are defined by the projections of the products $i_\alpha \times i_\beta$ onto the factors.

3 Exactness properties of the category of presheaves

Let C be a $\mathcal{U}$-category, and let $\widehat{C}$ be the category of presheaves on C. The exactness properties of $\widehat{C}$ are all deduced from the following proposition.

Proposition 3.1. The $\mathcal{U}$-projective and $\mathcal{U}$-inductive limits in $\widehat{C}$ are representable. For every object X of C, the functor on $\widehat{C}$

$$F \longmapsto F(X), \qquad F \in \widehat{C},$$

commutes with inductive and projective limits.

In other words, inductive and projective limits in $\widehat{C}$ are computed argument by argument. We list a few corollaries.

Corollary 3.2. Let γ be an algebraic structure defined by finite projective limits. The category of contravariant functors with values in $\mathcal{U}$-γ is equivalent to the category of γ-objects of $\widehat{C}$.

Corollary 3.3. A morphism of $\widehat{C}$ that is both a monomorphism and an epimorphism is an isomorphism. A morphism of $\widehat{C}$ factors uniquely as an epimorphism followed by a monomorphism. Representable inductive limits in $\widehat{C}$ are universal. Filtered $\mathcal{U}$-inductive limits in $\widehat{C}$ commute with finite projective limits. The canonical functor $C \to \widehat{C}$ commutes with representable projective limits.

More generally, one may say that the category $\widehat{C}$ inherits all the properties of the category of $\mathcal{U}$-sets that involve inductive and projective limits.

3.4.0

Let F be an object of $\widehat{C}$. We denote by C/F the following category. Its objects are pairs consisting of an object X of C and a morphism u from X to F. Given two objects (X, u) and (Y, v), a morphism from (X, u) to (Y, v) is a morphism $g : X \to Y$ such that the following diagram is commutative:

$$\begin{tikzcd} X \arrow[rr, "g"] \arrow[dr, "u"'] & & Y \arrow[dl, "v"] \\ & F & \end{tikzcd}$$

Proposition 3.4. With the notation of (3.4.0), the source functor $C/F \to \widehat{C}$ admits a representable inductive limit in $\widehat{C}$. The canonical morphism

$$\varinjlim_{C/F} \operatorname{source}(.) \longrightarrow F$$

is an isomorphism.

Corollary 3.5. Let F and H be two objects of $\widehat{C}$. There is a canonical isomorphism

$$\operatorname{Hom}(F, H) \xrightarrow{\sim} \varprojlim_{(X,u)\in \operatorname{Ob}(C/F)} H(X).$$

Proof. The corollary follows immediately from 3.4 and 1.2.

Proposition 3.6. Let C be a $\mathcal{U}$-category, let $\mathcal{V}$ be a universe containing $\mathcal{U}$, and let

$$i : \widehat{C}_{\mathcal{U}} \longrightarrow \widehat{C}_{\mathcal{V}}$$

be the natural inclusion functor between the corresponding categories of presheaves (1.3). The functor i commutes with inductive and projective limits. Moreover, for every object F of $\widehat{C}_{\mathcal{U}}$, every subobject of $i(F)$ is isomorphic to the image under i of a unique subobject of F; thus one obtains a bijection from the set of subobjects of F to the set of subobjects of $i(F)$.

4 Sieves

Definition 4.1. Let C be a category. A *sieve* of the category C is a full subcategory D of C having the following property: every object of C admitting a morphism to an object of D belongs to D. If X is an object of C, the sieves of the category C/X will be called, by abuse of language, *sieves of* X.

Let $\mathcal{U}$ be a universe such that C is a $\mathcal{U}$-category, and let $\widehat{C}$ be the corresponding category of presheaves. To every sieve of X one associates a subobject of X in $\widehat{C}$ as follows: to every object Y of C one assigns the set of morphisms $f : Y \to X$ such that the object (Y, f) belongs to the sieve.

Proposition 4.2. The map defined above establishes a bijection between the set of sieves of X and the set of subobjects of X in $\widehat{C}$.

Proof. We only indicate the inverse map. To every subfunctor R of X one associates the category C/R of objects of C over R (3.4). One checks immediately that C/R is a sieve of X.

Remark 4.2.1. One sees in the same way that the sieves of C are in canonical one-to-one correspondence with the set of subfunctors of the final functor on C, the final object of $\widehat{C}$.

4.3

Let C be a $\mathcal{U}$-category. By abuse of language, we shall also call the subobjects of X in the category $\widehat{C}$ the sieves of X. This abuse of language allows us, for every presheaf F and every sieve R of X, to define $\mathrm{Hom}_{\widehat{C}}(R, F)$ as the set of morphisms from the functor R to F. There is, moreover, an isomorphism canonical and functorial in F (3.5):

$$\mathrm{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_{C/R} F(.),$$

which also gives a direct definition.[1] Similarly, Proposition 4.2 allows us to transpose the usual operations on functors to sieves. We mention the following.

4.3.1 Base change

Let R be a sieve of X, and let $f : Y \to X$ be a morphism of objects of C. The fiber product $R \times_X Y$ is a sieve of Y, called the *sieve deduced from* R *by base change*. The corresponding subcategory of C/Y is the inverse image of the subcategory of C/X defined by R, under the canonical functor $C/Y \to C/X$ defined by f.

4.3.2 Order relation, intersection, union

The inclusion relation on subfunctors of X is an order relation. One may define the union and intersection of a family of sieves indexed by an arbitrary set as the least upper bound and greatest lower bound of the corresponding family of subpresheaves.

[1] More simply, C/R identifies with R, so that one has the formula $\mathrm{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_R F$, where F abusively denotes the composite of F with the tautological source functor $R \to C/X \to C$.

4.3.3 Image, generated sieve

Let $(F_\alpha)_{\alpha\in A}$ be a family of presheaves, and for each $\alpha \in A$ let $f_\alpha : F_\alpha \to X$ be a morphism, where X is an object of C. The *image* of this family of morphisms is the union of the images of the f_α. The image of this family is therefore a sieve of X. In particular, if all the F_α are objects of C, the image sieve will be called the sieve generated by the morphisms f_α. The category C/R is the full subcategory of C/X formed by those objects $X' \to X$ over X for which there exists an X-morphism from X' to one of the F_α.

As an exercise, the reader may translate the relations and operations just defined on subfunctors into terms of the categories C/R. He will then see that these relations and operations do not depend on the universe $\mathcal{U}$ such that C is a $\mathcal{U}$-category, and that they are therefore defined for every category, without requiring one to specify the universe to which the sets of morphisms belong. This was in any case already foreseeable *a priori* from 3.6.

5 Functoriality of categories of presheaves

5.0

Let C, C', and D be three categories, and let $u : C \to C'$ be a functor. We denote by u^* the functor obtained by composition with u:

$$u^* : \mathrm{Hom}((C')^{\mathrm{op}}, D) \longrightarrow \mathrm{Hom}(C^{\mathrm{op}}, D), \qquad G \longmapsto G \circ u.$$

The functor u^* commutes with inductive and projective limits.

Proposition 5.1. Suppose that C is small and that, in D, $\mathcal{U}$-inductive limits (respectively projective limits) are representable. The functor u^* admits a left adjoint $u_!$ (respectively a right adjoint u_*). Thus one has an isomorphism

$$\mathrm{Hom}_{\mathrm{Hom}(C^{\mathrm{op}},D)}(F, u^*G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Hom}((C')^{\mathrm{op}},D)}(u_!F, G),$$
$$\big(\text{respectively } \mathrm{Hom}_{\mathrm{Hom}(C^{\mathrm{op}},D)}(u^*G, F) \simeq \mathrm{Hom}_{\mathrm{Hom}((C')^{\mathrm{op}},D)}(G, u_*F)\big).$$

Proof. We indicate only the proof of the existence of the left adjoint. The corresponding part of the proposition follows formally from the isomorphisms

$$\mathrm{Hom}(C^{\mathrm{op}}, D) \xrightarrow{\sim} \mathrm{Hom}(C, D^{\mathrm{op}})^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Hom}((C^{\mathrm{op}})^{\mathrm{op}}, D^{\mathrm{op}})^{\mathrm{op}}.$$

Let Y be an object of C'. Denote by I_u^Y the following category. Its objects are pairs (X, m), where X is an object of C and m is a morphism $Y \to u(X)$. If (X, m) and (X', m') are two objects of I_u^Y, a morphism from (X, m) to (X', m') is a morphism $\xi : X \to X'$ such that $m' = u(\xi)m$. Composition is defined in the evident way.

Let $f : Y \to Y'$ be a morphism of C'. By composition, f defines a functor $I_u^f : I_u^Y \to I_u^{Y'}$.

We also have a functor $\mathrm{pr}_Y : I_u^Y \to C$ which sends the object (X, m) to the object X. Then the evident diagram

$$\begin{tikzcd} I_u^Y \arrow[r, "I_u^f"] \arrow[d, "\mathrm{pr}_Y"'] & I_u^{Y'} \arrow[dl, "\mathrm{pr}_{Y'}"] \\ C & \end{tikzcd} \tag{$*$}$$

is commutative.

Now let F be a presheaf on C and set

$$u_!F(Y) = \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y \,. \tag{5.1.1}$$

The commutativity of diagram $(*)$ and the functoriality of inductive limits make $u_!F$ into a presheaf on C'. Let us show that the functor $u_!$ is left adjoint to u^*. For this it is enough to show that, for every presheaf G on C', there is a functorial isomorphism

$$\mathrm{Hom}(u_!F, G) \xrightarrow{\sim} \mathrm{Hom}(F, u^*G).$$

Let $\xi \in \mathrm{Hom}(u_!F, G)$. For every object X of C, we have a morphism

$$\xi_{u(X)} : u_!F(u(X)) \longrightarrow G(u(X)).$$

But $(X, \mathrm{id}_{u(X)})$ is an object of $I_u^{u(X)}$, and by the definition of the inductive limit there is a canonical morphism

$$F(X) \longrightarrow u_!F(u(X)).$$

We obtain, for every object X of C, a morphism

$$\eta_X : F(X) \longrightarrow G(u(X)),$$

which is visibly functorial in X. Hence a morphism

$$\eta : F \longrightarrow u^*G.$$

Conversely, let $\eta \in \mathrm{Hom}(F, u^*G)$. One obtains, for every object Y of C', a morphism of functors

$$\eta_Y : F \circ \mathrm{pr}_Y \longrightarrow (u^*G) \circ \mathrm{pr}_Y,$$

and hence, by composing with the evident morphism from the functor $(u^*G) \circ \mathrm{pr}_Y$ to the constant functor $G(Y)$, a morphism

$$\xi_Y : u_!F(Y) \longrightarrow G(Y),$$

which is functorial in Y. Hence a morphism

$$\xi : u_!F \longrightarrow G.$$

The reader will check that the two maps thus defined are inverse to one another, completing the proof.

Proposition 5.2. Suppose that in D the $\mathcal{U}$-inductive limits are representable, the finite projective limits are representable, and filtered $\mathcal{U}$-inductive limits commute with finite projective limits. Suppose moreover that, in C, finite projective limits are representable and that the functor u is left exact (2.3.2). Then finite projective limits are representable in $\mathrm{Hom}(C^{\mathrm{op}}, D)$ and in $\mathrm{Hom}((C')^{\mathrm{op}}, D)$, and the functor $u_!$ is left exact.

Proof. The first assertion is trivial. Let us prove the second. By the proof of 5.1, for every presheaf F on C and every object Y of C' one has

$$u_!F(Y) \xrightarrow{\sim} \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y \,.$$

It is therefore enough to show that the category $(I_u^Y)^{\mathrm{op}}$ satisfies axioms PS 1 and PS 2 (2.7), and that this category is connected. The verification is left to the reader.

5.3

Specializing these results to the case where D is the category of $\mathcal{U}$-sets, one obtains a sequence of three functors

$$u_!, \quad u^*, \quad u_*,$$

which is a "sequence of adjoint functors" in the sense that, for two consecutive functors in the sequence, the one on the right is right adjoint to the other. Their essential properties are summarized in the following proposition.

Proposition 5.4. Let C be a small category, let C' be a $\mathcal{U}$-category, and let $u : C \to C'$ be a functor.

1) The functor $u^* : \widehat{C'} \to \widehat{C}$ commutes with inductive and projective limits.

2) The functor $u_* : \widehat{C} \to \widehat{C'}$ commutes with projective limits. For every presheaf F on C and every object Y of C', one has

$$u_*F(Y) \xrightarrow{\sim} \operatorname{Hom}_{\widehat{C}}(u^*(Y), F).$$

3) The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with inductive limits. The functor $u_!$ is defined only up to isomorphism, but it can always be chosen so that the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\ \widehat{C} \arrow[r, "u_!"'] & \widehat{C'} \end{tikzcd}$$

is commutative, where h and h' are the canonical inclusion functors.

For every presheaf F on C, one has

$$u_!F \xrightarrow{\sim} \varinjlim_{C/F} h' \circ u.$$

4) If finite projective limits are representable in C and if u is left exact (2.3.2), then the functor $u_!$ is left exact.

Proof. Assertion (1) is trivial. Assertion (2) follows from the fact that u_* is a right adjoint, by 2.11 and 1.4. The same applies to assertion (3), but one also uses 3.4. Finally, assertion (4) is precisely 5.2, which applies by 2.7.

Proposition 5.5. Let C and C' be two small categories, and let

$$C \underset{v}{\overset{u}{\rightleftarrows}} C'$$

be a pair of functors, where v is left adjoint to u. Then there exist isomorphisms, compatible with the adjunction isomorphisms,

$$v^* \xrightarrow{\sim} u_!, \qquad v_* \xrightarrow{\sim} u^*.$$

Proof. It is enough to exhibit an isomorphism $v^* \xrightarrow{\sim} u_!$; the other isomorphism will follow by adjunction. Let F be a presheaf on C and let Y be an object of C'. Then

$$v^*F(Y) \xrightarrow{\sim} \mathrm{Hom}(v(Y), F).$$

Using 3.4, we get

$$\mathrm{Hom}(v(Y), F) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(v(Y), .).$$

Since v is left adjoint to u, it follows that

$$\varinjlim_{C/F} \mathrm{Hom}(v(Y), .) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(Y, u(.)).$$

Using 5.4.3), we then obtain

$$\varinjlim_{C/F} \mathrm{Hom}(Y, u(.)) \xrightarrow{\sim} \mathrm{Hom}(Y, u_!F) \xrightarrow{\sim} u_!F(Y).$$

Thus, for every object Y of C', we have determined an isomorphism $v^*F(Y) \xrightarrow{\sim} u_!F(Y)$, visibly functorial in Y and in F.

Corollary 5.5.1. Let $u : C \to C'$ be a functor that admits a left adjoint. The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with projective limits; recall that it commutes with inductive limits by 5.4.3.

Remark 5.5.2. One thus obtains a "sequence of *four* adjoint functors" (cf. 5.3):

$$v_!, \quad v^* = u_!, \quad v_* = u^*, \quad u_*,$$

of which the first three (respectively the last three) commute with inductive limits (respectively projective limits).

Proposition 5.6. The hypotheses are those of 5.4. The following conditions are equivalent:

i) the functor u is fully faithful;

ii) the functor $u_!$ is fully faithful;

iii) the adjunction morphism $\mathrm{id}_{\widehat{C}} \to u^*u_!$ is an isomorphism;

iv) the functor u_* is fully faithful;

v) the adjunction morphism $u^*u_* \to \mathrm{id}_{\widehat{C}}$ is an isomorphism.

Proof. It is clear that ii) $\Leftrightarrow$ iii) and iv) $\Leftrightarrow$ v), by general properties of adjoint functors, and that ii) $\Rightarrow$ i), by 5.4.3). Let us prove that i) $\Rightarrow$ iii). The functors $\mathrm{id}_{\widehat{C}}$, u^*, and $u_!$ commute with inductive limits. By 3.4, it is therefore enough to prove that $H \to u^*u_!H$ is an isomorphism when H is representable, which is evident.

It remains to show that iii) is equivalent to v). For every object H (respectively K) of $\widehat{C}$, denote by $\Phi(H) : H \to u^*u_!H$ (respectively by $\Psi(K) : u^*u_*K \to K$) the adjunction morphism. Then one has a commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\widehat{C}}(H, u^*u_*K) \arrow[r, "\mathrm{Hom}(H,\Psi(K))"] \arrow[d, "\sim"'] & \mathrm{Hom}_{\widehat{C}}(H, K) \\
\mathrm{Hom}_{\widehat{C'}}(u_!H, u_*K) \arrow[d, "\sim"'] & \\
\mathrm{Hom}_{\widehat{C}}(u^*u_!H, K) \arrow[uur, "\mathrm{Hom}(\Phi(H),K)"'] &
\end{tikzcd}$$

Hence $\Phi(H)$ is an isomorphism for every H if and only if $\Psi(K)$ is an isomorphism for every K.

Remark 5.7.

a) The equivalences ii) $\Leftrightarrow$ iii) $\Leftrightarrow$ iv) $\Leftrightarrow$ v) are general facts about adjoint functors.

b) The explicit form of $u_!$, respectively u_*, given in the proof of 5.1 shows immediately that, under the general hypotheses of 5.1, if u is fully faithful, then the adjunction morphism $\mathrm{id} \to u^*u_!$ (respectively $u^*u_* \to \mathrm{id}$) is an isomorphism; equivalently, $u_!$ (respectively u_*) is fully faithful.

5.8.0

Let γ be a species of algebraic structure defined by finite projective limits. Let $\mathcal{U}$-γ-**Set** be the category of γ-objects of $\mathcal{U}$-**Set**, and let

$$\mathrm{und} : \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

be the underlying-set functor. For simplicity, we assume that the species of structure under consideration has a single underlying set. Let C be a category. Composition with und gives a functor denoted

$$\widehat{\mathrm{und}} : \mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}) \longrightarrow \widehat{C}.$$

Since in $\widehat{C}$ projective limits are computed argument by argument, the functor $\widehat{\mathrm{und}}$ factors as an equivalence

$$\mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}) \xrightarrow{\approx} \widehat{C}_\gamma, \tag{5.8.1}$$

where $\widehat{C}_\gamma$ is the category of γ-objects of $\widehat{C}$, followed by a functor again denoted

$$\widehat{\mathrm{und}} : \widehat{C}_\gamma \longrightarrow \widehat{C},$$

and called the *underlying presheaf of sets* functor.

5.8.2

Suppose that the functor

$$\mathcal{U}\text{-}\gamma\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

admits a left adjoint

$$\mathrm{Free} : \mathcal{U}\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set},$$

the functor of "freely generated γ-objects." Examples are the free group, free commutative group, free A-module, and so on. In fact one can show that this condition is always satisfied. Composition with Free gives a functor

$$\widehat{\mathrm{Free}} : \widehat{C} \longrightarrow \mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}),$$

and, by composing with the equivalence (5.8.1), a functor again denoted

$$\widehat{\mathrm{Free}} : \widehat{C} \longrightarrow \widehat{C}_\gamma,$$

and called the functor "presheaf of freely generated γ-objects." The functor $\widehat{\mathrm{Free}}$ is left adjoint to $\widehat{\mathrm{und}}$.

Proposition 5.8.3. Let γ be a species of algebraic structure defined by finite projective limits such that, in the category of γ-objects of $\mathcal{U}$-**Set**, $\mathcal{U}$-inductive limits are representable.[2] Let C be a category belonging to $\mathcal{U}$, let C' be a $\mathcal{U}$-category, and let $u : C \to C'$ be a functor. Denote by $\widehat{C}_\gamma$ (respectively $\widehat{C'}_\gamma$) the category of γ-objects of $\widehat{C}$ (respectively $\widehat{C'}$), and by $(u^*)^\gamma$ the functor on γ-objects deduced from u^*. It follows from 5.1 and the equivalence 5.8.1 that there exists a functor left adjoint (respectively right adjoint) to $(u^*)^\gamma$. This functor is denoted $u_{!\gamma}$ (respectively $u_{*\gamma}$).

1) The functor $(u^*)^\gamma$ commutes with inductive and projective limits. The diagram

$$\begin{tikzcd} \widehat{C'}_\gamma \arrow[r, "(u^*)^\gamma"] \arrow[d, "\widehat{\mathrm{und}'}"'] & \widehat{C}_\gamma \arrow[d, "\widehat{\mathrm{und}}"] \\ \widehat{C'} \arrow[r, "u^*"'] & \widehat{C} \end{tikzcd} \qquad (*)$$

is commutative; here $\widehat{\mathrm{und}'}$ and $\widehat{\mathrm{und}}$ denote the underlying-set functors.

2) The functor $u_{*\gamma}$ commutes with projective limits. The diagram

$$\begin{tikzcd} \widehat{C}_\gamma \arrow[r, "u_{*\gamma}"] \arrow[d, "\widehat{\mathrm{und}}"'] & \widehat{C'}_\gamma \arrow[d, "\widehat{\mathrm{und}'}"] \\ \widehat{C} \arrow[r, "u_*"'] & \widehat{C'} \end{tikzcd} \qquad (**)$$

is commutative up to isomorphism.

[2] One can show that this condition is always satisfied.

3) The functor $u_{!\gamma}$ commutes with inductive limits. Suppose that $\widehat{\mathrm{und}}$ (respectively $\widehat{\mathrm{und}'}$) admits a left adjoint $\widehat{\mathrm{Free}}$ (respectively $\widehat{\mathrm{Free}'}$), as in (5.8.2). The diagram

$$\begin{tikzcd} \widehat{C} \arrow[r, "u_!"] \arrow[d, "\widehat{\mathrm{Free}}"'] & \widehat{C'} \arrow[d, "\widehat{\mathrm{Free}'}"] \\ \widehat{C}_\gamma \arrow[r, "u_{!\gamma}"'] & \widehat{C'}_\gamma \end{tikzcd} \tag{***}$$

is commutative up to isomorphism.

Suppose that $u_!$ commutes with finite projective limits (5.4 and 5.6). Then the diagram

$$\begin{tikzcd} \widehat{C}_\gamma \arrow[r, "u_{!\gamma}"] \arrow[d, "\widehat{\mathrm{und}}"'] & \widehat{C'}_\gamma \arrow[d, "\widehat{\mathrm{und}'}"] \\ \widehat{C} \arrow[r, "u_!"'] & \widehat{C'} \end{tikzcd} \tag{****}$$

is commutative up to isomorphism, and $u_{!\gamma}$ commutes with finite projective limits.

Proof. Assertion (1) is evident. Assertion (2) is also evident, since u_* is a right adjoint and hence, by 2.11, commutes with projective limits and in particular with finite projective limits; this gives the commutativity of diagram (**). The commutativity of diagram (***) follows from the uniqueness, up to isomorphism, of the left adjoint functor. Finally, the commutativity of diagram (****) follows immediately from the fact that $u_!$ commutes with finite projective limits.

Notation 5.9. By abuse of notation, the functors $(u^*)^\gamma$ and $u_{*\gamma}$ will often be denoted by u^* and u_*; this risks no confusion in view of the commutativity of diagrams $(*)$ and $(**)$. In contrast, when $u_!$ does not commute with finite projective limits, diagram $(* * **)$ is not commutative up to isomorphism, and the notations $u_!$ and $u_{!\gamma}$ must be used to avoid confusion.

5.10

Let C be a small category. For every object X of $\widehat{C}$, let C/X denote the category of arrows whose target is X and whose source is an object of C. The source functor defines a functor $j_X : C/X \to C$. Let $m : Y \to X$ be a morphism of $\widehat{C}$. Composition of morphisms defines a functor $j_m : C/Y \to C/X$. The diagram

$$\begin{tikzcd} C/Y \arrow[rr, "j_m"] \arrow[dr, "j_Y"'] & & C/X \arrow[dl, "j_X"] \\ & C & \end{tikzcd}$$

is commutative. It follows from 5.1 that, for every object F of $\widehat{C/X}$ and every object Y of C, one has

$$j_{X!}F(Y) = \coprod_{u \in \mathrm{Hom}_{\widehat{C}}(Y,X)} F(u). \tag{5.10.1}$$

Formula (5.10.1) allows one to define $j_{X!}$ when C is a $\mathcal{U}$-category, and one verifies that the functor $j_{X!}$ thus defined is always left adjoint to the functor

$$j_X^* : \widehat{C} \longrightarrow \widehat{C/X}.$$

Proposition 5.11. Let C be a $\mathcal{U}$-category, and let X be a presheaf on C.

1) The functor

$$j_{X!} : \widehat{C/X} \longrightarrow \widehat{C}$$

factors through the category $\widehat{C}/X$:

$$\widehat{C/X} \xrightarrow{e_X} \widehat{C}/X \longrightarrow \widehat{C}.$$

The functor e_X is an equivalence of categories.

2) The functor

$$e_X \circ j_X^* : \widehat{C} \longrightarrow \widehat{C}/X$$

is canonically isomorphic to the functor

$$H \longmapsto (H \times X \xrightarrow{\mathrm{pr}_2} X).$$

Proof.

1) Let f be the final object of $\widehat{C/X}$. One has a canonical isomorphism

$$f \xrightarrow{\sim} \varinjlim_{Y \in \mathrm{Ob}(C/X)} Y,$$

and hence $j_{X!}(f) = \varinjlim_{Y \in \mathrm{Ob}(C/X)} j_X(Y)$ by 5.4. But $j_{X!}(f) \simeq X$, which gives the factorization. To show that e_X is an equivalence, we merely exhibit a quasi-inverse: to every object $H \to X$ of $\widehat{C}/X$ associate the presheaf on C/X

$$(Y \to X) \longmapsto \mathrm{Hom}_{\widehat{C}/X}((Y \to X), (H \to X)).$$

2) The functor $e_X \circ j_X^*$ is right adjoint to the forgetful functor, and therefore $e_X \circ j_X^*$ is canonically isomorphic to the functor $H \mapsto (H \times X \xrightarrow{\mathrm{pr}_2} X)$.

5.12

Let $m : Y \to X$ be a morphism of $\widehat{C}$. By (5.11), the morphism m is canonically isomorphic to the image under e_X of an object of $\widehat{C/X}$, which we shall denote by $[m]$. By restriction to subcategories, the functor e_X defines an equivalence

$$(C/X)/[m] \xrightarrow{e_m} C/Y.$$

The diagram

$$\begin{tikzcd} (C/X)/[m] \arrow[rr, "e_m"] \arrow[dr, "j_{[m]}"'] & & C/Y \arrow[dl, "j_m"] \\ & C/X & \end{tikzcd}$$

is commutative up to canonical isomorphism.

5.13

We record a result that will be useful in Exposé VI. Let $u : C \to C'$ be a functor between small categories. For every object H of $\widehat{C}$, denote by

$$u/H : C/H \longrightarrow C'/u_!H$$

the functor which associates to every morphism $m : X \to H$ the morphism

$$u_!X \xrightarrow{u_!m} u_!H$$

(one knows from 5.4 that one may always set $u_!X = uX$). The following diagram is commutative:

$$\begin{tikzcd} C/H \arrow[r, "u/H"] \arrow[d, "j_H"'] & C'/u_!H \arrow[d, "j_{u_!H}"] \\ C \arrow[r, "u"'] & C'. \end{tikzcd} \tag{5.13.1}$$

We therefore have a diagram commutative up to isomorphism; and, since $(u/H)_!$ carries the final object of $\widehat{C/H}$ to the final object of $\widehat{C'/u_!H}$, the diagram

$$\begin{tikzcd} \widehat{C/H} \arrow[r, "(u/H)_!"] \arrow[d, "e_H"'] & \widehat{C'/u_!H} \arrow[d, "e_{u_!H}"] \\ \widehat{C}/H \arrow[r, "u_!/H"'] & \widehat{C'}/u_!H \end{tikzcd} \tag{5.13.3}$$

is commutative up to isomorphism.

Proposition 5.14. Let $u : C \to C'$ be a functor between small categories. Suppose that u has the following property:

(PPF) For every object X of C, the functor

$$u/X : C/X \longrightarrow C'/uX$$

is fully faithful.

Then:

1) Let f be the final object of $\widehat{C}$. The functor u factors as

$$C = C/f \xrightarrow{u/f} C'/u_!f \xrightarrow{j_{u_!f}} C'.$$

The functor u/f is fully faithful.

2) The functor $u_! : \widehat{C} \to \widehat{C'}$ factors as

$$\widehat{C} = \widehat{C/f} \xrightarrow{(u/f)_!} \widehat{C'/u_!f} \xrightarrow{e_{u_!f}} \widehat{C'}/u_!f \longrightarrow \widehat{C'},$$

where the functor $(u/f)_!$ is fully faithful, the functor $e_{u_!f}$ is an equivalence, and $\widehat{C'}/u_!f \to \widehat{C'}$ is the forgetful functor.

3) In particular, the functor $u_!$ is faithful and consequently the adjunction morphism $\mathrm{id} \xrightarrow{\Phi} u^* u_!$ is a monomorphism. Moreover, for every morphism $\alpha : H \to K$ of $\widehat{C}$, the diagram

$$\begin{tikzcd} H \arrow[r, hook, "\Phi(H)"] \arrow[d, "\alpha"'] & u^* u_! H \arrow[d, "u^* u_!(\alpha)"] \\ K \arrow[r, hook, "\Phi(K)"'] & u^* u_! K \end{tikzcd}$$

is cartesian.

Proof.

1) The factorization comes from diagram (5.13.1). The functor u is faithful. Hence u/f is faithful. Let us show that it is fully faithful. Let X and Y be two objects of C, let $\mathrm{can}_X : uX \to u_! f$ and $\mathrm{can}_Y : uY \to u_! f$ be the canonical morphisms, and let

$$\begin{tikzcd} uX \arrow[rr, "m"] \arrow[dr, "\mathrm{can}_X"'] & & uY \arrow[dl, "\mathrm{can}_Y"] \\ & u_! f & \end{tikzcd}$$

be a morphism of $C'/u_! f$. We have $u_! f = \varinjlim_{Z \in \mathrm{Ob}\, C} uZ$, hence by 3.1,

$$\mathrm{Hom}_{\widehat{C'}}(uX, u_! f) = \varinjlim_{Z \in \mathrm{Ob}\, C} \mathrm{Hom}_{C'}(uX, uZ).$$

By the definition of the inductive limit, to say that $\mathrm{can}_Y\, m = \mathrm{can}_X$ is equivalent to saying that there exist

a) a finite sequence of objects X_i of C, $i \in [0, n]$, with $X_0 = X$ and $X_n = Y$;

b) for every i, a morphism $m_i : uX \to uX_i$, with $m_0 = \mathrm{id}_X$ and $m_n = m$;

c) for every pair $(i, i+1)$, a morphism $f_i : X_i \to X_{i+1}$ or a morphism $f_i : X_{i+1} \to X_i$;

such that the corresponding triangular diagrams

$$\begin{tikzcd} & uX \arrow[dl, "m_i"'] \arrow[dr, "m_{i+1}"] & \\ uX_i \arrow[rr, "u(f_i)"'] & & uX_{i+1} \end{tikzcd} \quad \text{or} \quad \begin{tikzcd} & uX \arrow[dl, "m_i"'] \arrow[dr, "m_{i+1}"] & \\ uX_i & & uX_{i+1} \arrow[ll, "u(f_i)"] \end{tikzcd}$$

are commutative. One then proves immediately, by induction on i and using property (PPF), that m_i is of the form $u(p_i)$. In particular $m = u(p)$, and hence u/f is fully faithful.

2) The factorization is immediate. The functor $(u/f)_!$ is fully faithful by 5.6. The other assertions follow from 5.11.

3) The functor $u_!$ is the composite of the forgetful functor, which is faithful, and fully faithful functors. It is therefore faithful. It follows, from the general properties of adjoint functors, that the adjunction morphism $\mathrm{id} \to u^* u_!$ is a monomorphism. By 2), the functor $u_!$ appears

as the composite of a fully faithful functor $v : \widehat{C} \to \widehat{C'}/u_! f$ and the forgetful functor. The functor u^*, right adjoint to $u_!$, is therefore the composite of the "product by $u_! f$" functor, right adjoint to the forgetful functor, and a functor w right adjoint to v. Moreover, since v is fully faithful, the adjunction morphism $\mathrm{id} \to wv$ is an isomorphism. The last assertion follows easily.

Exposé I. Presheaves

Sections 6–7

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 6–7. Obvious typographical slips in the working French TEX source have been corrected where the mathematics fixes the reading; they are recorded in the accompanying worklog.

6 Faithful functors and conservative functors

Definition 6.1. Let E be a category, and let

$$(\varphi_i : E \to F_i)_{i \in I}$$

be a family of functors. We say that the family (φ_i) is *faithful* if, for every pair of objects X, Y of E and every pair of arrows $u, v : X \rightrightarrows Y$, the equalities $\varphi_i(u) = \varphi_i(v)$ for all $i \in I$ imply $u = v$. In other words, for every X, Y the map

$$\mathrm{Hom}_E(X, Y) \longrightarrow \prod_i \mathrm{Hom}_{F_i}(\varphi_i(X), \varphi_i(Y))$$

defined by the φ_i is injective. We say that (φ_i) is *conservative* if every arrow u of E such that $\varphi_i(u)$ is an isomorphism for all $i \in I$ is itself an isomorphism. We say that (φ_i) is conservative for monomorphisms, respectively for epimorphisms, etc., if the preceding condition is verified whenever u is a monomorphism, respectively an epimorphism, etc.

6.1.1

If one introduces the single functor

$$\varphi : E \longrightarrow F = \prod_{i \in I} F_i$$

defined by the family (φ_i), then it is clear that the family is faithful, respectively conservative, respectively conservative for monomorphisms, etc., if and only if φ is faithful, respectively conservative, etc. Here, of course, this means that the one-object family consisting of φ has the corresponding property. Thus there would be no serious loss in subsequently restricting to the case of a family consisting of a single functor. For convenience of later reference, however, we give the following statements for families.

The notions of 6.1 are especially useful when the φ_i satisfy suitable exactness properties; in that case they tend to coincide.

Proposition 6.2. The notation is that of 5.1.

(i) Suppose that kernels of pairs of arrows, or cokernels of pairs of arrows, are representable in E, and that the φ_i commute with them. Then

$$(\varphi_i) \text{ conservative} \quad \Longrightarrow \quad (\varphi_i) \text{ faithful}.$$

(ii) Suppose that fiber products, respectively amalgamated sums, are representable in E, and that the φ_i commute with them. Suppose that (φ_i) is faithful or conservative. Then, for every arrow u of E, the arrow u is a monomorphism, respectively an epimorphism, if and only if, for every $i \in I$, the arrow $\varphi_i(u)$ has the corresponding property.

(iii) Suppose that in E fiber products and amalgamated sums are representable, that the φ_i commute with them, and that every arrow in E which is a bimorphism, i.e. both a monomorphism and an epimorphism, is an isomorphism. Then

$$(\varphi_i) \text{ faithful} \quad \Longrightarrow \quad (\varphi_i) \text{ conservative.}$$

(iv) Suppose that in E fiber products, respectively amalgamated sums, are representable, and that the φ_i commute with them. If (φ_i) is conservative for monomorphisms, respectively for epimorphisms, then (φ_i) is conservative.

(v) Let D be a diagram type, let $F : d \mapsto F(d)$ be a diagram of type D in E, let X be an object of E, and let $u = (u_d)_{d \in D}$ be a family of arrows

$$X \longrightarrow F(d) \qquad (\text{respectively } F(d) \longrightarrow X).$$

Suppose that (φ_i) is conservative, that projective limits, respectively inductive limits, of type D are representable in E, and that the φ_i commute with them. Then u makes X a projective limit, respectively an inductive limit, of F in E if and only if, for every $i \in I$, $\varphi_i(u)$ makes $\varphi_i(X)$ a projective limit, respectively an inductive limit, of $\varphi_i(F)$ in F_i.

Proof. For (i), in the non-dual statement, given a pair of arrows $u, v : X \rightrightarrows Y$, it suffices to express the equality $u = v$ by the condition that the inclusion

$$\mathrm{Ker}(u, v) \longrightarrow X$$

is an isomorphism. Here, and below, we do not repeat the dual argument for the dual statement.

For (ii), if (φ_i) is faithful, one expresses the condition that $u : X \to Y$ be a monomorphism by the equality $\mathrm{pr}_1 = \mathrm{pr}_2$ for the fiber product $X \times_Y X$. If (φ_i) is conservative, one expresses it by the condition that the diagonal morphism

$$\delta : X \longrightarrow X \times_Y X$$

be an isomorphism.

For (iv), in the latter argument the morphism δ is a monomorphism, so it would actually have sufficed to assume that (φ_i) is conservative for monomorphisms. This then implies that (φ_i) is conservative tout court. Indeed, if $u \in \mathrm{Fl}(E)$ is such that the $\varphi_i(u)$ are isomorphisms, then by what precedes they are monomorphisms, and hence are isomorphisms by the hypothesis on (φ_i).

Assertion (iii) is a trivial consequence of (ii), and (v) is a trivial consequence of the definitions.

We note the following consequence of (i), (ii), and (iv).

Corollary 6.3. Suppose that in E fiber products and amalgamated sums are representable and that the φ_i commute with them, and that kernels of pairs of arrows or cokernels of pairs of arrows are representable and that the φ_i commute with them. It suffices, for example, that finite projective limits and finite inductive limits be representable in E, and that the φ_i be exact functors. Then the following conditions are equivalent:

a) (φ_i) is faithful;

b) (φ_i) is conservative;

c) (φ_i) is conservative for monomorphisms;

(c') (φ_i) is conservative for epimorphisms.

We also record the following fact.

Proposition 6.4. Let $\varphi : E \to F$ be a functor admitting a right adjoint ψ, so that

$$\mathrm{Hom}_F(\varphi(X), Y) \simeq \mathrm{Hom}_E(X, \psi(Y)).$$

For φ, respectively ψ, to be faithful it is necessary and sufficient that, for every object X of E, respectively every object Y of F, the adjunction morphism

$$X \longrightarrow \psi\varphi(X)$$

be a monomorphism. For φ, respectively ψ, to be fully faithful it is necessary and sufficient that the preceding adjunction morphism be an isomorphism.

Indeed, if X', X are two objects of E, the map

$$\mathrm{Hom}_E(X', X) \longrightarrow \mathrm{Hom}_F(\varphi(X'), \varphi(X)) \tag{$*$}$$

is identified with the map obtained from

$$X \longrightarrow \psi\varphi(X) \tag{$**$}$$

by applying the functor $\mathrm{Hom}_E(X', -)$. Thus, for $(*)$ to be a monomorphism, respectively an isomorphism, for all X', with X fixed, it is necessary and sufficient that the adjunction morphism $(**)$ be a monomorphism, respectively an isomorphism.

Proposition 6.5. Let $p : E' \to E$ be a functor. The following conditions are equivalent:

(i) p is faithful, conservative, and fibred *(SGA 1, VI, 6.1)*.

(ii) p is a fibred functor whose fibers are discrete categories.

(iii) For every $X' \in \mathrm{Ob}(E')$, the functor

$$E'_{/X'} \longrightarrow E_{/p(X')}$$

induced by p is an equivalence of categories, surjective on objects.

(iv) If E is a $\mathcal{U}$-category, there exists an object $F \in \mathrm{Ob}(\widehat{E})$ and an equivalence of categories over E *(SGA 1, VI, 4.3)*

$$E' \xrightarrow{\sim} E_{/F},$$

where $E_{/F}$ is the full subcategory of $\widehat{E}_{/F}$ consisting of the arrows $X \to F$ whose source is in E.

6.5.1

Recall that a category C is said to be *discrete* if it is a *groupoid*, i.e. every arrow is invertible, and if it is *rigid*, i.e. the automorphism group of every object is reduced to the unit group. Equivalently, the category is equivalent to the category C' defined by a set I, with $\mathrm{Ob}(C') = I$ and with identity arrows as its only arrows. When C is already assumed to be a groupoid, saying that C is discrete amounts to saying that, for any two objects X, Y of C, there is at most one arrow from X to Y, i.e. that C is isomorphic to the category defined by a preordered set.

The equivalence of conditions (i) and (ii) in 6.5 is an immediate consequence of these reminders and of the following lemma.

Lemma 6.5.2. Let $p : E' \to E$ be a fibred functor. Then:

(i) for p to be conservative it is necessary and sufficient that its fiber categories be groupoids;

(ii) for p to be faithful it is necessary and sufficient that its fiber categories be ordered categories.

Proof. [Proof of 6.5.2] Suppose first that p is conservative. For every arrow u in a fiber E'_X, one has $p(u) = \mathrm{id}_X$, an isomorphism; hence u is an isomorphism in E', and also in E'_X, since an inverse of u in E' is evidently an inverse in E'_X. Thus E'_X is a groupoid. Conversely, suppose that the E'_X are groupoids, and let u' be an arrow of E' such that $p(u')$ is an isomorphism. We prove that u' is an isomorphism. Since p is fibred, one can factor

$$u' : X' \longrightarrow Y'$$

as a composite

$$X' \longrightarrow u^*(Y') \longrightarrow Y',$$

where the first arrow is an X-morphism, with $X = p(X')$ and $u = p(u')$, and the second is a cartesian morphism over u. The first arrow is an isomorphism because E'_X is a groupoid, and the second is an isomorphism because a cartesian morphism in a fibred category is evidently an isomorphism as soon as its projection is an isomorphism.

Now suppose that p is faithful, and let X', Y' be two objects of a fiber category E'_X. Two arrows from X' to Y' lie over the same arrow id_X of E, hence are identical; therefore E'_X is ordered. Conversely, suppose that the fiber categories are ordered, and prove that p is faithful. Let $u', v' : X' \rightrightarrows Y'$ be arrows of E' over the same arrow $u : X \to Y$ of E. They factor as

$$X' \rightrightarrows u^*(Y') \longrightarrow Y',$$

where the two arrows $X' \rightrightarrows u^*(Y')$ are arrows of E'_X with the same source and the same target. These are therefore equal, and hence $u' = v'$. □

We return to the proof of 6.5. We have proved (i) $\Leftrightarrow$ (ii). On the other hand, (ii) $\Leftrightarrow$ (iv) is fairly clear: indeed, on one hand the category $E_{/F}$ is fibred over E with fibers the discrete categories defined by the sets $F(X)$, as follows at once from the definitions; on the other hand, if p is as in (ii), then by the sorites of SGA 1, VI, 8, the category E' fibred over E is E-equivalent to the split category over E defined by the functor $F : E^{\mathrm{op}} \to \mathbf{Set}$ which sends an object X to the set of isomorphism classes of objects of E'_X. This split category is E-isomorphic to $E_{/F}$.

Since (iv) $\Rightarrow$ (iii) is clear, it remains only to prove (iii) $\Rightarrow$ (i). But it is clear that, for p to be faithful, respectively conservative, it is necessary and sufficient that the induced functors

$$E'_{/X'} \longrightarrow E_{/p(X')}$$

be faithful, respectively conservative; *a fortiori*, it suffices that these functors be fully faithful. It remains only to prove that (iii) implies that p is fibred. But again one sees that a functor p is fibred if and only if the induced functors $E'_{/X'} \to E_{/p(X')}$ are fibred. This is so, in particular, if they are equivalences of categories surjective on objects.

7 Generating and cogenerating subcategories

Definition 7.1. Let E be a category and let C be a full subcategory of E. We say that C is a subcategory of E *generating by strict epimorphisms*, respectively *generating by epimorphisms*, if, for every object X of E, the family of arrows of E with target X and with source $X' \in \mathrm{Ob}(C)$ is strictly epimorphic, respectively epimorphic (1.3). We say that C is a *generating subcategory*, respectively *generating for monomorphisms*, respectively *generating for strict monomorphisms*, if, for every arrow $u : Y \to X$, respectively every monomorphism of E, respectively every strict monomorphism of E (10.5), such that for every $X' \in \mathrm{Ob}(C)$ the corresponding map

$$\mathrm{Hom}_E(X', Y) \longrightarrow \mathrm{Hom}_E(X', X)$$

is bijective, u is an isomorphism. Finally, we say that a family (X_i) of objects of E generates by strict epimorphisms, respectively etc., if the full subcategory C of E generated by this family generates by strict epimorphisms, respectively etc.

7.1.1

In terms of the family $(h_{X'})_{X' \in \mathrm{Ob}(C)}$ of functors represented by the objects $X' \in \mathrm{Ob}(C)$,

$$h_{X'} : E \longrightarrow \mathbf{Set} \qquad (X' \in \mathrm{Ob}(C)),$$

one can express the condition that C be generating, respectively generating for monomorphisms, respectively generating for strict monomorphisms, by saying that the family $(h_{X'})$ is conservative, respectively conservative for monomorphisms, respectively conservative for strict monomorphisms (6.1). It follows immediately from the definitions as well that C generates by epimorphisms if and only if the family $(h_{X'})_{X' \in \mathrm{Ob}(C)}$ is faithful (6.1). We shall also give below, in 7.2(i), an analogous interpretation of the condition that C generate by strict epimorphisms.

7.1.2

As with the notions introduced in 6.1, the notions in 7.1 are especially useful when E has suitable exactness properties; then the various notions introduced have a marked tendency all to be equivalent (7.3). For this reason the question of which of the notions in 7.1 should be considered the most important hardly arises: in the most important cases they coincide, and the term "generating subcategory" may therefore be interpreted indifferently as referring to any of the properties considered in 7.1, for example the first one, which is the strongest of all, as we shall see in 7.2(ii).

7.1.3

Suppose that E is a $\mathcal{U}$-category, and consider the canonical functor

$$\varphi : E \longrightarrow \widehat{C} = \mathrm{Hom}(C, \mathcal{U}\text{-}\mathbf{Set}), \tag{7.1.3.1}$$

obtained by composing $E \to \widehat{E} \to \widehat{C}$, where the first functor is the canonical functor (1.3.3), and the second is the restriction functor to C. It is evident that saying that φ is conservative, respectively faithful, is the same as saying that the family of functors

$$h_{X'} : X \longmapsto \mathrm{Hom}_E(X', X) = \varphi(X)(X')$$

for variable $X' \in \mathrm{Ob}(C)$ is a conservative, respectively faithful, family, i.e. also, by 7.1.2, that C is generating, respectively generating by epimorphisms. Similarly, φ is conservative for monomorphisms, respectively for strict monomorphisms, if and only if the family of the $h_{X'}$ is conservative for monomorphisms, respectively for strict monomorphisms; equivalently, if and only if the subcategory C of E is generating for monomorphisms, respectively for strict monomorphisms.

Proposition 7.2. Let E be a $\mathcal{U}$-category, and let C be a full subcategory.

(i) The following conditions are equivalent:

a) C is a subcategory generating by strict epimorphisms.

b) For every $X \in \mathrm{Ob}(E)$, if $C_{/X}$ denotes the full subcategory of $E_{/X}$ formed by the arrows $X' \to X$ with $X' \in \mathrm{Ob}(C)$, then the natural arrow from the inclusion functor $j : C_{/X} \to E$ to the constant functor on $C_{/X}$ with value X makes X an inductive limit of j:

$$X \xleftarrow{\sim} \varinjlim_{C_{/X}} X'.$$

c) The canonical functor φ of (7.1.3.1) is fully faithful.

(ii) Among the notions of 7.1 one has the following implications:

$$\begin{array}{ccc} & \text{1) } C \text{ generates by strict epimorphisms } (\varphi \text{ fully faithful}) & \\ & \Downarrow \quad \searrow & \\ \text{2) } C \text{ generates by epimorphisms } (\varphi \text{ faithful}) & & \text{3) } C \text{ is generating } (\varphi \text{ conservative}) \\ & & \Downarrow \\ & \text{4) } C \text{ generates for monomorphisms } (\varphi \text{ conservative for monomorphisms}) & \\ & \Downarrow & \\ & \text{5) } C \text{ generates for strict monomorphisms } (\varphi \text{ conservative for strict monomorphisms}). & \end{array}$$

(iii) One has the following conditional implications:

a) If in E epimorphic families of arrows are strictly epimorphic, then 2) $\Rightarrow$ 1). If in E monomorphisms are strict, then 5) $\Rightarrow$ 4).

b) If in E kernels of pairs of arrows, respectively fiber products, are representable, then 3) $\Rightarrow$ 2), respectively 4) $\Rightarrow$ 3).

c) If in E every family of morphisms $X_i \to X$ with common target X factors as a strictly epimorphic, respectively epimorphic, family $X_i \to Y$ followed by a monomorphism, respectively by a strict monomorphism, $Y \to X$, then 4) $\Rightarrow$ 1), respectively 5) $\Rightarrow$ 2).

We immediately record the following corollary.

Corollary 7.3. All the notions considered in 7.1, and recalled in the implication diagram of 7.2(ii), are equivalent in each of the following two cases:

(i) In E, kernels of pairs of arrows and fiber products are representable, monomorphisms are strict, and epimorphic families of arrows are strictly epimorphic.

(ii) In E, every family $(X_i \to X)_{i \in I}$ of arrows with common target X factors as an epimorphic family $(X_i \to Y)$ followed by a monomorphism $Y \to X$, every monomorphism of E is strict, and every epimorphic family of arrows of E is strict.

Indeed, in case (i) one has 4) $\Rightarrow$ 3) and 3) $\Rightarrow$ 2) by (b), and 5) $\Rightarrow$ 4) and 2) $\Rightarrow$ 1) by (a). In case (ii) one has, by (c), the implications 4) $\Rightarrow$ 1) and 5) $\Rightarrow$ 2). The conclusion follows from the implication diagram 7.2(ii).

Proof. [Proof of 7.2] For (i), the implication b) $\Rightarrow$ a) follows at once from the definitions. We prove a) $\Rightarrow$ b). Under hypothesis a), one must prove that, for every $X, Y \in \mathrm{Ob}(E)$, every system of arrows

$$u_{X'} : X' \longrightarrow Y$$

indexed by the $X' \in \mathrm{Ob}(C_{/X})$ such that $u_{X'} f = u_{X''}$ for every arrow $f : X'' \to X'$ in $C_{/X}$ factors through an arrow, necessarily unique by hypothesis a), $X \to Y$. By a), it suffices to check that, for every object Z of $E_{/X}$ and every pair of morphisms $v' : Z \to X'$ and $v'' : Z \to X''$ in $E_{/X}$, with X' and X'' in $C_{/X}$, one has $u_{X'} v' = u_{X''} v''$. But by hypothesis a), the family of arrows $w : X''' \to Z$, with $X''' \in \mathrm{Ob}(C)$, is epimorphic; hence it suffices to check that for every such w one has

$$(u_{X'} v')w = (u_{X''} v'')w,$$

which is also written $u_{X'}(v'w) = u_{X''}(v''w)$ and follows immediately from the hypothesis on the family u.

We now prove the equivalence of b) and c). For every $X \in \mathrm{Ob}(E)$, the object $\varphi(X)$ of $\widehat{C}$ is the inductive limit in $\widehat{C}$ of the canonical functor $\widehat{C}_{/\varphi(X)} \to \widehat{C}$ (3.4). But $\widehat{C}_{/\varphi(X)}$ is canonically isomorphic to $C_{/X}$, so that in $\widehat{C}$ one has

$$\varphi(X) = \varinjlim_{C_{/X}} X',$$

where the object X' of C is identified with the functor in $\widehat{C}$ that it represents. Consequently, for a second object Y of E, there is a canonical isomorphism

$$\begin{aligned} \mathrm{Hom}_{\widehat{C}}(\varphi(X), \varphi(Y)) &\simeq \varprojlim_{C_{/X}} \mathrm{Hom}_{\widehat{C}}(X', \varphi(Y)) \\ &\simeq \varprojlim_{C_{/X}} \varphi(Y)(X') \\ &\simeq \varprojlim_{C_{/X}} \mathrm{Hom}_E(X', Y). \end{aligned} \qquad (*)$$

Thus the map

$$\mathrm{Hom}_E(X,Y) \longrightarrow \mathrm{Hom}_{\widehat{C}}(\varphi(X),\varphi(Y))$$

defined by φ is, through the isomorphism between the extreme terms of $(*)$, exactly the map

$$\mathrm{Hom}_E(X,Y) \longrightarrow \varprojlim_{C_{/X}} \mathrm{Hom}_E(X',Y)$$

induced by the inductive system of arrows $X' \to X$ indexed by $C_{/X}$ considered in 7.2(i)b). Therefore the first map is bijective for every Y, with X fixed, if and only if X is an inductive limit of the inclusion functor $j : C_{/X} \to E$. This proves the equivalence of b) and c).

For (ii), the implications 1) $\Rightarrow$ 2) and 3) $\Rightarrow$ 4) $\Rightarrow$ 5) are trivial by the definitions. The implication 1) $\Rightarrow$ 3) is obtained by interpreting property 1) as the full faithfulness of φ, by (i), and observing that a fully faithful functor is conservative. We have already observed in 7.1.1 that 3) means that φ is conservative.

For (iii), assertion (a) is a tautology. Assertion (b) follows from 6.2(i), respectively 6.2(iv), applied to the functor φ, noting that φ is left exact. Finally we prove (c). For $X \in \mathrm{Ob}(E)$, consider the family of all morphisms $X_i \to X$ with $X_i \in \mathrm{Ob}(C)$. By the hypothesis on E, it factors as a strictly epimorphic, respectively epimorphic, family $X_i \to Y$ followed by a monomorphism, respectively by a strict monomorphism, $Y \to X$. Then hypothesis 4), respectively 5), implies that $Y \to X$ is an isomorphism. Thus the family in question is strictly epimorphic, respectively epimorphic. □

Proposition 7.4. Let E be a category, let C be a full subcategory of E, and let X be an object of E. Suppose that C is generating in E, respectively that C generates for monomorphisms and that the fiber product over X of two subobjects of X is representable in E. Then a strict subobject (10.11), respectively a subobject, X' of X is known once, for every $T \in \mathrm{Ob}(C)$, one knows the subset of $\mathrm{Hom}_E(T,X)$ which is the image of $\mathrm{Hom}_E(T,X')$. Consequently, the cardinal of the set of strict subobjects, respectively of the set of subobjects, of X is bounded above by

$$\prod_{T\in\mathrm{Ob}(C)} 2^{\mathrm{card}\,\mathrm{Hom}_E(T,X)},$$

and if $X \in \mathrm{Ob}(C)$, it is bounded above by $2^{\mathrm{card}\,\mathrm{Fl}(C)}$.

We first prove the statement with "respectively". Let X' and X'' be two subobjects of X such that, for every $T \in \mathrm{Ob}(C)$, the images of $\mathrm{Hom}_E(T,X')$ and $\mathrm{Hom}_E(T,X'')$ in $\mathrm{Hom}_E(T,X)$ are equal. They are therefore also equal to the image of $\mathrm{Hom}_E(T,X''')$, where X''' is the fiber product of X' and X'' over X. Since C generates for monomorphisms, it follows that the monomorphisms $X''' \to X'$ and $X''' \to X''$ are isomorphisms. Hence X' and X'' are equal, each being equal to the subobject X''' of X.

We now prove the non-respective assertion. By the definition of strict subobject, it suffices to check that knowing the subset $\mathrm{Hom}_E(T,X')$ of $\mathrm{Hom}_E(T,X)$ for every $T \in \mathrm{Ob}(C)$ determines the set of pairs of arrows $u,v : X \rightrightarrows T$ such that $ui = vi$, where $i : X' \to X$ is the canonical injection. Since C is generating, the relation $ui = vi$ is equivalent to the relation $(ui)f = (vi)f$ for every $f \in \mathrm{Hom}_E(T,X')$, i.e. to $ug = vg$ for every $g \in \mathrm{Hom}_E(T,X)$ coming from $\mathrm{Hom}_E(T,X')$ (that is, of the form if with $f \in \mathrm{Hom}_E(T,X')$). This proves the assertion.

Corollary 7.5. Let E be a category, let C be a full generating subcategory, and let X be an object of C. Then a strict quotient (10.8) X' of X is known once, for every $T \in \mathrm{Ob}(C)$, one knows the subset of $\mathrm{Hom}_E(T, X)^2$ formed by the pairs (u, v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. Thus the cardinal of the set of strict quotients of X is bounded above by

$$\prod_{T \in \mathrm{Ob}(C)} 2^{\mathrm{card}\,\mathrm{Hom}_E(T,X)^2}.$$

Indeed, by definition, a strict quotient X' of X is known once, for every object Y of E, one knows the subset of $\mathrm{Hom}_E(Y, X) \times \mathrm{Hom}_E(Y, X)$ formed by the pairs (u, v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. But the relation $qu = qv$ is equivalent to $(qu)f = (qv)f$ for every morphism $f : T \to Y$ whose source T lies in C, since C is generating. This relation may also be written $q(uf) = q(vf)$, which proves the first assertion of 7.5. The second follows immediately.

7.5.1

One can generalize 7.5 by introducing, for a family of objects $(X_i)_{i\in I}$ of E, the notion of a *strict quotient in E of the family*. By this we mean a strictly epimorphic family $(p_i : X_i \to X')_{i\in I}$ of morphisms of E, with the convention, as for ordinary quotients, that two such families $(p_i : X_i \to X')$ and $(q_i : X_i \to X'')$ are identified if there exists an isomorphism $v : X' \to X''$ (necessarily unique) such that $vp_i = q_i$ for every $i \in I$. With this terminology, the proof of 7.5 applies *mutatis mutandis* to give the following variant.

Variant 7.5.2. Let E and C be as in 7.5, and let $(X_i)_{i\in I}$ be a family of objects of E. Then a strict quotient X' of $(X_i)_{i\in I}$ in E (7.5.1) is known once, for every $T \in \mathrm{Ob}(C)$ and every pair $(i, j) \in I \times I$, one knows the subset of $\mathrm{Hom}_E(T, X_i) \times \mathrm{Hom}_E(T, X_j)$ formed by the pairs (u, v) such that $p_i u = p_j v$, where $p_i : X_i \to X'$ denotes the canonical morphism for each $i \in I$. Consequently, the cardinal of the set of strict quotients of $(X_i)_{i\in I}$ in E is bounded above by

$$\prod_{T\in \mathrm{Ob}(C)} \prod_{i,j\in I} 2^{\mathrm{card}\,\mathrm{Hom}_E(T,X_i)\times \mathrm{card}\,\mathrm{Hom}_E(T,X_j)}.$$

7.5.3

One sees at once that the analogous conclusion is true if one assumes only that C generates for strict monomorphisms, provided one assumes that the products $X_i \times X_j$ are representable and restricts to *effective* quotients of the family $(X_i)_{i\in I}$, i.e. to strict quotients X' such that the fiber products $X_i \times_{X'} X_j$ are representable in E; this is no restriction if E is stable under fiber products.

Proposition 7.6. Let E be a category, let C be a full subcategory of E generating by epimorphisms (7.1), let Π_0 be an infinite cardinal $\geq \mathrm{card}\,\mathrm{Fl}(C)$, let $\Pi \geq \Pi_0$ be a cardinal, and let $(u_i : X_i \to X)_{i\in I}$ be a universal epimorphic family (10.3) in E formed by squareable morphisms (10.7), with sources $X_i \in \mathrm{Ob}(C)$, and such that $\mathrm{card}(I) \leq \Pi$. Then

$$\mathrm{card}\,\mathrm{Fl}(C_{/X}) \leq \Pi^{\Pi_0}.$$

Let $T \in \mathrm{Ob}(C)$. It will be enough to prove that

$$\mathrm{card}\,\mathrm{Hom}_E(T, X) \leq \Pi^{\Pi_0}. \tag{7.6.1}$$

for then one successively obtains

$$\operatorname{card}\operatorname{Ob}(C_{/X}) \leq (\Pi^{\Pi_0}) \times \operatorname{card}\operatorname{Ob}(C) \leq (\Pi^{\Pi_0}) \times \Pi_0 = \Pi^{\Pi_0},$$

and finally

$$\operatorname{card}\operatorname{Fl}(C_{/X}) \leq ((\Pi^{\Pi_0})^2) \times \Pi_0 = \Pi^{\Pi_0},$$

since for two objects S, T of $C_{/X}$ one has

$$\operatorname{card}\operatorname{Hom}_{C_{/X}}(S,T) \leq \operatorname{card}\operatorname{Hom}_C(S,T) \leq \Pi_0.$$

To prove (7.6.1), first note the following lemma.

Lemma 7.6.2. Let J be a set such that $\operatorname{card}(J) = \Pi_0$. For every object T of C and every morphism $f : T \to X$, there exists an epimorphic family $(v_j : S_j \to T)_{j\in J}$, with sources $S_j \in \operatorname{Ob}(C)$, and for every $j \in J$ an element $i(j) \in I$ and a morphism $g_j : S_j \to X_{i(j)}$, such that

$$u_{i(j)} g_j = f v_j.$$

Indeed, the family of projections

$$X_i \times_X T \longrightarrow T$$

is epimorphic by hypothesis. On the other hand, since C generates by epimorphisms, for every i the family of arrows $S \to X_i \times_X T$ with source $S \in \operatorname{Ob}(C)$ is epimorphic. Thus, by transitivity, the family of arrows $v : S \to T$ with source in C which factor through one of the $X_i \times_X T$ is epimorphic. These are precisely the arrows $v : S \to T$ with source in C for which there exist $i \in I$ and $g : S \to X_i$ such that $u_i g = fv$. Since the set of these arrows $v : S \to T$ is contained in $\operatorname{Fl}(C)$ and therefore has cardinal $\leq \Pi_0$, it can be indexed by the index set J. This proves the lemma.

Now a morphism $f : T \to X$ is known once the fv_j are known, and these are known once the g_j are known. Thus $\operatorname{card}\operatorname{Hom}_E(T, X)$ is bounded above by the cardinal of the set of families $(i(j), S_j, v_j, g_j)_{j\in J}$. The cardinal of the set of maps $j \mapsto i(j)$ from J to I is $\leq \Pi^{\Pi_0}$; and for such a map fixed, the cardinal of the set of the corresponding families S_j, g_j, v_j is bounded above by the cardinal of the set of maps from J to $\operatorname{Fl}(C) \times \operatorname{Fl}(C)$, which is $\leq \Pi_0^{\Pi_0}$, since $\operatorname{card}(\operatorname{Fl}(C) \times \operatorname{Fl}(C)) = \Pi_0^2 = \Pi_0$. Thus the left hand side of (7.6.1) is bounded above by

$$\Pi^{\Pi_0} \times \Pi_0^{\Pi_0} = \Pi^{\Pi_0},$$

which completes the proof of 7.6.

Proposition 7.7. Let E be a category in which fiber products are representable, let $(X_\alpha)_{\alpha\in A}$ be a generating family of objects of E, and let $(\varphi_i)_{i\in I}$ be a family of functors $\varphi_i : E \to E_i$ commuting with fiber products. For (φ_i) to be conservative (6.1), it is necessary and sufficient that for every $\alpha \in A$ and every subobject X' of X_α distinct from X_α, there exist $i \in I$ such that the morphism

$$\varphi_i(X') \longrightarrow \varphi_i(X_\alpha)$$

is not an isomorphism. In this case, if E is a $\mathcal{U}$-category and A is $\mathcal{U}$-small, there exists a $\mathcal{U}$-small subset J of I such that $(\varphi_j)_{j\in J}$ is already a conservative family of functors.

The necessity of the condition of conservativity under consideration is evident; we prove its sufficiency. By 6.2(iv), it suffices to prove that (φ_i) is conservative for monomorphisms. Let $u : Y' \to Y$ be a monomorphism in E which is not an isomorphism. We must prove that there is an $i \in I$ such that $\varphi_i(u)$ is not an isomorphism. By the hypothesis on (X_α), there exists an α and a morphism $X_\alpha \to Y$ which does not factor through Y'. In other words, the inverse image X' of Y' is a subobject of X_α distinct from X_α. By hypothesis, there exists an $i \in I$ such that

$$\varphi_i(X') \longrightarrow \varphi_i(X_\alpha)$$

is not an isomorphism. Since

$$\varphi_i(X') \simeq \varphi_i(X_\alpha) \times_{\varphi_i(Y)} \varphi_i(Y'),$$

it follows that $\varphi_i(Y') \to \varphi_i(Y)$ is not an isomorphism.

The second assertion of 7.7 follows at once from the first, taking 7.4 into account.

Corollary 7.7.1. Let E be a $\mathcal{U}$-category in which fiber products are representable, and suppose that E admits a generating family of objects which is $\mathcal{U}$-small. Then, for every generating family $(Y_i)_{i\in I}$ of E, there exists a $\mathcal{U}$-small generating subfamily $(Y_j)_{j\in J}$, with $\mathrm{card}(J) \in \mathcal{U}$.

It suffices to apply 7.7 to a $\mathcal{U}$-small generating family $(X_\alpha)_{\alpha\in A}$ of E and to the family of functors φ_i represented by the Y_i.

Proposition 7.8. Let E be a category, let C be a full subcategory generating by strict epimorphisms, let D be a category, and let $\mathrm{Hom}'(E, D)$ be the full subcategory of $\mathrm{Hom}(E, D)$ consisting of the functors which commute with inductive limits of type $C_{/X}$, where X ranges over the objects of E. Then the functor $F \mapsto F|_C$ induces a fully faithful functor

$$\mathrm{Hom}'(E, D) \longrightarrow \mathrm{Hom}(C, D).$$

Consequently, if $\mathrm{Hom}(C, D)$ is a $\mathcal{U}$-category (1.1), for example by (1.1.1 b)) if C is $\mathcal{U}$-small and D is a $\mathcal{U}$-category, then $\mathrm{Hom}'(E, D)$ is also a $\mathcal{U}$-category.

Let $F, G : E \to D$ be two functors in the source category, and let $u : F \to G$ be a morphism. If $X = \varinjlim X_i$ in E and if F and G commute with the inductive limit in question, then $u(X) : F(X) \to G(X)$ is identified with the limit of the morphisms $u(X_i) : F(X_i) \to G(X_i)$, and hence is known once the $u(X_i)$ are known. This shows that the functor considered in 7.8 is faithful, by the implication a) $\Rightarrow$ b) in 7.2(i). Conversely, let $v : F|_C \to G|_C$ be a morphism; we prove that it comes from a morphism $u : F \to G$. For every $X \in \mathrm{Ob}(E)$ define

$$u(X) : F(X) = \varinjlim_{C_{/X}} F(X_i) \longrightarrow G(X) = \varinjlim_{C_{/X}} G(X_i)$$

as the inductive limit of the $v(X_i)$. It is immediate that this gives a morphism functorial in X, hence a morphism $u : F \to G$, and finally that the morphism induced by u from $F|_C$ to $G|_C$ is v. This completes the proof.

7.1 Families and subcategories cogenerating

Let E be a category, and let C be a full subcategory of E. We say that C is cogenerating by strict monomorphisms, respectively cogenerating by monomorphisms, respectively cogenerating, respectively cogenerating for epimorphisms, respectively cogenerating for strict epimorphisms, if the full subcategory C^{op} of E^{op} generates by strict epimorphisms, respectively etc. The terminology for families is analogous. Thus all the results in this section concerning the notion of generating subcategory and its variants (7.1) give corresponding results for the dual notions, which we leave to the reader to formulate for personal satisfaction.

Proposition 7.10. Let E be a category, let C be a full generating subcategory (7.1), and let D be a full subcategory of E. For D to be cogenerating (7.9), it is necessary and sufficient that, for every pair of arrows

$$T \underset{v}{\overset{u}{\rightrightarrows}} X$$

in E with source $T \in \mathrm{Ob}(C)$ and with $u \neq v$, there exist an arrow $w : X \to I$ with target $I \in \mathrm{Ob}(D)$ such that $wu \neq wv$.

By definition, to say that D is cogenerating means that for every pair of arrows $Y \rightrightarrows X$ in E such that $u \neq v$, there exists an arrow $w : X \to I$, with $I \in \mathrm{Ob}(D)$, such that $wu \neq wv$. Thus 7.10 simply says that it suffices to test this property when $Y \in \mathrm{Ob}(C)$. Since C is generating, the hypothesis $u \neq v$ implies that there exists $f : T \to Y$ such that $uf \neq vf$. By the hypothesis there exists $w : X \to I$, with target $I \in \mathrm{Ob}(D)$, such that $w(uf) \neq w(vf)$, and hence $wu \neq wv$. $\square$

Corollary 7.11. With the notation of 7.10, suppose that the objects I of D are injective objects of E, i.e. that for every monomorphism $X \to Y$ in E, the corresponding map

$$\mathrm{Hom}_E(Y, I) \longrightarrow \mathrm{Hom}_E(X, I)$$

is surjective. Suppose moreover that every pair of arrows

$$T \underset{v}{\overset{u}{\rightrightarrows}} X$$

in E factors as an epimorphic, respectively effective epimorphic, pair $T \rightrightarrows X'$ followed by a monomorphism $i : X' \to X$. Then in the criterion 7.10 for D to be cogenerating, one may restrict to pairs (u, v) which are epimorphic, respectively effective epimorphic.

We conclude:

Corollary 7.12. Let E be a $\mathcal{U}$-category admitting a small generating subcategory C, and such that every object of E is the source of a monomorphism into an injective object of E. Suppose moreover that, for every $T \in \mathrm{Ob}(C)$, the sum $T \amalg T$ in E is representable, and that every morphism with source $T \amalg T$ factors as an effective epimorphism followed by a monomorphism. Then E admits a small full cogenerating subcategory D. More precisely, one may take D such that

$$\mathrm{card}\,\mathrm{Ob}(D) \leq \prod_{T, T' \in \mathrm{Ob}(C)} 2^{\mathrm{card}(\mathrm{Hom}_E(T', T \amalg T))^2}.$$

Indeed, by 7.11 it is enough, for every $T \in \mathrm{Ob}(C)$ and every effective quotient X of $T \amalg T$, to choose an embedding of X into an injective object I of E, and to take for D the full subcategory of E generated by these I. The conclusion then follows from 7.5.

Examples 7.13. To construct small cogenerating subcategories in terms of small generating subcategories, one is therefore led to look for conditions ensuring that a $\mathcal{U}$-category E has "enough injectives", i.e. that every object embeds into an injective object by a monomorphism. It is well known [Tohoku] that this condition is satisfied in a $\mathcal{U}$-abelian category with exact small filtered inductive limits (axiom AB5 of *loc. cit.*) admitting a small generating family. The construction of *loc. cit.* is not, moreover, essentially tied to abelian categories, and also works in the category of sheaves of $\mathcal{U}$-sets on a topological space $X \in \mathcal{U}$. We shall not state here the exactness properties which make the construction in question work, and shall limit ourselves to observing that in the special case of the category of sheaves of sets on X, one immediately reduces to the case of *loc. cit.* as follows.

Note that if $\mathcal{O}_X$ is a ring on X, then every injective object I of the category of $\mathcal{O}_X$-modules is also injective as an object of the category of sheaves of sets. Indeed, if F is a sheaf of sets and $\mathcal{O}_X[F]$ is the "free $\mathcal{O}_X$-module generated by F", then by definition there is a homomorphism of sheaves of sets

$$F \longrightarrow \mathcal{O}_X[F] \tag{$*$}$$

giving an isomorphism, functorial in F,

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X[F], I) \xrightarrow{\sim} \mathrm{Hom}(F, I).$$

Since the functor $F \mapsto \mathcal{O}_X[F]$ plainly carries monomorphisms to monomorphisms, it follows at once that, if I is an injective $\mathcal{O}_X$-module, then it is also an injective sheaf of sets. It follows that if the morphism $(*)$ is a monomorphism, as happens if one takes for $\mathcal{O}_X$ a constant ring with value a ring $A \neq 0$, then an embedding of $\mathcal{O}_X[F]$ into an injective $\mathcal{O}_X$-module I gives an embedding of F into the underlying injective sheaf of sets of I.

The same argument applies, without change, to the category of sheaves of $\mathcal{U}$-sets on a $\mathcal{U}$-site, which will be introduced in the next exposé.

The interest of the existence of a small cogenerating subcategory lies chiefly in the representability criterion 8.12.7 below.

Exposé I. Presheaves

Section 8.1–8.6

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 8.1 through Section 8.6. The source has one visibly truncated sentence in the proof of Proposition 8.6.4; the translation marks the mathematically forced completion as referring to the immediately following Corollary 8.6.5.

8 Ind-objects and pro-objects

8.1 Cofinal functors and cofinal subcategories

Definition 8.1.1. Let

$$\varphi : I \longrightarrow I' \tag{8.1.1.1}$$

be a functor. We say that φ is a *cofinal functor* if, for every category C and every functor $u : I' \to C$, considering $\varinjlim u$ and $\varinjlim u \circ \varphi$ as objects of $\mathrm{Hom}(G(\mathcal{V}\text{-}\mathbf{Set}))^{\mathrm{op}}$ (2.1, 2.3.1), where $\mathcal{V}$ is a universe such that $I, I' \in \mathcal{V}$ and C is a $\mathcal{V}$-category, the canonical morphism

$$\varinjlim u \circ \varphi \longrightarrow \varinjlim u \tag{8.1.1.2}$$

is an isomorphism. When φ is the inclusion functor of a subcategory of I', we say that I is a *cofinal subcategory* if φ is cofinal.

8.1.2

Notice that, for fixed φ and u, the bijectivity of (8.1.1.2) does not depend on the choice of the universe $\mathcal{V}$.

Returning to the definition of the terms appearing in (8.1.1.2), one sees that to say that φ is cofinal is also to say that, for every universe $\mathcal{V}$ such that I and I' are $\mathcal{V}$-small, and for every functor

$$v : I'^{\mathrm{op}} \longrightarrow (\mathcal{V}\text{-}\mathbf{Set}),$$

the canonical homomorphism

$$\varprojlim v \longrightarrow \varprojlim v \circ \varphi \tag{8.1.2.1}$$

is an isomorphism, that is, a bijection. To see that this condition is indeed necessary, observe, putting $v = v^{\mathrm{op}}$, that it also says that the condition of Definition 8.1.1 is fulfilled when one takes $C = (\mathcal{V}\text{-}\mathbf{Set})^{\mathrm{op}}$; this implies that the inductive limits considered in (8.1.1.2) are representable in C.

8.1.2.2

It is immediate from the definition that *the composite of two cofinal functors is cofinal.*

Proposition 8.1.3. Let $\varphi : I \to I'$ be a functor.

a) For φ to be cofinal, it is necessary that φ satisfy the condition:

F 1) For every object i' of I', there exists an object i of I such that $\varphi(i)$ dominates i', i.e. such that $\mathrm{Hom}(i', \varphi(i)) \neq \emptyset$.

b) Suppose that I is filtered. For φ to be cofinal, it is necessary and sufficient that φ satisfy condition (F 1) and the following condition:

F 2) For every object i of I and every double arrow

$$i' \underset{g'}{\overset{f'}{\rightrightarrows}} \varphi(i)$$

in I' with target $\varphi(i)$, there exists an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$.

Moreover, if φ is cofinal, then I' is filtered.

c) Suppose that I' is filtered and that φ is fully faithful. For φ to be cofinal, it is necessary and sufficient that it satisfy condition F 1) of a). This implies that I is filtered.

Proof. For necessity in a) and b), one uses Definition 8.1.1 only in the case where u is the canonical inclusion functor

$$u : I' \hookrightarrow \widehat{I'}$$

(choosing a universe $\mathcal{U}$ such that I and I' are $\mathcal{U}$-small). Then (8.1.1.2) may be interpreted as an arrow of $\widehat{I'}$, whose target is the *final* functor on I'. One sees immediately that condition F 1) expresses that this arrow is an epimorphism of $\widehat{I'}$, i.e. that it is surjective on each argument, since inductive limits in $\widehat{I'}$ are calculated argument by argument. This proves a).

Now suppose that I is filtered and that φ is cofinal. Then I' is filtered: indeed, the condition that two objects of I' be dominated by a third follows at once from the corresponding condition on I and from F 1). It remains only to prove condition PS 2) of 2.7, i.e. that for every double arrow

$$f', g' : i' \rightrightarrows j'$$

in I', there exists an arrow $h' : j' \to k'$ in I' such that $h'f' = h'g'$. By F 1) we may suppose that k' is of the form $\varphi(i)$. But by the hypothesis that φ is cofinal, for every object i' of I' the set $\varinjlim_i \mathrm{Hom}(i', \varphi(i))$ has one element. It follows from the standard description of filtered inductive limits in **Set** (2.8.1) that there is an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$. This finishes the proof that I' is filtered, and proves F 2) at the same time.

To prove that the conditions stated in b) are sufficient for φ to be cofinal, use the form 8.1.2 of the definition. One immediately checks that F 1) implies that (8.1.2.1) is a monomorphism, i.e. injective on each argument, while F 2), together with F 1), ensures that it is bijective, since projective limits in $\widehat{I'^{\mathrm{op}}}$ are calculated argument by argument. This proves b).

Finally, if I' is filtered and φ is fully faithful, it is immediate that F 1) implies that I is filtered and implies F 2). Thus c) follows from a) and b).

8.1.4

When I' is a filtered category and I is a full subcategory of I', Proposition 8.1.3 c) shows that the condition that I be cofinal in I' depends only on the subset $\mathrm{Ob}\, I$ of the preordered set $\mathrm{Ob}\, I'$ (for the preorder relation "$x \leq y$" iff "$\mathrm{Hom}(x, y) \neq \emptyset$"). If J' is a preordered set and J is a subset of J', we shall sometimes say that J is a *cofinal subset of* J' when every element of J' is dominated by an element of J. When J' is filtered, this therefore means that the inclusion functor $\mathcal{J} \hookrightarrow \mathcal{J}'$ of the associated categories is cofinal.

8.1.5

In what follows, we shall use the notion of cofinal functor only in cases where the categories I and I' are filtered. Classically, one even restricted oneself to categories associated with preordered sets, i.e. categories in which there is at most one arrow with given source and target. It turns out, however, that this restriction is inconvenient in applications, because the "natural" filtered categories which arise in many applications are *not* ordered categories. The following result, due to P. Deligne, nevertheless shows that there is no essential difference between the two points of view.

Proposition 8.1.6. Let I be a small filtered category. Then there exists a small ordered set E, and a cofinal functor

$$\varphi : \mathcal{E} \longrightarrow I,$$

where $\mathcal{E}$ denotes the category associated with E.

First suppose that the preordered set $\operatorname{Ob} I$ has no greatest element. By a subdiagram of I we mean a pair $D = (\mathcal{O}, F)$ consisting of a subset F of $\operatorname{Fl}(I)$ and a subset $\mathcal{O}$ of $\operatorname{Ob} I$, such that for every $f \in F$, the source and target of f belong to $\mathcal{O}$. An element e of $\mathcal{O}$ is called a *final object* of the subdiagram D if, for every $x \in D$, the set $\operatorname{Hom}(x, e) \cap F$ of arrows of D with source x and target e has exactly one element f_x, if for every arrow $g : x \to y$ of D one has $f_x = f_y \circ g$, and if $f_e = \mathrm{id}_e$.

Let E be the set of *finite* subdiagrams D of I having a *unique* final element $\varphi(D)$, and order E by inclusion. If D, D' are elements of E and $D \subset D'$, then there is a unique arrow $\varphi(D', D)$ of D' with source $\varphi(D)$ and target $\varphi(D')$; if $D \subset D' \subset D''$ are inclusions in E, then of course

$$\varphi(D'', D) = \varphi(D'', D')\varphi(D', D),$$

and also $\varphi(D, D) = \mathrm{id}_{\varphi(D)}$. Thus one obtains a functor $\varphi : \mathcal{E} \to I$, and it remains to prove that E is filtered and that φ is cofinal. By Proposition 8.1.3 b), three verifications are required.

1) Condition F 1) of 8.1.3. For every $i \in I$, take D to be the subdiagram of I reduced to the object i and its identity arrow. Then $i \leq \varphi(D) = i$. This condition F 1) also shows that $E \neq \emptyset$.

2) Condition F 2). For every $i \in \operatorname{Ob} I$, every $D \in E$, and every double arrow $i \rightrightarrows \varphi(D)$, one must find a diagram $D' \in E$, containing D, such that

$$\varphi(D', D) : \varphi(D) \longrightarrow \varphi(D')$$

equalizes the double arrow. Since I is filtered, there exists an object j of I and an arrow $f : \varphi(D) \to j$ equalizing the double arrow. Since I has no greatest element, we may suppose that j is a *strict* upper bound of $\varphi(D)$, which implies that it is distinct from all the other objects of D. Let D' be the subdiagram of I whose set of objects is $\mathcal{O} \cup \{j\}$, and whose set of arrows consists of the arrows of D, together with the composites

$$x \xrightarrow{f_x} \varphi(D) \xrightarrow{f} j,$$

where x is an object of D and f_x is the unique arrow of D with source x and target $\varphi(D)$, and together with id_j. It is then clear that D' is a finite subdiagram of I, that it has j as its unique final object, hence that $D' \in E$, and that D' satisfies the required condition.

3) Let D and D' be two subdiagrams in E. They are contained in some common $D'' \in E$. Indeed, one can find a strict upper bound j of $\varphi(D)$ and $\varphi(D')$,

$$\begin{tikzcd} \varphi(D) \arrow[dr, "f"] & \\ & j \\ \varphi(D') \arrow[ur, "f'"'] & \end{tikzcd}$$

and take D'' to be the subdiagram whose set of objects is the union of the sets of objects of D and D' and of $\{j\}$, and whose set of arrows is the union of the sets of arrows of D and D', the composites $f \circ f_x$ with x an object of D, the composites $f' \circ f'_{x'}$ with x' an object of D', and $\{\mathrm{id}_j\}$. This is indeed a finite subdiagram of I. We show that, after replacing j, f, f' by j', gf, gf' with a suitable $g : j \to j'$, we may arrange for j to be a final object of D''. This is the same as saying that, for every x which is an object of both D and D', one has

$$f f_x = f' f'_{x'}.$$

But this is only a *finite* set of double arrows to be equalized by some $g : j \to j'$, and this is possible because I is filtered.

This completes the proof in the case considered. The general case is reduced to it by introducing the filtered category $\mathbb{N}$ associated with the ordered set $\mathbb{N}$ of natural numbers, and observing that the category $\mathbb{N} \times I$ is filtered, has no greatest element, and that the projection $\mathbb{N} \times I \to I$ is a cofinal functor.

Corollary 8.1.7. Let I be a $\mathcal{U}$-category. For there to exist a small filtered ordered set E and a cofinal functor $\mathcal{E} \to I$, it is necessary and sufficient that I be filtered and that $\mathrm{Ob}\, I$ admit a small cofinal subset I' (8.1.4).

This is necessary by 8.1.3 a), and sufficient by 8.1.6 applied to the full subcategory of I defined by I'.

Definition 8.1.8. Let I be a filtered category. We say that I is *essentially small* if I is a $\mathcal{U}$-category and if it satisfies the equivalent conditions of 8.1.7.

8.2 Ind-objects and ind-representable functors

8.2.1

In what follows we fix a $\mathcal{U}$-category C, which we always regard as embedded in the category $\widehat{C}$ of $\mathcal{U}$-presheaves by means of the canonical functor (1.3.1)

$$h : C \hookrightarrow \widehat{C}. \tag{8.2.1.1}$$

We shall call an *ind-object of* C (or an *inductive system* of C) any functor

$$\varphi : I \longrightarrow C, \tag{8.2.1.2}$$

where I is a *filtered* category (2.7), called the *index category* of the ind-object. Apart from this, I is arbitrary; in particular we do not require I to be a $\mathcal{U}$-category. It will often be convenient to denote an ind-object φ by the indexed notation $(X_i)_{i \in \operatorname{Ob} I}$, or even, by a further abuse of notation, by $(X_i)_{i \in I}$, $(X_i)_i$, or (X_i), where $X_i = \varphi(i)$ for $i \in \operatorname{Ob} I$. This notation is no more and no less abusive than the classical notation for inductive systems indexed by filtered ordered sets, where the transition morphisms are likewise not mentioned. The X_i are called the *component objects* of the ind-object $\mathcal{X}$. Notice that, in the general case considered here, the "*transition morphisms*" $\varphi(f)$ of the inductive system are indexed by the set $\operatorname{Fl}(I)$ of arrows of I, which in general is not identified with a subset of $\operatorname{Ob} I \times \operatorname{Ob} I$. In other words, for given i and j, with j dominating i, there may exist more than one transition morphism from X_i to X_j.

8.2.2

The most useful ind-objects $\mathcal{X} = (X_i)_{i \in I}$ of C are those for which the index category I is essentially small (8.1.8); such an ind-object is called *essentially small.* If $(X_i)_{i \in I}$ is such an object, using the fact that small inductive limits in $\widehat{C}$ are representable (3.1), one may consider

$$L(\mathcal{X}) := \varinjlim h \cdot \mathcal{X} = \varinjlim_i X_i \in \operatorname{Ob} \widehat{C}, \tag{8.2.2.1}$$

where the last limit is taken *in* $\widehat{C}$. Thus $L(\mathcal{X})$ is the presheaf

$$L(\mathcal{X}) : Y \longmapsto \varinjlim_i \operatorname{Hom}(Y, X_i) \tag{8.2.2.2}$$

on C. We say that this presheaf is the *presheaf ind-represented by the ind-object* $\mathcal{X} = (X_i)_{i \in I}$ *of* C. A presheaf F on C is called an *ind-representable presheaf* if it is isomorphic to a presheaf $L(\mathcal{X})$ ind-represented by an essentially small ind-object. It follows from 8.1.6 that, in this definition, one may suppose that I is a small category, and even that I is associated with an ordered set belonging to $\mathcal{U}$.

8.2.3

Consider an ind-object $\mathcal{X} = (X_i)_{i \in I}$, given by a functor $\varphi : I \to C$, and let

$$u : I' \longrightarrow I$$

be a functor, where I' is a second filtered category. Then the functor $\varphi \circ u$ is an ind-object $\mathcal{X}'$ of C with index category I', written in indexed notation as $(X_{u(i')})_{i' \in I'}$. It is called the ind-object of C *deduced from* $(X_i)_{i \in I}$ *by change of index categories* using the functor u. If I and I', equivalently $\mathcal{X}$ and $\mathcal{X}'$, are essentially small, one obtains a canonical morphism

$$L(\mathcal{X}') = \varinjlim_{i'} X_{u(i')} \longrightarrow L(\mathcal{X}) = \varinjlim_i X_i \tag{8.2.3.1}$$

between the presheaves ind-represented by $\mathcal{X}'$ and $\mathcal{X}$. When u is a cofinal functor (8.1.1), $\mathcal{X}$ is essentially small if and only if $\mathcal{X}'$ is so (8.1.7), and the preceding homomorphism (8.2.3.1) is an isomorphism

$$L(\mathcal{X}') \simeq L(\mathcal{X}). \tag{8.2.3.2}$$

8.2.4

Let

$$\mathcal{X} = (X_i)_{i\in I}, \qquad \mathcal{Y} = (Y_j)_{j\in J} \tag{8.2.4.1}$$

be two essentially small ind-objects of C, indexed by filtered essentially small categories I and J. A *morphism* from the ind-object $\mathcal{X}$ to the ind-object $\mathcal{Y}$ is by definition a morphism

$$L(\mathcal{X}) \longrightarrow L(\mathcal{Y}) \tag{8.2.4.2}$$

between the presheaves they ind-represent. We put

$$\mathrm{Hom}_{\text{ind-ob}}(\mathcal{X}, \mathcal{Y}) = \mathrm{Hom}_{\widehat{C}}(L(\mathcal{X}), L(\mathcal{Y})). \tag{8.2.4.3}$$

We suppress the subscript "ind-ob" on Hom if no confusion is likely. Composition of morphisms of ind-objects of C is defined as composition of morphisms of the presheaves they ind-represent. If $\mathcal{V} \supset \mathcal{U}$ is a universe containing $\mathcal{U}$, and if one restricts to essentially small ind-objects which belong to $\mathcal{V}$, these form the set of objects of a category whose arrows are triples $(\mathcal{X}, \mathcal{Y}, u)$, where $\mathcal{X}$ and $\mathcal{Y}$ are essentially small ind-objects belonging to $\mathcal{V}$ and $u : \mathcal{X} \to \mathcal{Y}$ is a morphism of ind-objects, i.e. a morphism $L(\mathcal{X}) \to L(\mathcal{Y})$. The category so obtained is denoted

$$\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U}). \tag{8.2.4.4}$$

When $\mathcal{V} = \mathcal{U}$, it is denoted simply

$$\mathrm{Ind}_{\mathcal{U}}(C) \quad \text{or} \quad \mathrm{Ind}(C), \tag{8.2.4.5}$$

and is called the *category of ind-objects of* C relative to $\mathcal{U}$, the mention of $\mathcal{U}$ being suppressed when no confusion is likely.

Of course, if $\mathcal{V} \subset \mathcal{V}'$ are two universes containing $\mathcal{U}$, then $\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U})$ is a full subcategory of $\mathrm{Ind}_{\mathcal{V}'}(C,\mathcal{U})$. It follows from 8.2.3 that the inclusion functor is an *equivalence of categories*

$$\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U}) \xrightarrow{\approx} \mathrm{Ind}_{\mathcal{V}'}(C,\mathcal{U}). \tag{8.2.4.6}$$

This shows in particular that *every essentially small ind-object of* C *defines an element of* $\mathrm{Ind}(C)$*, uniquely up to isomorphism.* This observation justifies the fairly common practical abuse of language which consists in identifying any essentially small ind-object of C with an object of $\mathrm{Ind}(C)$.

It follows at once from the definitions that for every universe $\mathcal{V}$ we have a canonical functor

$$L : \mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U}) \longrightarrow \widehat{C} \tag{8.2.4.7}$$

which is fully faithful. These functors are of course known up to unique isomorphism, by (8.2.4.6), once one of them is known, in particular once one knows the canonical functor

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}. \tag{8.2.4.8}$$

Since this functor is fully faithful, it is often used to identify an object of the first category with the functor it pro-represents, or even to identify $\mathrm{Ind}(C)$ with its essential image in $\widehat{C}$, consisting of the ind-representable presheaves. Notice, however, that this identification has distinctly more drawbacks than the analogous identification of C with a full subcategory of $\widehat{C}$ via the functor h (8.2.1.1), because unlike the latter functor, the functor (8.2.4.8) is not in general injective on objects.

8.2.5

Let us spell out definition (8.2.4.3) in terms of the indexed expressions (8.2.4.1) of the ind-objects under consideration. Taking account of the definition (8.2.2.1), one obtains

$$\mathrm{Hom}(\mathcal{X},\mathcal{Y}) \simeq \varprojlim_i \mathrm{Hom}(X_i,\mathcal{Y}) := \varprojlim_i \mathrm{Hom}(X_i,L(\mathcal{Y})) = \varprojlim_i \varinjlim_j \mathrm{Hom}_C(X_i,Y_j),$$

the last equality coming from (8.2.2.2) applied to $\mathcal{Y}$. Thus there is a canonical bijection

$$\mathrm{Hom}((X_i)_{i\in I},(Y_j)_{j\in J}) \simeq \varprojlim_{i\in I} \varinjlim_{j\in J} \mathrm{Hom}_C(X_i,Y_j). \tag{8.2.5.1}$$

We leave to the reader the task of making the composition of morphisms of ind-objects explicit using this formula. Notice that it follows at once from this formula that the set $\mathrm{Hom}(\mathcal{X},\mathcal{Y})$ of homomorphisms of ind-objects is *$\mathcal{U}$-small*; equivalently, the categories (8.2.4.4) are $\mathcal{U}$-categories. Equivalently again, by (8.2.4.6), the *category* $\mathrm{Ind}(C)$ *is a $\mathcal{U}$-category*; that is, the right hand side of (8.2.5.1) is small when $I, J \in \mathcal{U}$, which follows immediately from the fact that C is a $\mathcal{U}$-category, hence the $\mathrm{Hom}_C(X_i,Y_j)$ are small.

8.2.6

Let I be an essentially small filtered category. Then one sees either from (8.2.5.1), or from the functoriality of the inductive limit of a functor $I \to \widehat{C}$ with respect to that functor, that there is a canonical functor

$$\mathrm{Hom}(I,C) \longrightarrow \mathrm{Ind}(C). \tag{8.2.6.1}$$

Notice that this functor is *not* in general fully faithful, or even faithful.

Remark 8.2.7. The definitions of 8.2.4 make clear the notion of *isomorphism* of two essentially small ind-objects, which also means the isomorphism of the presheaves they ind-represent. Notice that this is a very weak notion of isomorphism when compared with the notion of isomorphism in categories of the form $\mathrm{Hom}(I,C)$. Thus many fairly natural relations one can impose on an ind-object, such as having its components in a given strictly full subcategory, or being strict (8.12 below), are *not* stable under isomorphism. Therefore, if one wants to work with notions stable under isomorphism of ind-objects, one must "saturate" the notions under consideration by passing to the strictly full subcategory of $\mathrm{Ind}(C)$ generated by the subcategory of $\mathrm{Ind}(C)$ consisting of objects satisfying the condition in question. See, for example, the notion of essentially constant ind-object introduced below (8.4).

Exercise 8.2.8. Let $\mathcal{U} \subset \mathcal{V}$ be two universes and let C be a $\mathcal{U}$-category belonging to $\mathcal{V}$. Let $\mathrm{SysInd}(C)$ denote the following category.

a) The objects of $\mathrm{SysInd}(C)$ are functors $\varphi : I \to C$, where I is an essentially $\mathcal{U}$-small filtered category belonging to $\mathcal{V}$.

b) If (I,φ) and (J,ψ) are two objects of $\mathrm{SysInd}(C)$, a morphism in $\mathrm{SysInd}(C)$ from (I,φ) to (J,ψ) is a pair (m,u), where $m : I \to J$ is a functor and $u : \varphi \to \psi \circ m$ is a morphism of functors. The composition of morphisms in $\mathrm{SysInd}(C)$ is defined in the evident way.

Let S be the set of morphisms $(m,u):(I,\varphi)\to(J,\Psi)$ of $\mathrm{SysInd}(C)$ such that m is cofinal and u is an isomorphism. Let (I,φ) be an object of $\mathrm{SysInd}(C)$. The category $\mathrm{Fl}(I)$ of arrows of I maps to I by two natural functors: source and target. Moreover these two functors are related by the canonical morphism of functors $v:\text{source}\to\text{target}$. Hence there are two morphisms in $\mathrm{SysInd}(C)$ from $(\mathrm{Fl}(I),\varphi\circ\text{source})$ to (I,φ):

$$p_1(I,\varphi)=(\text{source},\mathrm{id}),$$

and

$$p_2(I,\varphi)=(\text{target},\varphi^*v:\varphi\circ\text{source}\to\varphi\circ\text{target}).$$

Let $p:\mathrm{SysInd}(C)\to\mathrm{Ind}(C)$ be the evident functor. Let B be a category. Show that the functor $F\mapsto F\circ p$,

$$\mathrm{Hom}(\mathrm{Ind}(C),B)\longrightarrow\mathrm{Hom}(\mathrm{SysInd}(C),B),$$

is fully faithful, and that a functor $G:\mathrm{SysInd}(C)\to B$ belongs to its essential image if and only if it has the following two properties:

1) For every $s\in S$, $F(s)$ is an isomorphism of B.

2) For every object (I,φ) of $\mathrm{SysInd}(C)$, $F(p_1(I,\varphi))=F(p_2(I,\varphi))$.

8.3 Characterization of ind-representable functors

Proposition 8.3.1. Let F be an ind-representable presheaf on C. Then F is left exact, i.e. (2.3.2), for every finite category J and every functor $\varphi:J\to C$ such that $\varinjlim\varphi$ is representable (i.e. $\varprojlim\varphi^{\mathrm{op}}$ is representable), the canonical map

$$F(\varinjlim\varphi)\longrightarrow\varprojlim F\circ\varphi$$

is bijective.

Indeed, representable presheaves are plainly left exact; hence by 2.8 so is every filtered inductive limit of such functors. $\square$

8.3.2

Given a presheaf F on C, we shall often have to work with the category

$$C_{/F}\hookrightarrow\widehat{C}_{/F}, \tag{8.3.2.1}$$

which denotes the full subcategory of the second category consisting of arrows $X\to F$ *whose source lies in* C. It follows from 1.4 that the objects of this category are identified with pairs (X,u), where X is an object of C and $u\in F(X)$. A morphism from (X,u) to (X',u') is then interpreted as an arrow $f:X\to X'$ in C such that $F(f)(u')=u$. The "forget-the-target" functor from $\widehat{C}_{/F}$ to $\widehat{C}$ induces a functor, called the *canonical* functor,

$$C_{/F}\longrightarrow C, \tag{8.3.2.2}$$

already considered in 3.4, where it is proved that the inductive limit of this functor exists in $\widehat{C}$ (with no smallness condition on $C_{/F}$) and is canonically isomorphic to F.

8.3.2.3

Notice that if finite inductive limits are representable in C, and if F is left exact, i.e. carries them to finite projective limits in **Set**, then the same is true in $C_{/F}$, and *a fortiori* (2.7.1) $C_{/F}$ is filtered. More generally, if sums of two objects and cokernels of double arrows are representable in C, and if F carries them respectively to products and to kernels, then $C_{/F}$ is stable under the same types of finite inductive limits. Hence it is filtered if and only if it is nonempty, i.e. if and only if the functor F is not the constant functor with value $\emptyset$.

Theorem 8.3.3. Let C be a $\mathcal{U}$-category and let $F \in \operatorname{Ob} \widehat{C}$, say $F : C^{\mathrm{op}} \to (\mathcal{U}\text{-}\mathbf{Set})$ is a $\mathcal{U}$-presheaf on C. The following conditions are equivalent:

(i) F is ind-representable (8.2.2).

(ii) The category $C_{/F}$ (8.3.2) is filtered and essentially small.

(iii) If finite inductive limits are representable in C: the functor F is left exact, and $\operatorname{Ob} C_{/F}$ admits a small cofinal subset (8.1.4).

(iii bis) If sums of two objects and cokernels of double arrows are representable in C: the functor F carries sums of two objects of C to products and cokernels of double arrows of C to kernels, the category $C_{/F}$ is nonempty, i.e. the functor F is not the constant functor with value $\emptyset$, and finally there exists a small family of objects of $C_{/F}$ such that every object X of $C_{/F}$ is dominated by some X_i, i.e. admits an F-morphism $X_i \to X$.

(iv) If the category C is equivalent to a small category: the category $C_{/F}$ is filtered.

(v) If the category C is equivalent to a small category and if filtered inductive limits in C are representable: the functor F is left exact.

Proof. (i) $\Rightarrow$ (ii). Suppose that F is ind-represented by $(X_i)_{i \in I}$, with I small. We prove that $C_{/F}$ is filtered. Let X, X' be two objects of $C_{/F}$, i.e. objects of C equipped with morphisms $X \to F$ and $X' \to F$. We prove that they are dominated by a third object of $C_{/F}$. The morphisms $X \to F$ and $X' \to F$ come from morphisms $X \to X_i$ and $X' \to X'_{i'}$, and after replacing i, i' by a common upper bound in $\operatorname{Ob} I$, we may suppose $i = i'$. We then take as common upper bound of X and X' the object X_i equipped with the canonical morphism $X_i \to F$.

Now let

$$X \overset{f}{\underset{g}{\rightrightarrows}} X' \xrightarrow{h} F$$

be a double arrow in $C_{/F}$. We prove that it is equalized by an arrow $X' \to X''$ of $C_{/F}$. The morphism $h : X' \to F$ is given by a morphism $h_i : X' \to X_i$, and the condition $hf = hg$ says that there exists $\alpha : i \to j$ in I such that

$$\varphi(\alpha)(h_i f) = \varphi(\alpha)(h_i g).$$

Thus, after replacing h_i by $\varphi(\alpha)h_i$, we equalize f and g. This proves that $C_{/F}$ is filtered. It is then immediate that it is essentially small, since the objects $(X_i \to F)_{i \in \operatorname{Ob} I}$ form a small cofinal family in $\operatorname{Ob} C_{/F}$.

(ii) $\Rightarrow$ (i). Put $I = C_{/F}$ and consider the canonical functor (8.3.2.1)

$$\varphi : I = C_{/F} \longrightarrow C.$$

The presheaf represented by this ind-object of C is known to be F (3.4), so F is ind-representable by definition (8.2.2).

(ii) $\Leftrightarrow$ (iii). Indeed, (ii) $\Rightarrow$ (iii) because an ind-representable functor is left exact (8.3.1), and (iii) $\Rightarrow$ (ii) because we have observed (8.3.2.3) that left exactness of F implies that $C_{/F}$ is filtered, and one applies Definition 8.1.8 to conclude that this category is essentially small.

The equivalence (ii) $\Leftrightarrow$ (iii bis) is proved in the same way as (ii) $\Leftrightarrow$ (iii).

The equivalences (ii) $\Leftrightarrow$ (iv) and (iii) $\Leftrightarrow$ (v) are trivial, since if C is equivalent to a small category, then $C_{/F}$ is evidently also equivalent to a small category and *a fortiori* is automatically essentially small as soon as it is filtered. This completes the proof of 8.3.3.

Remark 8.3.4. Let $F : C' \to C''$ be a functor between $\mathcal{U}$-categories. It appears that the notion of left exactness (2.3.2) is of practical interest only when finite projective limits are representable in C'. In the case where $C'' = (\mathcal{U}\text{-}\mathbf{Set})$ and C' is $\mathcal{U}$-small, writing C' in the form C^{op} (so $C = C'^{\mathrm{op}}$), the "right notion" which in the general case replaces left exactness is that of ind-representability of F when it is considered as a presheaf on C^{op}; equivalently, it is the pro-representability of F when it is considered as a functor on C' (cf. 8.10). This notion does coincide with left exactness when finite projective limits are representable in C', i.e. when finite inductive limits are representable in C (8.3.3). When one no longer assumes that C' is $\mathcal{U}$-small, or at least equivalent to a $\mathcal{U}$-small category, the two notions for a functor F, left exactness and pro-representability, no longer necessarily coincide, even if finite projective limits are representable in C' (cf. 8.12.9). In fact, it seems that in this case left exact functors which are not ind-representable should be regarded as pathological; the "good" objects remain the ind-representable functors. Compare 8.13.3. For the case of functors $F : C' \to C''$, with C' and C'' again arbitrary $\mathcal{U}$-categories, we shall develop below (8.11.5), more generally, a notion which "improves" that of left exactness, namely the notion of a functor admitting a pro-adjoint functor.

8.4 Constant and essentially constant ind-objects

Choose a final category e, i.e. one such that $\mathrm{Ob}\, e$ and $\mathrm{Fl}\, e$ are each reduced to a single element; for example, one may take for e the full subcategory of $\mathbf{Set}$ consisting of the empty set, if one wants a canonical choice. This is plainly a filtered and small category. If X is an object of C, we associate with it the constant ind-object indexed by e, with value X, denoted $c(X)$. It is clear that, as X varies, one obtains a fully faithful functor

$$c : C \longrightarrow \mathrm{Ind}(C), \tag{8.4.1}$$

which is moreover injective on objects, and through which we shall identify C with a full subcategory of $\mathrm{Ind}(C)$. Notice that the composite functor

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C} \tag{8.4.2}$$

is the canonical functor h (8.2.1.1).

8.4.3

More generally, let I be a filtered category. The constant functor on I with value an object X of C is also called *the constant ind-object of value X indexed by I*. If I is essentially small, it is clear that this ind-object ind-represents the functor $h(X)$ represented by X, hence this ind-object is isomorphic to $c(X)$.

8.4.4

An ind-object $\mathcal{X}$ of C is called *essentially constant* if it is isomorphic (in a category $\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{V}')$ for suitable $\mathcal{V}, \mathcal{V}'$) to an ind-object of the form $c(X)$, with $X \in \mathrm{Ob}\, C$. The object X, which is then determined up to canonical isomorphism, is called the *value* of the essentially constant ind-object in question. Thus, by definition, the functor c of (8.4.1) induces an equivalence of C with the full subcategory of $\mathrm{Ind}(C)$ consisting of essentially constant ind-objects; this subcategory is the essential image of the functor c.

Obviously a constant ind-object is essentially constant; the converse is true only in the trivial case where C is empty or a one-point category.

8.5 Filtered inductive limits in $\mathrm{Ind}(C)$

Proposition 8.5.1. Let C be a $\mathcal{U}$-category. In $\mathrm{Ind}(C)$, small filtered inductive limits are representable, and the canonical functor (8.2.4.8)

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}$$

commutes with them.

Taking account of the fact that L is fully faithful, the assertions of the proposition are equivalent to the following one.

Corollary 8.5.2. Every small filtered inductive limit, in $\widehat{C}$, of ind-representable presheaves is ind-representable.

This follows easily from criterion 8.3.3(ii). The details of the verification are left to the reader.

8.5.3

The "tautological" case of filtered inductive limits in $\mathrm{Ind}(C)$ is the case where one starts from an essentially small ind-object $\mathcal{X} = (X_i)_{i \in I}$ of C. One then has, in $\widehat{C}$ and hence also in $\mathrm{Ind}(C)$,

$$\mathcal{X} = \varinjlim_{I} X_i. \tag{8.5.3.1}$$

In writing this formula, one should take care that this is not an inductive limit in C, and that even when the latter exists, it is not isomorphic in $\mathrm{Ind}(C)$ to $\mathcal{X}$: indeed, the canonical functor (8.4.1) $c : C \to \mathrm{Ind}(C)$ *does not in general commute with filtered inductive limits.* See Exercise 8.5.5 below.

To avoid this possible confusion in writing (8.5.3.1), "certain authors" (= P. Deligne) prefer to write it

$$\mathcal{X} = \text{“}\varinjlim_{I}\text{”}\, X_i, \tag{8.5.3.2}$$

the role of the quotation marks being to indicate that the inductive limit is taken in a category of ind-objects $\mathrm{Ind}(C)$. By extension, one should then denote by "$\lim$" every inductive-limit operation in $\mathrm{Ind}(C)$, without requiring the components of the inductive system considered in $\mathrm{Ind}(C)$ to lie in the image, or in the essential image, of $c : C \to \mathrm{Ind}(C)$.

8.5.4

To compute the inductive or projective limit of a functor

$$\varphi : J \longrightarrow \mathrm{Ind}(C), \tag{8.5.4.1}$$

where J is now an arbitrary category, it is often convenient to make φ explicit by means of a functor

$$\Phi : J \times I \longrightarrow C, \tag{8.5.4.2}$$

where I is an essentially small, or preferably small, filtered category. Such a Φ indeed defines a functor

$$j \longmapsto (i \longmapsto \Phi(j,i)) : J \longrightarrow \mathrm{Hom}(I,C),$$

hence, after composing with the canonical arrow (8.2.6.1) $\mathrm{Hom}(I,C) \to \mathrm{Ind}(C)$, a functor

$$j \longmapsto (i \longmapsto \Phi(j,i)) : J \longrightarrow \mathrm{Ind}(C). \tag{8.5.4.3}$$

We may say that Φ is an *indexed expression of φ, indexed by the filtered category I*, if the corresponding functor (8.5.4.3) is isomorphic to φ. We shall study below (8.8) general existence conditions for an indexed expression of a given functor φ. Here we shall merely start from a functor φ given in indexed form (8.5.4.3), in the case where J is a small filtered category, in order to indicate the calculation of

$$\text{“}\varinjlim\text{”}\,\varphi = \text{“}\varinjlim_{j}\text{”}\,\varphi(j).$$

Formula (8.5.3.2), together with the associativity formula for inductive limits (2.5.0), then immediately gives the *tautological calculation* of “$\varinjlim$” φ:

$$\text{“}\varinjlim_{J}\text{”}\,\varphi = \text{“}\varinjlim_{J\times I}\text{”}\,\Phi, \tag{8.5.4.4}$$

i.e.

$$\text{“}\varinjlim_{j}\text{”}\,(i \longmapsto \Phi(j,i)) = \text{“}\varinjlim_{j,i}\text{”}\,\Phi(j,i). \tag{8.5.4.5}$$

Thus the inductive system sought is none other than Φ itself, the index category being $J \times I$, which is indeed filtered since J and I are so.

Exercise 8.5.5. Let C be a $\mathcal{U}$-category.

a) Prove that the following conditions are equivalent:

(i) Small inductive limits are representable in C, and the functor $c : C \to \mathrm{Ind}(C)$ commutes with them.

(ii) Small inductive limits are representable in C, and for every $X \in \mathrm{Ob}\,C$, the functor $Y \mapsto \mathrm{Hom}(X,Y)$ commutes with these limits.

(iii) The functor c is an equivalence of categories.

(iv) Small filtered inductive limits are representable in C, and the functor $\lim c : \mathrm{Ind}\,C \to C$ (cf. 8.7.1.5) is fully faithful, or equivalently an equivalence.

b) If C is a finite category, prove that the preceding conditions are equivalent to the following one: for every projector in C (10.6), its image is representable in C.

8.6 Extension of a functor to ind-objects

8.6.1

Let

$$f : C \longrightarrow C' \tag{8.6.1.1}$$

be a functor between $\mathcal{U}$-categories. For every ind-object

$$\varphi : I \longrightarrow C$$

of C, where I is a filtered category, the composite functor $f \circ \varphi$ is an ind-object of C', also denoted $\mathrm{Ind}(f)(\varphi)$. In indexed notation, if φ is written as $\mathcal{X} = (X_i)_{i \in I}$, one obtains

$$\mathrm{Ind}(f)(X_i)_{i \in I} = (f(X_i))_{i \in I}. \tag{8.6.1.2}$$

Let $\mathcal{Y} = (Y_j)_{j \in J}$ be a second inductive system of C. Then one obtains an evident map, also denoted $u \mapsto \mathrm{Ind}(f)(u)$:

$$\begin{aligned} \mathrm{Hom}(\mathcal{X}, \mathcal{Y}) &= \varprojlim_i \varinjlim_j \mathrm{Hom}(X_i, Y_j) \\ &\longrightarrow \mathrm{Hom}(\mathrm{Ind}(f)(\mathcal{X}), \mathrm{Ind}(f)(\mathcal{Y})) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}(f(X_i), f(Y_j)). \end{aligned} \tag{8.6.1.3}$$

One immediately checks that these maps are compatible with the composition of morphisms of ind-objects, referring to the unwritten formula for composition of morphisms of ind-objects (8.2.5). We have thus obtained, for every universe $\mathcal{V}$, a functor

$$\mathrm{Ind}(f) : \mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U}) \longrightarrow \mathrm{Ind}_{\mathcal{V}}(C', \mathcal{U}) \tag{8.6.1.4}$$

(notation of (8.2.4.5)), and in particular a functor

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C'). \tag{8.6.1.5}$$

If one has a second functor

$$g : C' \longrightarrow C'',$$

then of course there is an identity of functors between categories of ind-objects, or of $\mathrm{Ind}_{\mathcal{V}}$-objects if desired,

$$\mathrm{Ind}(gf) = \mathrm{Ind}(g) \circ \mathrm{Ind}(f), \tag{8.6.1.6}$$

and for $C' = C$ one has the perennial formula

$$\mathrm{Ind}(\mathrm{id}_C) = \mathrm{id}_{\mathrm{Ind}(C)}\,. \tag{8.6.1.7}$$

Thus, figuratively speaking, one may say that the category $\mathrm{Ind}(C)$ *depends functorially on* C, covariantly.[1] We leave to the reader the task of making explicit even a 2-*functorial* dependence, by defining, for every pair of $\mathcal{U}$-categories C, C', a canonical functor

$$\mathsf{Ind} : \mathrm{Hom}(C, C') \longrightarrow \mathrm{Hom}(\mathrm{Ind}(C), \mathrm{Ind}(C')), \tag{8.6.1.8}$$

and by specifying the functorial nature of the identity isomorphisms (8.6.1.6) and (8.6.1.7).

[1] For a contravariant dependence, under certain conditions, cf. 8.11.

8.6.2

Return to the functor (8.6.1.1), and consider the corresponding functor (5.1)

$$f_! : \widehat{C} \longrightarrow \widehat{C'}, \tag{8.6.2.1}$$

and the diagram of functors

$$\begin{tikzcd} \mathrm{Ind}(C) \arrow[r, "\mathrm{Ind}(f)"] \arrow[d, "L_C"'] & \mathrm{Ind}(C') \arrow[d, "L_{C'}"] \\ \widehat{C} \arrow[r, "f_!"'] & \widehat{C'}. \end{tikzcd} \tag{8.6.2.2}$$

Here the vertical arrows denote the canonical functors L of (8.2.4.8). Since the functor $f_!$ "extends f" and commutes with inductive limits (5.4.3), and since the functors L_C and $L_{C'}$ commute with small filtered inductive limits (8.5.1), it follows from (8.5.3.2) that for every ind-object $(X_i)_{i\in I}$ of C there is, in $\widehat{C'}$, a canonical isomorphism

$$f_!(L_C((X_i)_{i\in I})) \simeq \varinjlim_i f_! L_C(X_i) \simeq \varinjlim_i L_{C'} f(X_i) = L_{C'}(f(X_i)_{i\in I}),$$

i.e. a canonical isomorphism

$$f_! L_C(\mathcal{X}) \simeq L_{C'}(\mathrm{Ind}(f)(\mathcal{X})).$$

We leave to the reader the verification that the latter is functorial, i.e. that it corresponds to a canonical isomorphism

$$f_! L_C \simeq L_{C'} \circ \mathrm{Ind}(f). \tag{8.6.2.3}$$

In other words, *the square (8.6.2.2) is commutative up to canonical isomorphism.* Taking account of the fact that $f_!$ commutes with small inductive limits, and of 8.5.1, we conclude as follows.

Proposition 8.6.3. Let $f : C \to C'$ be a functor between $\mathcal{U}$-categories. Then the functor

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$$

commutes with filtered inductive limits. Moreover, it makes the following diagram commutative:

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "c_C"'] & C' \arrow[d, "c_{C'}"] \\ \mathrm{Ind}(C) \arrow[r, "\mathrm{Ind}(f)"'] & \mathrm{Ind}(C'), \end{tikzcd}$$

where the vertical arrows $c_C, c_{C'}$ are the canonical functors (8.4.1).

The last assertion is trivial from the definitions and has been included for ease of reference. Note moreover that the properties stated in 8.6.3 *characterize the functor* $\mathrm{Ind}(f)$ up to unique isomorphism as being induced by the functor $f_!$ (8.6.2.1), as follows from the proof just given of (8.6.2.3).

Proposition 8.6.4. The notation is that of 8.6.3.

a) For $\mathrm{Ind}(f)$ to be faithful, respectively fully faithful, it is necessary and sufficient that f be so.

b) For $\operatorname{Ind}(f)$ to be an equivalence of categories, it is necessary and sufficient that f be fully faithful and that every object of C' be isomorphic to a direct factor (10.6) of an object in the image of f.

Proof.

a) Necessity follows immediately from the fact that f is induced by $\operatorname{Ind}(f)$. For sufficiency, it is enough to use the form (8.6.1.3) of $\operatorname{Ind}(f)$ on the sets $\operatorname{Hom}(\mathcal{X}, \mathcal{Y})$, remembering that filtered inductive limits of sets, and arbitrary projective limits, carry monomorphisms to monomorphisms and isomorphisms to isomorphisms.

b) We may already suppose f, hence $\operatorname{Ind}(f)$, fully faithful. Since every object of $\operatorname{Ind}(C')$ is a small filtered inductive limit of objects of C', full faithfulness of $\operatorname{Ind}(f)$ implies that for this functor to be essentially surjective it is equivalent that every object X' of C' lie in its essential image. If one has an isomorphism

$$X' \xrightarrow{\sim} \text{“}\varinjlim\text{''}\, f(X_i),$$

then this isomorphism factors through one of the $f(X_i)$; this shows that X' is a direct factor of $f(X_i)$, proving necessity in b). For sufficiency, using the full faithfulness of $\operatorname{Ind}(f)$, the assertion follows at once from the following corollary.

Corollary 8.6.5. In $\operatorname{Ind}(C)$ the images of projectors (10.6) are representable, and the functor $\operatorname{Ind}(f) : \operatorname{Ind}(C) \to \operatorname{Ind}(C')$ commutes with the formation of these images.

Indeed, this follows from the fact that in $\operatorname{Ind}(C)$ small filtered inductive limits are representable (8.5.1) and that $\operatorname{Ind}(f)$ commutes with them (8.6.3), taking account of the fact that the image of a projector $p : X \to X$ may be interpreted as the inductive limit of the filtered inductive system indexed by $\mathbb{N}$

$$X \xrightarrow{p} X \xrightarrow{p} X \longrightarrow \cdots,$$

or, equivalently, as the inductive limit of the evident functor on the filtered category P having a single object and a nonidentity arrow Π such that $\Pi^2 = \Pi$.

Exposé I. Presheaves

Section 8.7–8.8

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 8.7 through Section 8.8. A few evident typographical slips in the source notation are corrected where the mathematical context forces the correction.

8 Ind-objects and pro-objects

8.7 The functor $\varinjlim_C : \mathrm{Ind}(C) \to C$. Universal characterizations of $\mathrm{Ind}(C)$

8.7.1

Let C again be a $\mathcal{U}$-category, and let

$$c : C \longrightarrow \mathrm{Ind}(C) \tag{8.7.1.1}$$

be the canonical functor. Let $\mathcal{X} = (X_i)_{i\in I}$ be an object of $\mathrm{Ind}(C)$. It follows immediately from the definition of representable inductive limits (2.1, 2.1.1) that $\varinjlim_i X_i$ is *representable in C if and only if the left adjoint of c is defined at $\mathcal{X}$*; in that case $\varinjlim_i X_i$, computed in C, is precisely the value of that left adjoint at $\mathcal{X}$:

$$\mathrm{Hom}_C(\varinjlim_i X_i, Y) \simeq \mathrm{Hom}_{\mathrm{Ind}(C)}((X_i)_{i\in I}, c(Y)). \tag{8.7.1.2}$$

Let

$$\mathrm{Ind}(C)' \subset \mathrm{Ind}(C) \tag{8.7.1.3}$$

denote the full subcategory of $\mathrm{Ind}(C)$ formed by the ind-objects of C which admit an inductive limit *in* C. The preceding observation implies that this subcategory is *strictly full*, and that one has a canonical functor

$$\varinjlim_C : \mathrm{Ind}(C)' \longrightarrow C, \tag{8.7.1.4}$$

whose value on an object $(X_i)_{i\in I}$ of $\mathrm{Ind}(C)'$ is its inductive limit in C. In the special case in which small filtered inductive limits are representable in C, one therefore obtains a natural functor

$$\varinjlim_C : \mathrm{Ind}(C) \longrightarrow C. \tag{8.7.1.5}$$

8.7.1.6

Of course, the preceding functors may also be defined for categories of the form $\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U})'$ and $\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U})$; but, in view of (8.2.4.6), they are already determined, up to unique isomorphism, by the corresponding functors in the case $\mathcal{V} = \mathcal{U}$.

8.7.1.7

From the preceding construction of $\varinjlim_C$ as a left adjoint, it follows immediately that this functor *commutes with arbitrary inductive limits*, and in particular with small filtered inductive limits, the latter being representable in $\mathrm{Ind}(C)$ by 8.5.1. Notice also that $\mathrm{Ind}(C)'$ always contains the essential image of c, formed by the essentially constant ind-objects, and that one has a canonical isomorphism, functorial in $X \in \mathrm{Ob}\,\mathrm{Ind}(C)'$,

$$\varinjlim_{C} c(X) \simeq X. \tag{8.7.1.8}$$

Equivalently, in the favorable case where C is stable under small inductive limits, one has a canonical isomorphism

$$\varinjlim_{C} \circ c \simeq \mathrm{id}_C\,. \tag{8.7.1.9}$$

8.7.2

Now consider a functor

$$f : C \longrightarrow E, \tag{8.7.2.1}$$

where E is a $\mathcal{U}$-category in which small inductive limits are representable, so that there is a functor

$$\varinjlim_{E} : \mathrm{Ind}(E) \longrightarrow E.$$

Since (8.6) also defined

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(E),$$

one may consider the composite

$$\bar{f} = \varinjlim_{E} \circ \mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow E, \tag{8.7.2.2}$$

which is sometimes called the *canonical extension of f to ind-objects*, although it should not be confused with $\mathrm{Ind}(f)$. Since it is a composite of two functors commuting with small filtered inductive limits (8.6.3, 8.7.1.7), it also commutes with small filtered inductive limits. Moreover, from the isomorphism (8.7.1.9), applied to E, and from (8.6.2.3), it follows that $\bar{f}$ *extends* f up to isomorphism; that is, there is a canonical isomorphism

$$\bar{f} \circ c_C \simeq f. \tag{8.7.2.3}$$

It is also rather clear, taking into account the fact that every object of $\mathrm{Ind}(C)$ is a small filtered inductive limit of objects of C, that the two preceding properties still *characterize* $\bar{f}$ up to unique isomorphism. This can be made precise so as to obtain, at the same time, a universal characterization, up to equivalence, of $\mathrm{Ind}(C)$ among the $\mathcal{U}$-categories E in which small filtered inductive limits are representable. If E and F are two such $\mathcal{U}$-categories, let

$$\mathrm{Hom}(E, F)' \subset \mathrm{Hom}(E, F) \tag{8.7.2.4}$$

denote the full subcategory of $\mathrm{Hom}(E, F)$ formed by the functors which commute with small filtered inductive limits. Then one has the following result.

Proposition 8.7.3. Let C be a $\mathcal{U}$-category, and use the notation above (8.7.2.4). Then the canonical functor

$$c_C : C \longrightarrow \mathrm{Ind}(C)$$

is 2-universal among functors with source C and target a $\mathcal{U}$-category in which small filtered inductive limits are representable. In other words, $\mathrm{Ind}(C)$ is such a category (8.2.5, 8.5.1), and if E is such a category, the functor

$$g \longmapsto g \circ c_C : \mathrm{Hom}(\mathrm{Ind}(C), E)' \longrightarrow \mathrm{Hom}(C, E) \tag{8.7.3.1}$$

is an equivalence of categories.

To prove the last assertion, it is enough to verify that the assignment $f \mapsto \bar{f}$ defined by (8.7.2.2) can be made into a *functor*

$$f \longmapsto \bar{f} = \varinjlim_{E} \cdot \mathrm{Ind}(f) : \mathrm{Hom}(C, E) \longrightarrow \mathrm{Hom}(\mathrm{Ind}(C), E)', \tag{8.7.3.2}$$

and that this functor is a quasi-inverse of (8.7.3.1). The details of the verification, which are essentially trivial, are left to the reader. One may also invoke 7.8 to conclude first that (8.7.3.1) is fully faithful, and then use (8.7.2.3) to conclude that it is essentially surjective.

8.7.4

Let us return to a functor between $\mathcal{U}$-categories

$$f : C \longrightarrow E, \tag{8.7.4.1}$$

where E is a $\mathcal{U}$-category in which small filtered inductive limits are representable. This gives a functor

$$\bar{f} : (X_i)_{i \in I} \longmapsto \varinjlim_{i} f(X_i) : \mathrm{Ind}(C) \longrightarrow E. \tag{8.7.4.2}$$

We shall study conditions on f which ensure that $\bar{f}$ is fully faithful, respectively an equivalence of categories. Since f is isomorphic to the composite $\bar{f} \circ c_C$, and since $c_C : C \to \mathrm{Ind}(C)$ is fully faithful, it is *necessary*, for $\bar{f}$ to be fully faithful, that f be so. Replacing C by its essential image in E, we therefore lose no essential generality by assuming, as we shall do below in order to simplify notation, that f is the inclusion functor of a subcategory C of E.

Proposition 8.7.5. The notation is that of 8.7.4.

a) For the functor $\bar{f}$ to be fully faithful, it is necessary and sufficient that the following condition hold:

  (PF) Every object X of C satisfies the following property: for every small filtered inductive system $(Y_i)_{i \in I}$ in C, with inductive limit $\varinjlim_i Y_i$ in E, the canonical map

  $$\varinjlim_{i} \mathrm{Hom}(X, Y_i) \longrightarrow \mathrm{Hom}(X, \varinjlim_{i} Y_i) \tag{8.7.5.1}$$

  is bijective.

b) For the functor $\bar{f}$ to be an equivalence of categories, it is necessary and sufficient that C satisfy condition (PF) and the following two conditions:

(ii) C is a subcategory of E generating by strict epimorphisms (7.1).

(iii) For every object X of E, the full subcategory $C_{/X}$ of $E_{/X}$ formed by objects X' over X whose source is in $\operatorname{Ob} C$ is filtered and essentially small.

Condition (iii) holds in particular if C is equivalent to a small category, if finite inductive limits are representable in C, and if the inclusion functor $f : C \to E$ commutes with them; that is, for every finite category J and every functor $\varphi : J \to C$, the inductive limit of $f\varphi$ is representable and is isomorphic in E to an object of C.

Proof.

a) With the notation of condition (PF), if $\mathcal{Y}$ denotes the ind-object (Y_i) and $\mathcal{X}$ the constant ind-object defined by X, then (8.7.5.1) is precisely the canonical map

$$\operatorname{Hom}(\mathcal{X}, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}), \bar{f}(\mathcal{Y})).$$

Thus, if $\bar{f}$ is fully faithful, this map is bijective. Conversely, if $\mathcal{X} = (X_j)_{j \in J}$ and $\mathcal{Y} = (Y_i)_{i \in I}$ are arbitrary ind-objects of C, then the map

$$u : \operatorname{Hom}(\mathcal{X}, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}), \bar{f}(\mathcal{Y}))$$

is the projective limit over j of the maps

$$u_j : \operatorname{Hom}(\mathcal{X}_j, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}_j), \bar{f}(\mathcal{Y})),$$

where $\mathcal{X}_j$ is the constant ind-object with value X_j. Hence u is bijective if the u_j are, and condition (PF) implies that $\bar{f}$ is fully faithful.

b) Suppose that (PF), (ii), and (iii) hold, and prove that $\bar{f}$ is an equivalence. It remains only to prove that it is essentially surjective. Let $X \in \operatorname{Ob} E$. Hypothesis (ii) means precisely that

$$\varinjlim_{C_{/X}} Z \longrightarrow X$$

is an isomorphism, while (iii) says that the category $C_{/X}$ is filtered and essentially small. Thus X is the image of the ind-object of C defined by the natural functor $C_{/X} \to E$.

Conversely, suppose that $\bar{f}$ is an equivalence. In view of a), it remains to check (ii) and (iii). We may suppose that $E = \operatorname{Ind}(C)$, with C identified with the subcategory $c(C)$ of E. But it is known (8.3.3(ii)) that, for every $X \in \operatorname{Ind}(C)$, identified if desired with the functor F on C^{op} which it ind-represents, $C_{/X} = C_{/F}$ is a filtered essentially small category. This proves (iii). Moreover, the inductive limit in $\widehat{C}$ of the functor $C_{/F} \to \widehat{C}$ is F. A fortiori the same is true in the full subcategory E of $\widehat{C}$, since $F = X$ lies in E. This proves (ii).

It remains to prove the last assertion of b). But the hypothesis made on C plainly implies that finite inductive limits are representable in the category $C_{/X}$; a fortiori $C_{/X}$ is filtered.

Remark 8.7.5.2. The preceding argument shows more generally that, when condition (PF) holds, the essential image of the fully faithful functor $\bar{f}$ (8.7.4.2) consists of the objects $X \in \operatorname{Ob} E$ such that the category $C_{/X}$ is filtered and essentially small–a condition automatically satisfied if C satisfies the conditions stated at the end of b)–and such that the inductive limit of the canonical functor $C_{/X} \to E$ is X.

Corollary 8.7.6. Let E_{PF} be the full subcategory of E formed by the objects satisfying condition (PF) of Proposition 8.7.5 a). Then, for every functor $J \to E_{PF}$, with J a finite category, which admits an inductive limit X in E, one has $X \in \operatorname{Ob} E_{PF}$. The canonical functor

$$(X_i)_{i \in I} \longmapsto \varinjlim_{i,E} X_i,$$

obtained from the inclusion $g : E_{PF} \hookrightarrow E$,

$$\bar{g} : \operatorname{Ind}(E_{PF}) \longrightarrow E \tag{8.7.6.1}$$

is fully faithful. If finite inductive limits are representable in E and E_{PF} is equivalent to a small category, then the essential image of $\bar{g}$ is formed by the objects X of E for which the canonical morphism

$$\varinjlim_{(E_{PF})_{/X}} Y \longrightarrow X$$

is an isomorphism. If, moreover, the subcategory E_{PF} of E generates by strict epimorphisms (7.1), then the functor (8.7.6.1) is an equivalence of categories.

All the assertions are evident, in the order in which they are stated, from 8.7.5, using 2.8 for the first assertion.

Corollary 8.7.7. Suppose that finite inductive limits are representable in E. Let C be a full subcategory of E, equivalent to a small category, and consider the functor

$$\bar{f} : \operatorname{Ind}(C) \longrightarrow E, \qquad (X_i)_{i \in I} \longmapsto \varinjlim_{i,E} X_i. \tag{8.7.7.1}$$

Let E_{PF} be as in 8.7.6. The following conditions are equivalent:

(i) The functor $\bar{f} : \operatorname{Ind}(C) \to E$ is an equivalence.

(ii) The category C is a subcategory of E generating by strict epimorphisms, is contained in E_{PF}, and every object of E_{PF} is isomorphic in E (or in E_{PF}, which is the same thing by 8.7.6) to a direct factor of an object of C.

When the subcategory C of E is stable under direct factors (10.6), the preceding conditions are also equivalent to the following:

(ii bis) $C = E_{PF}$, and C generates in E by strict epimorphisms.

(iii) The subcategory C of E generates by strict epimorphisms, is contained in E_{PF}, and finite inductive limits are representable in C.

When, moreover, finite inductive limits are representable in E, these conditions are also equivalent to

(iii bis) The subcategory C of E generates by strict epimorphisms, is contained in E_{PF}, and is stable in E under finite inductive limits; equivalently, under finite sums and cokernels of double arrows.

We already know (8.7.5) that condition (i) implies that $C \subset E_{PF}$ and that C generates by strict epimorphisms. Let us also prove that then every object X of E_{PF} is isomorphic to a direct factor of an object of C. Indeed, X is a filtered inductive limit $\varinjlim_i X_i$ in E of objects of C; by the definition of E_{PF}, for a suitable i the canonical homomorphism $X_i \to X$ admits a left inverse, so that X is indeed a direct factor of the object X_i of C. This proves (i) $\Rightarrow$ (ii), without any further hypothesis on C or E.

Let us prove (ii) $\Rightarrow$ (i). Since C is equivalent to a small category and every object of E_{PF} is a direct factor of an object of C, it follows that E_{PF} is also equivalent to a small category. Hence, by 8.7.6, the functor (8.7.6.1) is an equivalence of categories, and one concludes by applying 8.6.4 b) to the inclusion $C \hookrightarrow E_{PF}$.

When every direct factor in E of an object of C lies in C, it is clear that (ii) is equivalent to (ii bis). On the other hand, (ii bis) implies (iii), respectively (iii bis), by 8.7.6. Finally, (iii), and *a fortiori* (iii bis), implies (i) by 8.7.5. This completes the proof.

Exercise 8.7.8 (Karoubi envelopes). Let C be a $\mathcal{U}$-category, let $E = \mathrm{Ind}(C)$, and let $E_{PF} \supset C$ be the full subcategory of E defined in 8.7.6.

a) Prove that images of projectors are representable in E_{PF}, and that every object of E_{PF} is a direct factor of an object of C.

b) Call a category F *Karoubian* if images of projectors are representable in it. Let

$$\varphi : C \longrightarrow K$$

be a fully faithful functor, with K Karoubian, such that every object of K is a direct factor of an object in the image of C. Prove that φ is 2-*universal* among functors from C to Karoubian categories, more precisely: for every Karoubian category F, the functor

$$g \longmapsto g \circ \varphi : \mathrm{Hom}(K, F) \longrightarrow \mathrm{Hom}(C, F)$$

is an equivalence of categories. Use the fact that every functor commutes with images of projectors. The category K equipped with φ, determined up to equivalence, itself determined up to unique isomorphism, by the preceding properties, is called the *Karoubi envelope* $\widetilde{C}$ of C. Compare also IV, 7.5.

c) Deduce from a) and b) that $\mathrm{Ind}(C)_{PF}$, equipped with the functor $C \to \mathrm{Ind}(C)_{PF}$ induced by $c : C \to \mathrm{Ind}(C)$, makes $\mathrm{Ind}(C)_{PF}$ a Karoubi envelope of C.

d) Show that every functor $f : C \to C'$ extends, essentially uniquely, to a functor

$$\widetilde{f} : \widetilde{C} \longrightarrow \widetilde{C'}$$

between the Karoubi envelopes, and that if one takes these Karoubi envelopes as in c), then $\widetilde{f}$ is the functor induced by $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$. Show that the conditions of 8.5.4 b) are also equivalent to the following condition: $\widetilde{f}$ is an equivalence of categories.

Exercise 8.7.9. Let E be a $\mathcal{U}$-category.

a) Show that the following conditions are equivalent:

(i) E is equivalent to a category of the form $\operatorname{Ind}(C)$, with C a $\mathcal{U}$-category.

(i bis) As in (i), with C moreover Karoubian (8.7.8 b)).

(ii) The subcategory E_{PF} of E (8.7.6) generates by strict epimorphisms, and for every object X of E, $(E_{PF})_{/X}$ is a filtered essentially small category.

Show that, if C is a *Karoubian* $\mathcal{U}$-category such that E is equivalent to $\operatorname{Ind}(C)$, then C is equivalent to E_{PF}, by an equivalence determined up to unique isomorphism. Establish a 2-equivalence between the 2-category formed by the categories E satisfying the preceding conditions, with functors between them *commuting with small filtered inductive limits*, and the 2-category formed by the *Karoubian* $\mathcal{U}$-categories and arbitrary functors between them.

b) Show that the following conditions are equivalent:

(i) E is equivalent to a category of the form $\operatorname{Ind}(C)$, where C is a $\mathcal{U}$-category in which finite inductive limits are representable.

(i bis) As in (i), but with C moreover Karoubian.

(ii) Finite inductive limits are representable in E, the subcategory E_{PF} of E generates by strict epimorphisms, and for every object X of E there exists a small subset I of $\operatorname{Ob} E_{PF/X}$ such that every object of $E_{PF/X}$ is dominated by an object of I. Use 8.9.5 b).

8.8 Indexed representation of a functor $J \to \operatorname{Ind}(C)$

8.8.1

Let
$$\varphi : J \longrightarrow \operatorname{Ind}(C), \qquad \varphi \in \operatorname{Ob} \operatorname{Hom}(J, \operatorname{Ind}(C)),$$
be a functor, where C is a $\mathcal{U}$-category. Recall (8.5.4) that an *indexed representation of* φ is, by definition, an ind-object of $\operatorname{Hom}(J, C)$,
$$\psi : I \longrightarrow \operatorname{Hom}(J, C),$$
where I is a filtered essentially small category, whose inductive limit in $\operatorname{Hom}(J, \operatorname{Ind}(C))$ is isomorphic to φ, by a specified isomorphism. Thus φ admits an indexed representation if and only if it is isomorphic to an essentially small filtered inductive limit in $\operatorname{Hom}(J, \operatorname{Ind}(C))$ of objects of the full subcategory $\operatorname{Hom}(J, C)$. We shall say that the category J is *admissible for* C if every functor $\varphi : J \to \operatorname{Ind}(C)$ admits an indexed representation.

Since small filtered inductive limits are representable in $\operatorname{Ind}(C)$ (8.5.1), the same is true in $\operatorname{Hom}(J, \operatorname{Ind}(C))$, and they are computed argument by argument. Consequently, the inclusion functor
$$\operatorname{Hom}(J, C) \hookrightarrow \operatorname{Hom}(J, \operatorname{Ind}(C)) \tag{8.8.1.1}$$
extends canonically to a functor
$$\operatorname{Ind}(\operatorname{Hom}(J, C)) \longrightarrow \operatorname{Hom}(J, \operatorname{Ind}(C)) \tag{8.8.1.2}$$

by 8.7.2. The elements of the essential image of this functor are then precisely the φ which admit an indexed representation. Thus saying that J is admissible for C is the same as saying that the functor (8.8.1.2) is essentially surjective. In this connection we record the following result.

Proposition 8.8.2. If the category J is equivalent to a finite category, then the canonical functor (8.8.1.2) is fully faithful. Hence J is admissible for C if and only if this functor is an equivalence of categories.

By 8.7.5 a), everything reduces to proving that, for every small filtered inductive system $(\varphi_i)_{i\in I}$ in $\mathrm{Hom}(J,C)$ and every object φ of $\mathrm{Hom}(J,C)$, the canonical map

$$\varinjlim_i \mathrm{Hom}(\varphi,\varphi_i) \longrightarrow \mathrm{Hom}(\varphi,\varinjlim_i \varphi_i) \tag{8.8.2.1}$$

is bijective, where $\varinjlim_i \varphi_i$ denotes the inductive limit taken in $\mathrm{Hom}(J,\mathrm{Ind}(C))$. But, if φ and ψ are two functors $J\to C$, one has an exact diagram of sets, functorial in φ and ψ,

$$\mathrm{Hom}(\varphi,\psi) \longrightarrow \prod_{X\in \mathrm{Ob}\, J} \mathrm{Hom}(\varphi(X),\psi(X)) \rightrightarrows \prod_{\substack{f\in \mathrm{Fl}\, J\\ f:X\to Y}} \mathrm{Hom}(\varphi(X),\psi(Y)).$$

Since filtered inductive limits commute with kernels of double arrows and with finite products, the bijectivity of (8.8.2.1) follows when J is finite. The case in which J is equivalent to a finite category immediately reduces to this one.

Proposition 8.8.3. Let $\varphi : J\to \mathrm{Ind}(C)$ be a functor, with J equivalent to a small category. For φ to admit an indexed representation, it is sufficient that the following two conditions hold; these conditions are also necessary when J is equivalent to a finite category.

a) The category $\mathrm{Hom}(J,C)_{/\varphi}$, formed by the arrows of $\mathrm{Hom}(J,\mathrm{Ind}(C))$ with target φ and source in $\mathrm{Hom}(J,C)$, is filtered.

b) For every object j of J, the functor

$$\psi \longmapsto \psi(j) : \mathrm{Hom}(J,C)_{/\varphi} \longrightarrow C_{/\varphi(j)} \tag{8.8.3.1}$$

is cofinal, i.e. satisfies conditions F 1) and F 2) of 8.1.3.

Proof. Suppose a) and b) hold, and prove that φ admits an indexed representation. We may of course suppose that J is small. Put

$$I = \mathrm{Hom}(J,C)_{/\varphi}, \tag{8.8.3.2}$$

which is a filtered category by hypothesis, and first suppose that I is essentially small. Then the "inclusion" of I into $\mathrm{Hom}(J,C)$ defines an ind-object of $\mathrm{Hom}(J,C)$, say $i\mapsto\psi_i$. Moreover, one has a canonical homomorphism

$$\varinjlim_i \psi_i \longrightarrow \varphi, \tag{8.8.3.3}$$

given argument by argument by the homomorphism

$$\varinjlim_I \psi_i(j) \longrightarrow \varphi(j) = \varinjlim_{C_{/\varphi(j)}} X$$

deduced from the functor (8.8.3.1). Since the latter functor is cofinal, it follows that (8.8.3.3) is an isomorphism, and hence the conclusion follows.

In the case where I is not assumed essentially small, it is enough to construct an essentially small full subcategory I' such that (8.8.3.3) remains an isomorphism when $\varinjlim_{I'}$ is used instead of $\varinjlim_{I}$. For this, it is enough that the composite functors

$$I' \longrightarrow I \longrightarrow C_{/\varphi(j)}$$

induced by the functors (8.8.3.1) all be cofinal. One then concludes by the following lemma, whose proof is immediate from criterion (8.1.3 b)) and is left to the reader.

Lemma 8.8.3.4. Let I be a filtered $\mathcal{U}$-category, and let

$$f_j : I \longrightarrow I_j \qquad (j \in J)$$

be a small family of cofinal functors from I to essentially small filtered categories. Then there exists a full subcategory I' of I which is filtered and essentially small, and such that the functors induced by the f_j are cofinal.

Let us finally prove the necessity in 8.8.3 when J is assumed equivalent to a finite category. An indexed representation of φ by means of an essentially small filtered index category I defines a functor ψ and, for each j, a functor ψ_j:

$$\begin{tikzcd} I \arrow[r, "\psi"] \arrow[dr, "\psi_j"'] & \mathrm{Hom}(J, C)_{/\varphi} \arrow[d] \\ & C_{/\varphi(j)}. \end{tikzcd} \tag{8.8.3.5}$$

It is enough to prove that ψ and each of the resulting functors ψ_j are cofinal. By 8.1.3 b), it will then follow that $\mathrm{Hom}(J, C)_{/\varphi}$ is filtered, and by Definition 8.1.1 that the vertical arrow in (8.8.3.5) is also cofinal. By 8.8.2 one may identify φ with an object of $\mathrm{Ind}(E)$, where $E = \mathrm{Hom}(J, C)$, and the fact that the functor

$$\psi : I \longrightarrow E_{/\varphi}$$

deduced from the ind-object φ of E indexed by I is cofinal is a general fact, verified immediately by means of the criteria F 1) and F 2) of 8.1.3. The same argument, with E and φ replaced by C and $\varphi(j)$, shows that ψ_j is cofinal. This completes the proof.

Remark 8.8.4.

a) In the case where J is equivalent to a finite category, if φ admits an indexed representation, we have seen that (8.8.3.5) is a cofinal functor. This implies that the category $\mathrm{Hom}(J, C)_{/\varphi}$ itself is essentially small.

b) Suppose that finite inductive limits are representable in C. Then the same is true in $E = \mathrm{Hom}(J, C)$, where these limits are computed argument by argument; they are also inductive limits in $\mathrm{Hom}(J, \widehat{C})$ and *a fortiori* in $\mathrm{Hom}(J, \mathrm{Ind}(C))$. It follows immediately that condition a) of 8.8.3 is then automatically satisfied, and everything reduces to condition b). I do not know whether this condition is automatically satisfied when, in addition, J is assumed small.

We now come to the principal result of the present subsection.

Proposition 8.8.5. Let J be a category equivalent to a finite category, and suppose moreover that J is rigid, i.e. that for every $j \in \operatorname{Ob} J$, every endomorphism of j is the identity. Then J is admissible for C, whatever the $\mathcal{U}$-category C may be; equivalently, every functor $J \to \operatorname{Ind}(C)$ admits an indexed representation.

Replacing J by an equivalent category, we may suppose that J is *reduced*, i.e. that two isomorphic objects of J are identical. Then J is finite. We proceed by induction on $\operatorname{card} \operatorname{Ob} J$, the case where this cardinal is ≤ 0 being trivial; hence we suppose it is at least 1. Let j_0 be a *maximal* object of J, i.e. such that for every arrow $j_0 \to j$ there exists an arrow $j \to j_0$. Since J is rigid, this implies that $j_0 \to j$ is an isomorphism, and since J is reduced, this implies $j_0 = j$. Let J' be the full subcategory obtained from J by removing the object j_0, and let J'' be the full subcategory reduced to the object j_0. The proposition then follows from the following lemma.

Lemma 8.8.5.1. Let J be a finite category, and let J' and J'' be two full subcategories such that

$$\operatorname{Ob} J = \operatorname{Ob} J' \cup \operatorname{Ob} J'',$$

and such that, for every $j' \in \operatorname{Ob} J'$ and $j'' \in \operatorname{Ob} J''$, one has

$$\operatorname{Hom}(j'', j') = \varnothing.$$

If J' and J'' are admissible for C, then so is J.

Everything reduces to proving criteria a) and b) of 8.8.3. This amounts to six elementary verifications: the last two conditions for filtered categories for $\operatorname{Hom}(J, C)_{/\varphi}$, and the conditions F 1) and F 2) for the functors (8.8.3.1), successively in the case $j \in \operatorname{Ob} J'$ and in the case $j \in \operatorname{Ob} J''$; these latter conditions also imply that $\operatorname{Hom}(J, C)_{/\varphi}$ is nonempty. The rather tedious verification presents no difficulty and is left to the reader. The editor searched in vain for an elegant proof bypassing these unpleasant verifications.

Exercise 8.8.6.

a) Let J and J' be two categories admissible for C, one of them finite. Prove that $J \times J'$ is admissible for C. Use 8.8.2.

b) Prove that a small discrete category is admissible for every category C.

c) Let J be a category satisfying the following conditions: (i) J is rigid; (ii) $\operatorname{Ob} J$ is countable; (iii) for any two objects j, j' of J, $\operatorname{Hom}(j, j')$ is finite; (iv) for every object j of J, the set of objects of J which are dominated by j, i.e. which are sources of an arrow with target j, is finite. Show that J is admissible for every $\mathcal{U}$-category C. First show that J can be written as a filtered union of finite full subcategories J_n such that every object of J dominated by an object of J_n lies in J_n; then verify criteria a) and b) of 8.8.3.

Exercise 8.8.7. In the notation, we identify an ordered set with the category it defines.

a) Let C be an ordered set. Let $\widetilde{C}$ be the set of subsets A of C which are filtered and such that $x \in A$ and $y \leq x$ imply $y \in A$. Let $L_0 : C \to \widetilde{C}$ be the map which sends $x \in C$ to the set $L_0(x)$ of all $y \in C$ such that $y \leq x$. Show that L_0 is injective and that the order on C

is induced by that on $\widetilde{C}$. For every $A \in \widetilde{C}$, consider the inclusion $f(A) : A \to C$, which is an ind-object of C. For every ind-object $\varphi : I \to C$ of C, let $g(\varphi) \in \widetilde{C}$ be the subset of C formed by the elements of C dominated by an element of the form $\varphi(i)$. Show that in this way one obtains two quasi-inverse equivalences

$$\operatorname{Ind}(C) \underset{g}{\overset{f}{\rightleftarrows}} \widetilde{C}$$

which transform the functor L_0 into the canonical functor $L : C \to \operatorname{Ind}(C)$.

b) Suppose that, for every $x \in C$, the set of elements majorizing, respectively minorizing, x is finite. Show that every ind-object, respectively pro-object, of C is essentially constant; more precisely, for every such $\varphi : I \to C$, respectively $\varphi : I^{\mathrm{op}} \to C$, there exists $i_0 \in \operatorname{Ob} I$ such that the induced functor on $I_{/i_0}$ is constant, i.e. sends every arrow to an isomorphism.

c) Let $C \subset \mathbb{N}^{\mathrm{op}} \times \mathbb{N}$ be the ordered set formed by the pairs of natural numbers (j, i) such that $i \geq j$. Show that $\widetilde{\mathbb{N}}$ is isomorphic to the ordered set obtained from $\mathbb{N}$, identified with a subset of $\widetilde{\mathbb{N}}$ as in a), by adjoining a greatest element ∞. Show that $\widetilde{C}$ is isomorphic to $\mathbb{N}^{\mathrm{op}} \times \widetilde{\mathbb{N}}$. Consider the functor

$$\varphi : \mathbb{N}^{\mathrm{op}} \longrightarrow \widetilde{C}, \qquad \varphi(j) = (j, \infty).$$

Show that:

1) the projective limit of φ is not representable in $\widetilde{C}$, although filtered projective limits in C are representable, and are essentially constant by b);

2) for every functor $\psi : \mathbb{N}^{\mathrm{op}} \to C$, one has $\operatorname{Hom}(\psi, \varphi) = \varnothing$, and *a fortiori* the functor φ admits no representation in indexed form.

Exercise 8.8.8. Let $X = \mathbb{N} \amalg \mathbb{N}$ be the disjoint union of two copies of $\mathbb{N}$, let s be the symmetry of X, and for every $i \in \mathbb{N}$, let X_i be the subset $\Delta_{i+1} \amalg \Delta_i$ of X, where Δ_i denotes the subset of $\mathbb{N}$ formed by the j such that $j \leq i$. Let C be the subcategory of the category of subsets of X whose objects are the X_i and whose arrows are the maps between the X_i induced either by the identity of X or by its symmetry s. Let $\widetilde{C}$ be the category defined analogously, but in which the object X is also allowed.

a) Show that $\operatorname{Ind}(C)$ is equivalent to $\widetilde{C}$, with the canonical functor $C \to \operatorname{Ind}(C)$ corresponding to the inclusion $C \to \widetilde{C}$. Show that there is no pair (Y, f), consisting of an object Y of C and an endomorphism f of Y, together with a morphism of objects-with-endomorphism in $\operatorname{Ind}(C)$ from (Y, f) to (X, s).

b) Conclude that, if J is the category with one object and, besides the identity arrow, a single arrow of order 2, then the functor $J \to \operatorname{Ind}(C)$ defined by (X, s) admits no indexed representation.

Exposé I. Presheaves

Sections 8.9–8.13

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 8.9 through 8.13. A few evident typographical slips in the source notation are corrected where the mathematical context forces the correction.

8 Ind-objects and pro-objects

8.9 Exactness properties of $\mathrm{Ind}(C)$

Proposition 8.9.1. Let C be a $\mathcal{U}$-category.

a) The canonical functors L (8.2.4.8) and c (8.4.1)

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C}$$

commute with projective limits.

b) Suppose that finite inductive limits are representable in C, and that C is equivalent to a small category. Then small projective limits are representable in $\mathrm{Ind}(C)$.

c) If small projective limits (respectively finite projective limits) are representable in C, then the same is true in $\mathrm{Ind}(C)$. Let J be a category which is finite and rigid, or discrete. If projective limits of type J are representable in C, the same type of projective limits is representable in $\mathrm{Ind}(C)$.

d) Small filtered inductive limits in $\mathrm{Ind}(C)$, which are representable by 8.5.1, are left exact, that is, they commute with finite projective limits.

Proof.

a) The fact that L commutes with projective limits follows from the fact that it is fully faithful and from the computation of projective limits in $\widehat{C}$ argument by argument. Thus, in order to verify that a projective system of morphisms $F \to F_i$ $(i \in I)$ in $\widehat{C}$ makes F the projective limit of the F_i, it is enough to verify the analogous assertion for the projective systems of sets

$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(X, F_i)$$

for all $X \in \mathrm{Ob}\, C$. But $C \subset \mathrm{Ind}(C)$.

The fact that c commutes with projective limits follows formally from the fact that L does so, together with the fact that the composite $L \circ c : C \to \widehat{C}$ does so.

b) In view of a), the assertion says that every projective limit of ind-representable presheaves is ind-representable. This follows immediately from criterion 8.3.3(v), since a projective limit of left exact functors is left exact; this is just the fact that projective limits commute with projective limits (2.5.0).

c) This follows formally from the analogous property of $\widehat{C}$ (3.3), together with the facts that L is conservative, being fully faithful, and that it commutes with the limits of the type in question, by a) and 8.5.1.

d) It is well known (cf. 2.3) that representability of finite projective limits is equivalent to representability of projective limits of the special finite ordered types consisting of the empty type and the elementary fiber-product type. Representability of small projective limits is equivalent to representability of products and finite projective limits. Hence the second assertion made in c) implies the first.

First suppose that I is finite. Using 8.8.5 on the representability of functors $\varphi : J \to \mathrm{Ind}(C)$ in indexed form (8.5.4)

$$\Phi : J \times I \longrightarrow C, \tag{8.9.1.1}$$

the existence of $\varprojlim \varphi$ is a special case of the following more precise and more general result.

Corollary 8.9.2. Let $\varphi : J \to \mathrm{Ind}(C)$ be a functor given in indexed form Φ as in (8.9.1.1), where J is a finite category. If, for every $i \in \mathrm{Ob}\, I$, the partial functor $j \mapsto \Phi(j,i)$ has a projective limit (respectively an inductive limit) representable in C, then φ has a projective limit (respectively an inductive limit) representable in $\mathrm{Ind}(C)$, and there is a canonical isomorphism

$$\varprojlim \varphi \simeq \text{“}\varinjlim_i\text{”} \varprojlim_j \Phi(j,i), \tag{8.9.2.1}$$

respectively

$$\varinjlim \varphi \simeq \text{“}\varinjlim_i\text{”} \varinjlim_j \Phi(j,i), \tag{8.9.2.2}$$

where $\varprojlim_j$ (respectively $\varinjlim_j$) is computed in C.

For the first formula, one uses only that in $\mathrm{Ind}(C)$ filtered inductive limits commute with $\varprojlim_j$, and that c commutes with $\varprojlim_J$ by 8.9.1(d) and a):

$$\begin{aligned}\varprojlim \varphi &= \varprojlim_j \text{“}\varinjlim_i\text{”}\, \Phi(j,i)\\ &\xleftarrow{\sim} \text{“}\varinjlim_i\text{”} \varprojlim_j{}^{\mathrm{Ind}(C)} \Phi(j,i) \simeq \text{“}\varinjlim_i\text{”} \overset{C}{\varprojlim_j} \Phi(j,i).\end{aligned}$$

The second is proved in the same way, using the commutation of inductive limits of type I with inductive limits of type J (2.5.0) and the fact that c commutes with inductive limits of type J, proved in 8.9.4(a) below.

The case where J is discrete. This case is contained in the following more precise assertion.

Corollary 8.9.3. Let I_α be essentially small filtered categories indexed by a small set A, and, for each $\alpha \in A$, let

$$X(\alpha) = (X(\alpha)_{i_\alpha})_{i_\alpha \in I_\alpha}$$

be an ind-object of C indexed by I_α. Let I be the product category of the I_α, and suppose that, for every $\mathbf{i} = (i_\alpha)_{\alpha \in A} \in I$, the product

$$\prod_{\alpha \in A} X(\alpha)_{i_\alpha}$$

is representable in C. Then the product $\prod_{\alpha\in A} X(\alpha)$ is representable in $\mathrm{Ind}(C)$, and is canonically isomorphic to the ind-object indexed by I given by the formula

$$\prod_{\alpha\in A} \text{“}\varinjlim_{i_\alpha\in I_\alpha}\text{”}\, X(\alpha)_{i_\alpha} \simeq \text{“}\varinjlim_{\mathbf{i}=(i_\alpha)_{\alpha\in A}\in I}\text{”} \left(\prod_{\alpha\in A} X(\alpha)_{i_\alpha}\right), \tag{8.9.3.1}$$

where the product on the right is computed in C. Notice that I is filtered and essentially small because the I_α are.

Indeed, it is well known, and immediate after reducing to the category of sets, that the displayed formula is valid when the limits are computed in $\widehat{C}$. The conclusion then follows from the fact that L commutes with the limits under consideration (8.9.1(a) and 8.5.1).

Remarks 8.9.4. The proof just given for 8.9.2 and 8.9.3 shows, more generally, that if, for every $i \in \mathrm{Ob}\, I$, the limit $\varprojlim_j \Phi(j,i)$ (respectively $\varinjlim_j \Phi(j,i)$), computed in $\mathrm{Ind}(C)$, is representable, then the corresponding limit of φ is representable. This and the argument in c) show that, for a given category J arising from a finite ordered set or a discrete category, or more generally one which is C-admissible in the sense of 8.3.1, projective limits (respectively inductive limits) of type J are representable in $\mathrm{Ind}(C)$ if and only if, for every functor $\varphi : J \to C$, the projective limit (respectively inductive limit) of φ, computed in $\mathrm{Ind}(C)$, is representable. In the non-parenthesized case this also means, by a), that every projective limit of type J of representable presheaves on C is ind-representable.

Proposition 8.9.5. Let C be a $\mathcal{U}$-category.

a) The canonical functor
$$c : C \longrightarrow \mathrm{Ind}(C)$$
is right exact, hence exact in view of 8.9.1(a).

b) If finite inductive limits (respectively finite sums) are representable in C, then small inductive limits (respectively small sums) are representable in $\mathrm{Ind}(C)$. Let J be a finite or discrete preordered set. If inductive limits of type J are representable in C, then the same is true in $\mathrm{Ind}(C)$.

Proof.

a) Suppose that in C one has $X \simeq \varinjlim_J X_j$, with J finite and with X and the X_j in C. Then, for every ind-object $\mathcal{Y} = (Y_i)_{i\in I}$ of C,
$$\begin{aligned}\mathrm{Hom}(X,\mathcal{Y}) &\simeq \varinjlim_i \mathrm{Hom}(X,Y_i) \simeq \varinjlim_i \varprojlim_j \mathrm{Hom}(X_j,Y_i)\\ &\simeq \varprojlim_j \varinjlim_i \mathrm{Hom}(X_j,Y_i) \simeq \varprojlim_j \mathrm{Hom}(X_j,\mathcal{Y}),\end{aligned}$$
where the middle isomorphism is 2.8. The desired conclusion follows by comparing the extreme terms.

b) It was already observed in 8.8.4 that the assertions follow from a) and 8.9.2, at least for finite inductive limits. To prove the conclusions in the infinite cases, one is reduced to the case of sums, which may be interpreted as a filtered inductive limit of finite sums; the conclusion follows from 8.5.1.

Remark 8.9.6. We have already observed that the functor c does not in general commute with filtered inductive limits, and therefore not with infinite sums either, in contrast with what happens for $L : \mathrm{Ind}(C) \to \widehat{C}$. Conversely, apart from commutation with filtered inductive limits (8.5.1), L practically never has commutation properties with any other type of inductive limit: initial object, sum of two objects, or cokernels of double arrows. Thus, if C has an initial object $\varnothing_C$, which is then an initial object of $\mathrm{Ind}(C)$ by a), $\varnothing_C$ is never an initial object of $\widehat{C}$, that is, never identical with the constant presheaf of value $\varnothing$, since $\mathrm{Hom}(\varnothing_C, \varnothing_C) \neq \varnothing$.

Proposition 8.9.7. Let

$$f : C \longrightarrow C'$$

be a functor between $\mathcal{U}$-categories, and let J be a finite and rigid category (8.8.5). If inductive limits (respectively projective limits) of type J are representable in C and if f commutes with them, then

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$$

also commutes with this type of limit.

This follows immediately from 8.8.5 and from the calculation, in 8.9.2, of finite limits in a category of ind-objects, for a functor represented in indexed form.

Corollary 8.9.8. If finite inductive limits (respectively finite projective limits) are representable in C, and if f is right exact (respectively left exact), then the same is true of $\mathrm{Ind}(f)$. In the non-parenthesized case, $\mathrm{Ind}(f)$ even commutes with arbitrary small inductive limits.

The first assertion follows from 8.9.7. The second follows from the first, taking into account that $\mathrm{Ind}(f)$ commutes with filtered inductive limits by 8.6.3.

Exercise 8.9.9. Let C be a $\mathcal{U}$-category.

a) Suppose finite sums are representable in C. Show that if finite sums in C are disjoint (respectively universal; cf. II 4.5), then small sums in $\mathrm{Ind}(C)$ are disjoint (respectively universal).

b) Suppose every morphism in C factors as an epimorphism followed by a monomorphism (respectively as an effective epimorphism followed by a monomorphism, respectively as an epimorphism followed by an effective monomorphism, respectively as an effective epimorphism followed by an effective monomorphism). Show that $\mathrm{Ind}(C)$ satisfies the same property. In the last parenthesized case one concludes that every epimorphism of $\mathrm{Ind}(C)$ is effective, every monomorphism of $\mathrm{Ind}(C)$ is effective, and every bimorphism of $\mathrm{Ind}(C)$ is an isomorphism. Show that, in this case, every epimorphism (respectively monomorphism) of $\mathrm{Ind}(C)$ can be represented by an inductive system of epimorphisms (respectively monomorphisms) of C. If, moreover, every epimorphism (respectively every monomorphism) in C is universal, then the same property holds in $\mathrm{Ind}(C)$.

c) Suppose C is additive (respectively abelian). Then $\mathrm{Ind}(C)$ is additive (respectively abelian).

d) Let $X = \text{“}\varinjlim\text{”}_I X_i$ be an ind-object of C, and let $R \rightrightarrows X$ be an equivalence relation in X. For every $i \in \mathrm{Ob}\, I$, let $R_i \rightrightarrows X_i$ be the equivalence relation in X_i, considered as an object of $\mathrm{Ind}(C)$, induced by R via $X_i \to X$. Here one assumes that the fiber product

$$R_i = R \times_{X \times X} (X_i \times X_i)$$

in $\mathrm{Ind}(C)$ is representable by an object of $\mathrm{Ind}(C)$, which is the case if finite projective limits are representable in C. Show that, for R to be effective, it is enough that the R_i be effective.

e) Let X be an object of C, and let R be an equivalence relation in $c(X)$, that is, in X considered as an object of $\mathrm{Ind}(C)$. Suppose fiber products are representable in C. For R to be effective, it is necessary that R be of the form "$\varinjlim$"$_i R_i$, where the R_i are equivalence relations in X, viewed as an object of C; this condition is sufficient if equivalence relations are effective in C.

f) Suppose every morphism in C factors as an effective epimorphism followed by an effective monomorphism, and suppose finite projective limits are representable. Let X be an object of C, and let R be a subobject of $X \times X$, viewed as an object of $\mathrm{Ind}(C)$. Write R in the form "$\varinjlim$"$_i Z_i$, where $(Z_i)_{i\in I}$ is a filtered increasing family of subobjects of $X \times X$ in C; this is possible by b). Show that, for R to be an equivalence relation in X, it is necessary and sufficient that for every $i \in \mathrm{Ob}\, I$ there exist $j \in \mathrm{Ob}\, I$ containing $s(Z_i)$ and $Z_i \circ Z_i$, where s is the symmetry of $X \times X$.

g) Suppose finite projective limits and finite sums are representable in C, every equivalence relation in C is effective, and every morphism in C factors as an effective epimorphism followed by an effective monomorphism. Show that equivalence relations in $\mathrm{Ind}(C)$ are universal if and only if C satisfies the following condition:

ST) For every object X of C and every subobject Z of $X \times X$ in C, if one defines recursively a sequence of subobjects Z_n $(n \geq 0)$ of $X \times X$ in C by $Z_0 = Z$ and

$$Z_{n+1} = \mathrm{Sup}(Z_n, s(Z_n), Z_n \circ Z_n),$$

where the supremum is taken in the set of subobjects of $X \times X$, which exists by the hypotheses on C, then the sequence $(Z_n)_{n\geq 0}$ is stationary.

k) Show that in $\mathrm{Ind}(\mathbf{Set})$ equivalence relations are not necessarily effective.

*NB. For criteria ensuring that $\mathrm{Ind}(C)$ is a topos, see VI 8.9.9.$_*$

Exercise 8.9.10. Let C be a $\mathcal{U}$-category. Put $\mathrm{Pro}(C) = (\mathrm{Ind}(C^{\mathrm{op}}))^{\mathrm{op}}$ (cf. 8.11).

a) Suppose finite sums (respectively small sums) are representable in C. Show that if they are disjoint (cf. II 4.5), then $\mathrm{Pro}(C)$ satisfies the same condition.

b) Let J be a small set such that sums indexed by J are representable in C, hence also in $\mathrm{Pro}(C)$. Let $(X(\alpha))_{\alpha\in J}$ be a family of elements of $\mathrm{Pro}(C)$, with

$$X(\alpha) = \text{“}\varprojlim_{I_\alpha}\text{”} X_{i_\alpha}.$$

Show that, for the sum of the $X(\alpha)$ in $\mathrm{Pro}(C)$ to be universal, it is enough that the same be true for each of the families $\varprojlim_J X_{i_\alpha}$, where $(i_\alpha)_{\alpha\in J} \in \prod_{\alpha\in J} I_\alpha$. Deduce that, if sums of type J are universal in C, then for the same to be true in $\mathrm{Pro}(C)$ it is necessary and sufficient that, for every family $(X(\alpha))_{\alpha \in J}$ as above with $I_\alpha = I$ for all $\alpha \in J$, the canonical homomorphism in $\mathrm{Pro}(C)$

$$“\varprojlim_{I}” \amalg_\alpha X(\alpha)_i \longleftarrow “\varprojlim_{I^J}” \amalg_{\alpha \in J} X(\alpha)_{i_\alpha}$$

be an isomorphism. Deduce that in $\mathrm{Pro}(\mathbf{Set})$ sums of type J are universal if and only if J is finite.

Exercise 8.9.11. Let C be a $\mathcal{U}$-category.

a) Let $(X_i \to X)_{i \in I}$ be a family of morphisms in C. Show that for it to be epimorphic in $\mathrm{Ind}(C)$, it is enough that it admit a finite subfamily which is epimorphic in C. Prove that this condition is also necessary if one assumes that every finite family of morphisms with target X in C factors as an epimorphic family followed by an effective monomorphism (10.5). For the latter assertion, let P be the set of finite subsets of I, ordered by inclusion, let $X' = (X'_J)_{J \in P}$ be the ind-object formed by the images of the finite subfamilies of the given family, and finally let

$$X'' = X \amalg_{X'} X = “\varinjlim”_J \, X \amalg_{X'_J} X.$$

Show that the two canonical morphisms $X \rightrightarrows X''$ are distinct but coincide on the X'_J.

b) Suppose that in C every finite family of morphisms with the same target factors as an epimorphic family followed by an effective monomorphism. Prove that $\mathrm{Ind}(C)$ has a small subcategory generating by epimorphisms (7.1) if and only if C has a small subcategory C' such that every object of C is the target of a finite epimorphic family with source in C'. If one assumes that C has a small generating subcategory (7.1), then $\mathrm{Ind}(C)$ has a small subcategory generating by epimorphisms if and only if C is equivalent to a small category. For this last statement, use 7.5.2.

c) $\mathrm{Ind}(\mathbf{Set})$ has no small generating subcategory.

Exercise 8.9.12. Let C be a $\mathcal{U}$-category, and consider

$$\mathrm{Pro}(C) = \mathrm{Ind}(C^{\mathrm{op}})^{\mathrm{op}}.$$

a) Show that if $\mathrm{Pro}(C)$ has a small subcategory P' generating by epimorphisms (7.1), then there is a small subcategory C' of C which generates by epimorphisms in $\mathrm{Pro}(C)$. Take the category of components of the objects of P'.

b) Show that the category $\mathrm{Pro}(\mathbf{Set})$ has no small subcategory generating by epimorphisms. If C' is as in a), choose a set X whose cardinal strictly dominates the cardinals of the elements of C', and consider the pro-set $\mathcal{X}$ formed by the complements in X of the subsets of cardinal $< \mathrm{card}(X)$. Show that, for every nonempty set Y of cardinal $< \mathrm{card}(X)$, one has $\mathrm{Hom}(Y, \mathcal{X}) = \varnothing$, but that $\mathcal{X}$ is not isomorphic to the constant pro-set $\varnothing$.

8.10 Dual notions: pro-objects and pro-representable functors

Let C be a $\mathcal{U}$-category. A *pro-object* of C is a functor

$$\varphi : I^{\mathrm{op}} \longrightarrow C, \tag{8.10.1}$$

where I is an essentially small filtered category, called the index category, and where, as usual, I^{op} denotes the opposite category. Let $\mathcal{V}$ be a universe such that $\mathcal{V} \supset \mathcal{U}$. Notice that pro-objects of C indexed by $I \in \mathcal{V}$ are in bijective correspondence with ind-objects of C^{op} indexed by $I \in \mathcal{V}$: to such an ind-object $\psi : I \to C^{\mathrm{op}}$ one associates the pro-object $\varphi = \psi^{\mathrm{op}} : I^{\mathrm{op}} \to C$. Following this correspondence, one defines, by transport of structure and reversal of arrows, the notion of morphism between pro-objects of C from the analogous notion (8.2.4.2, 8.2.5.1) for ind-objects. One accordingly obtains the category of pro-objects of C indexed by $I \in \mathcal{V}$, together with a canonical isomorphism

$$\mathrm{Pro}_{\mathcal{V}}(C,\mathcal{U}) \simeq \mathrm{Ind}_{\mathcal{V}}(C^{\mathrm{op}},\mathcal{U})^{\mathrm{op}}. \tag{8.10.2}$$

For $\mathcal{V} = \mathcal{U}$ one simply speaks of the *category of pro-objects of* C, denoted $\mathrm{Pro}_{\mathcal{U}}(C)$ or simply $\mathrm{Pro}(C)$; it is therefore defined by

$$\mathrm{Pro}(C) = \mathrm{Pro}_{\mathcal{U}}(C,\mathcal{U}) \simeq \mathrm{Ind}(C^{\mathrm{op}})^{\mathrm{op}}. \tag{8.10.3}$$

If two pro-objects are given in indexed form

$$\mathcal{X} = (X_i)_{i\in I}, \qquad \mathcal{Y} = (Y_j)_{j\in J}, \tag{8.10.4}$$

then formula (8.2.5.1) takes here the form

$$\mathrm{Hom}(\mathcal{X},\mathcal{Y}) \simeq \varprojlim_j \varinjlim_i \mathrm{Hom}(X_i, Y_j). \tag{8.10.5}$$

The functor (8.4.1), applied to C^{op}, gives, after passage to opposite categories, a canonical functor, denoted by the same letter c when no confusion is possible, which allows one to identify C with a full subcategory of $\mathrm{Pro}(C)$:

$$c : C \longrightarrow \mathrm{Pro}(C). \tag{8.10.6}$$

The arrows were reversed in the definition of morphisms of pro-objects precisely in order to obtain such a functor, rather than a functor $C \to \mathrm{Pro}(C)^{\mathrm{op}}$. Pro-objects in the essential image of (8.10.6) are again called *essentially constant pro-objects*.

Put

$$\check{C} = (C^{\mathrm{op}})\widehat{\ } = \mathrm{Hom}(C, \mathbf{Set}). \tag{8.10.7}$$

Then the functor (8.2.4.7) may be regarded as a canonical fully faithful functor

$$L : \mathrm{Pro}_{\mathcal{V}}(C,\mathcal{U}) \longrightarrow \check{C}^{\mathrm{op}}, \tag{8.10.8}$$

and in particular one obtains

$$L : \mathrm{Pro}(C) \hookrightarrow \check{C}^{\mathrm{op}}. \tag{8.10.9}$$

We shall leave to the reader the task of translating, as needed, the results of the preceding sections and of the following ones from the language of ind-objects into that of pro-objects. Depending on the mathematical context, one or the other language is the more useful, not to mention the cases in which both notions occur simultaneously, for instance when one has to consider complex categories such as $\mathrm{Pro}(\mathrm{Ind}(C))$ or $\mathrm{Ind}(\mathrm{Pro}(C))$. We content ourselves with giving a few additional indications, in order to fix terminology and notation and to clarify the yoga.

8.10.10

A *covariant functor* $F : C \to \mathbf{Set}$ is called a *pro-representable functor* if it lies in the essential image of (8.10.9), or equivalently of 8.10.8. The full subcategory of $\check{C}$ consisting of such functors is therefore equivalent to $\mathrm{Pro}(C)^{\mathrm{op}}$. One generally prefers to work in the opposite category, as the essential image as a subcategory of (8.10.9), hence equivalent to $\mathrm{Pro}(C)$ itself. In accord with this usage, it is often preferable to regard covariant functors

$$F : C \longrightarrow \mathbf{Set}$$

as objects of $\mathrm{Hom}(C, \mathbf{Set})^{\mathrm{op}} = \check{C}^{\mathrm{op}}$, and consequently to write the category of morphisms

$$X \longrightarrow F \quad \text{in } \check{C}^{\mathrm{op}},$$

that is, the pairs (X, u) consisting of an object X of C and an element $u \in F(X)$, as

$$_F\backslash C. \tag{8.10.11}$$

With this convention one obtains a covariant functor, the source functor,

$$_F\backslash C \longrightarrow C, \tag{8.10.12}$$

and 3.4 is rewritten in the form

$$F \xrightarrow{\sim} \text{“}\varprojlim_{(_F\backslash C)^{\mathrm{op}}}\text{”} X. \tag{8.10.13}$$

8.10.14

Criterion 8.3.3, applied to C^{op}, becomes a criterion of pro-representability for F: it is necessary and sufficient that $(_F\backslash C)^{\mathrm{op}}$ be filtered and essentially small. The latter condition is superfluous if C is equivalent to a small category. If, moreover, finite projective limits are representable in C, this is equivalent to saying that F commutes with them, that is, that F is *left exact*.

8.10.15

In $\mathrm{Pro}(C)$, small filtered projective limits are representable, and the functor (8.10.9) commutes with them. In other words, this functor transforms projective limits in $\mathrm{Pro}(C)$ into inductive limits in $\check{C} = \mathrm{Hom}(C, \mathbf{Set})$, just as does its composite $C \to \check{C}^{\mathrm{op}}$ with (8.10.6), which shares with it the unfortunate property of being contravariant when it is regarded as taking values in $\check{C}$. Notice carefully that pro-representable functors from C to $\mathbf{Set}$ are functors which are small filtered inductive limits of representable functors, and not projective limits, as the terminology might perhaps suggest.

8.11 Ind-adjoints and pro-adjoints

8.11.1

Let

$$f : C \longrightarrow C' \tag{8.11.1.1}$$

be a functor between $\mathcal{U}$-categories. It induces the functor $F \mapsto F \circ f$

$$f^* : \widehat{C'} \longrightarrow \widehat{C}. \tag{8.11.1.2}$$

We say that f *admits an ind-adjoint* if this functor sends ind-representable functors to ind-representable functors. Since f^* commutes with inductive limits, and since the full subcategory of $\widehat{C'}$ formed by the ind-representable functors is stable under small filtered inductive limits, it is equivalent to say that f admits an ind-adjoint, or that f^* sends the representable objects of $\widehat{C'}$ to ind-representable functors on C, that is,

$$X \longmapsto \operatorname{Hom}(f(X), Y') \quad \text{is ind-representable for every } Y' \in \operatorname{Ob} C'. \tag{8.11.1.3}$$

Another way to express the condition that f admit an ind-adjoint is to say that there is a functor

$$g : \operatorname{Ind}(C') \longrightarrow \operatorname{Ind}(C) \tag{8.11.1.4}$$

which is essentially induced by f^*, that is, such that there is an isomorphism of bifunctors

$$\operatorname{Hom}_{\operatorname{Ind}(C)}(X, g(\mathcal{Y}')) \simeq \operatorname{Hom}_{\operatorname{Ind}(C')}(f(X), \mathcal{Y}'), \tag{8.11.1.5}$$

where $X \in \operatorname{Ob} C$ and $\mathcal{Y}' \in \operatorname{Ob} \operatorname{Ind}(C')$. In fact it is enough to have a functor

$$g_0 : C' \longrightarrow \operatorname{Ind}(C) \tag{8.11.1.6}$$

and an isomorphism of bifunctors

$$\operatorname{Hom}_{\operatorname{Ind}(C)}(X, g_0(Y')) \simeq \operatorname{Hom}_{C'}(f(X), Y') \tag{8.11.1.7}$$

for $X \in \operatorname{Ob} C$ and $Y' \in \operatorname{Ob} C'$. The functor g, respectively g_0, is plainly determined by f up to unique isomorphism, and conversely f can be reconstructed up to canonical isomorphism from g, respectively from g_0. Formula (8.11.1.5) makes it clear that g commutes with inductive limits and extends g_0; this determines it up to unique isomorphism in terms of g_0 by 8.7.2. The functor g, and sometimes also the functor g_0 that it extends, is called the *ind-adjoint functor of* f. There is no need here to specify "right", since the other one, if it exists, will be called the pro-adjoint (cf. 8.11.5 below).

Of course, if f admits a right adjoint

$$f^{\mathrm{ad}} : C' \longrightarrow C,$$

then it admits an ind-adjoint g, and this is canonically isomorphic to the canonical extension of f^{ad} to ind-objects:

$$g \simeq \operatorname{Ind}(f^{\mathrm{ad}}). \tag{8.11.1.8}$$

The notion of ind-adjoint is therefore a natural generalization of the notion of right adjoint.

Now consider the canonical extension

$$\operatorname{Ind}(f) : \operatorname{Ind}(C) \longrightarrow \operatorname{Ind}(C'). \tag{8.11.1.9}$$

We then have the following proposition.

Proposition 8.11.2. For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary and sufficient that the functor $\operatorname{Ind}(f)$ in (8.11.1.9) admit a right adjoint. This right adjoint is canonically isomorphic to the ind-adjoint (8.11.1.4).

The sufficiency and the last assertion are trivial from the ind-adjunction formula (8.11.1.5). For necessity, note that, by passage to the projective limit in this formula over the X_i, one deduces an adjunction isomorphism for the ind-objects $\mathcal{X} = (X_i)_{i\in I}$ and $\mathcal{Y}'$:

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(\mathcal{X}, g(\mathcal{Y}')) \simeq \mathrm{Hom}_{\mathrm{Ind}(C')}(\mathrm{Ind}(f)(\mathcal{X}), \mathcal{Y}'). \tag{8.11.2.1}$$

□

Corollary 8.11.3. If f admits an ind-adjoint, then $\mathrm{Ind}(f)$ commutes with inductive limits, and the ind-adjoint g commutes with projective limits.

Proposition 8.11.4. For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary that f be right exact, and this condition is sufficient when C is equivalent to a small category.

This follows from criterion (8.11.1.3), and from 8.3.1 and 8.3.3(iv). For another criterion in terms of the notion of accessible functor, see 8.13.3.

8.11.5

Now consider the functor $F \mapsto F \circ f$

$$f^{\mathrm{op}*} : \check{C}' \longrightarrow \check{C} \tag{8.11.5.1}$$

induced by f. We shall say that f *admits a pro-adjoint* if the preceding functor sends pro-representable functors to pro-representable functors, that is, if there exists a functor, called the *pro-adjoint functor of* f,

$$g : \mathrm{Pro}(C') \longrightarrow \mathrm{Pro}(C), \tag{8.11.5.2}$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Pro}(C)}(g(\mathcal{Y}'), \mathcal{X}) \simeq \mathrm{Hom}_{\mathrm{Pro}(C')}(\mathcal{Y}', \mathrm{Pro}(f)(\mathcal{X})). \tag{8.11.5.3}$$

Of course, saying that f admits a pro-adjoint means that f^{op} admits an ind-adjoint, so the notions and results for ind-adjoints translate trivially into terms of pro-adjoints. Let us merely point out that f admits a pro-adjoint if and only if

$$\mathrm{Pro}(f) : \mathrm{Pro}(C) \longrightarrow \mathrm{Pro}(C')$$

admits a left adjoint, and that in this case the preceding functor g is such a left adjoint of $\mathrm{Pro}(f)$; and that for this it is necessary that f be left exact, this condition also being sufficient when C is equivalent to a small category. In this case, f is exact if and only if it admits both an ind-adjoint and a pro-adjoint.

Example 8.11.6. Consider the case of a functor

$$f : C \longrightarrow \mathbf{Set}.$$

If this functor admits a pro-adjoint, then it is pro-representable. The converse is true if and only if the full subcategory of $\check{C}$ formed by pro-representable functors is stable under small products; this is the case in particular if C is equivalent to a small category (8.10.14), or if small products are representable in C (8.9.5(b) applied to C^{op}). Thus, up to minor qualifications, one may say that, for a functor $f : C \to C'$, the existence of a pro-adjoint is the *natural generalization of pro-representability of* f, which is defined when $C' = \mathbf{Set}$.

8.12 Strict ind-objects and pro-objects. Application to a representability criterion

8.12.1

Let
$$\mathcal{X} = (X_i)_{i \in I}$$
be an ind-object of the $\mathcal{U}$-category C, and let F be the presheaf which it ind-represents. It is immediate that the following two conditions are equivalent: the canonical morphisms
$$X_i \longrightarrow F = \text{“}\varinjlim_i\text{”} X_i$$
are monomorphisms of $\mathrm{Ind}(C)$, or equivalently of $\widehat{C}$, that is, monomorphisms of functors argument by argument; and, for every arrow $i \to j$ of I, the corresponding transition morphism
$$X_i \longrightarrow X_j$$
is a monomorphism. When these conditions hold, and when moreover I is an ordered category, $\mathcal{X}$ is called a *strict ind-object*. Notice that this condition is not invariant under isomorphism of ind-objects. An ind-object will be called *essentially strict* if it is isomorphic to a strict ind-object. A presheaf will be called *strictly ind-representable* if it is ind-representable by a strict ind-object; thus $F = \text{“}\varinjlim_i\text{”} X_i$ is strictly ind-representable if and only if $\mathcal{X} = (X_i)_{i \in I}$ is essentially strict.

8.12.1.1. Let F be a presheaf on C, and consider the full subcategory of $C_{/F}$ formed by the arrows $X \to F$ with source in C which are monomorphisms. We shall call it the *category of representable subfunctors of* F; it is the category associated with the ordered set of representable subfunctors of F, ordered by the order induced from the set of subobjects of F. The following then follows immediately from the definitions.

Proposition 8.12.2. For a presheaf F on C to be strictly ind-representable, it is necessary and sufficient that the ordered category I of representable subfunctors of F be filtered and essentially small and that one have
$$\varinjlim_I X_i \xrightarrow{\sim} F. \tag{8.12.2.1}$$
When, for every object X of C, the set of subobjects of X in C is small, for example when C admits a small generating subcategory (7.4), this implies that the category of representable subfunctors is even small.

8.12.2.2. If F is strictly ind-representable, there is therefore a preferred way to ind-represent it by an ind-object, and that ind-object is even strict: take the representation (8.12.2.1). A strict ind-object $\mathcal{Y}$ is called *saturated* if it is isomorphic to an ind-object of the form (X_i) appearing in (8.12.2.1), for a suitable F. Thus there exists, up to unique isomorphism, only one saturated strict ind-object isomorphic to the given strict ind-object, namely the one considered in 8.12.2, taking $F = \varinjlim \mathcal{Y}$ as a limit in $\widehat{C}$.

8.12.2.3. Suppose one already knows that a small cofinal subset can be found in $\mathrm{Ob}\, C_{/F}$, which is the case in particular if F is ind-representable. Then the same condition follows in the full subcategory I considered in 8.12.2, so in the criterion 8.12.2 one may omit the condition that I be essentially small.

8.12.3

Let F be a presheaf on C, and consider an object

$$u : X \longrightarrow F$$

of $C_{/F}$. We say that u, or the pair (X, u), is *minimal* if, for every factorization of u as

$$X \xrightarrow{p} X' \xrightarrow{u'} F, \tag{8.12.3.1}$$

with p a strict epimorphism (10.2), the morphism p is an isomorphism. Considering u as an element of $F(X)$ (1.4), to say that u is minimal therefore means that every strict epimorphism $p : X \to X'$ such that

$$u \in \operatorname{Im}\bigl(F(p) : F(X') \to F(X)\bigr)$$

is an isomorphism. This notion is clarified by part a) of the following lemma.

Lemma 8.12.4. Let F be a presheaf on C.

a) Suppose that F transforms cokernels into kernels, and consider a morphism

$$u : X \longrightarrow F$$

with $X \in \operatorname{Ob} C$. For u to be a monomorphism, it is sufficient, when cokernels of double arrows are representable in C, that u be minimal; this condition is also necessary if fiber products are representable in C.

b) Suppose finite inductive limits are representable in C. For the subcategory of representable subfunctors of F (8.12.1.1) to be filtered, not necessarily small, and to have inductive limit F, it is sufficient that F be left exact and that every morphism $u : X \to F$, with $X \in \operatorname{Ob} C$, factor as

$$X \longrightarrow X' \xrightarrow{u'} F$$

with u' minimal. This condition is also necessary if fiber products are representable in C.

c) Suppose that C admits a small generating subcategory (7.1), and that I, the category of representable subfunctors of F, is filtered. Then I is small.

Proof.

a) Suppose u is minimal; we prove that u is a monomorphism, that is, that for every double arrow $v, v' : Y \rightrightarrows X$ such that $uv = uv'$, one has $v = v'$. Indeed, if $X' = \operatorname{Coker}(v, v')$, then u factors as $X \xrightarrow{p} X' \xrightarrow{u'} F$, since F transforms cokernels into kernels. As p is a strict epimorphism by construction, it follows that p is an isomorphism, that is, $v = v'$. Conversely, suppose u is a monomorphism, and prove that it is minimal. Consider a factorization (8.12.3.1). Since p is a strict epimorphism, it is the cokernel of the canonical double arrow

$$v, v' : X \times_{X'} X \rightrightarrows X.$$

Since $(u'p)v = (u'p)v'$ and $u'p = u$ is a monomorphism, we have $v = v'$, hence p is an isomorphism.

b) Sufficiency: since F is left exact, $C_{/F}$ is filtered. Since we know that $F = \varinjlim_{C_{/F}} X$, we are reduced by 8.1.3(c) to proving that the full subcategory I of $C_{/F}$ formed by the subobjects of F is cofinal in $C_{/F}$. By the sufficiency assertion in a), this is exactly what is ensured by the hypothesis that every object of $C_{/F}$ is dominated by a minimal object. Necessity: since every filtered inductive limit of left exact functors is again left exact, the first condition is trivially necessary. The second then follows from the necessity assertion in a).

c) A representable subfunctor $X \hookrightarrow F$ of F is known once one knows the full subcategory $C'_{/X}$ of $C'_{/F}$, where C' is a fixed small generating subcategory of C. Use the hypothesis that I is filtered. Since $C'_{/F}$ is small, the set of its full subcategories is small, whence the conclusion.

Proposition 8.12.5. Let C be a $\mathcal{U}$-category in which finite inductive limits are representable and which admits a small generating subcategory (7.1). Let F be a presheaf on C. For F to be strictly ind-representable, it is sufficient that it satisfy the following two conditions; and these conditions are also necessary if fiber products are representable in C:

a) F is left exact.

b) Every pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in F(X)$, is dominated in $C_{/F}$ by a minimal pair (8.12.3). In other words, there is a minimal pair (X', u'), with $X' \in \operatorname{Ob} C$ and $u' \in F(X')$, and a morphism $f : X \to X'$ such that $u = F(f)(u')$.

Sufficiency follows from 8.12.2 and 8.12.4(b),(c); necessity follows from 8.3.1 and 8.12.4(a).

Remark 8.12.6. When F transforms amalgamated sums in C into fiber products, one sees immediately that, for a given pair (X, u) as in b), the set of strict quotients X' of X such that

$$u \in \operatorname{Im}\big(F(X') \to F(X)\big)$$

is decreasing filtered. It follows that if the set of strict quotients of X is Artinian, then condition b) of 8.12.5 is automatically satisfied. Thus, if the preceding condition on X is satisfied for every object X of C – in which case one sometimes also says that the objects of C^{op} are *Artinian* – then 8.12.5 implies that F is strictly ind-representable if and only if F is left exact. In this case, F is strictly ind-representable as soon as it is ind-representable (8.3.1).

Corollary 8.12.7. Let C be a $\mathcal{U}$-category in which small projective limits are representable, and which admits a small cogenerating subcategory (7.9, 7.13). Then a functor $F : C \to \mathbf{Set}$ is representable if and only if F commutes with small projective limits.

Necessity is clear. For sufficiency, apply 8.12.5 to the opposite category C^{op}. To prove first that F is strictly pro-representable, it remains to prove that every pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in F(X)$, is dominated by a minimal pair. But the family $(X_i)_{i \in I}$ of strict subobjects of X such that

$$u \in \operatorname{Im}\big(F(X_i) \to F(X)\big)$$

is small (7.5 in its dual form). Since F is left exact, this family is cofiltered (8.12.6), and since F commutes with small projective limits one sees that $X' = \varprojlim_i X_i$ is a smallest object of this family. Use the fact that F, being left exact, transforms monomorphisms into monomorphisms.

If $u' \in F(X')$ is the unique element whose image in $F(X)$ is u, it follows that (X', u') is a minimal pair dominating (X, u). This proves that F is pro-representable, and the conclusion follows from the following lemma.

Lemma 8.12.8.1. Let $F : C \to \mathbf{Set}$ be a functor, where C is a $\mathcal{U}$-category in which small projective limits are representable. For F to be representable, it is necessary and sufficient that it be pro-representable and commute with small projective limits.

Necessity is clear. We prove sufficiency. To say that F is representable plainly means that ${}_F\backslash C$ has an initial object; indeed, the initial objects of ${}_F\backslash C$ are precisely the isomorphisms $F \to X$, that is, the data of a representation for F. Since F is pro-representable, $({}_F\backslash C)^{\mathrm{op}}$ is filtered and equivalent to a small category. Thus

$$X = \varprojlim_{({}_F\backslash C)^{\mathrm{op}}} X$$

is representable in C, and since F commutes with the limit in question, it follows that X is an initial object of ${}_F\backslash C$. □

Corollary 8.12.8. With C satisfying the hypotheses of 8.12.7, let $f : C \to C'$ be a functor from C to a $\mathcal{U}$-category C'. For f to admit a left adjoint, it is necessary and sufficient that f commute with small projective limits.

This reduces trivially to 8.12.7 by applying that statement to the composite functors of the form

$$X \longmapsto \mathrm{Hom}(Y', f(X)).$$

Examples 8.12.9. As noted in 7.13, the hypotheses on C in 8.12.7 and 8.12.8 are satisfied when C is the category of $\mathcal{U}$-sheaves of sets on a topological space $X \in \mathcal{U}$, or more generally on a $\mathcal{U}$-site (II 3.0.2). We give an instructive example, due to H. Bass, showing that the hypothesis of existence of a small cogenerating subcategory D of C is not redundant in 8.12.7. Take C to be the category of groups belonging to $\mathcal{U}$. Let J be the set of isomorphism classes of simple groups in C. For each $j \in J$, choose a simple group G_j in the class j, and let I be the filtered ordered set of $\mathcal{U}$-small subsets of J. For $i \in I$, let

$$X_i = \amalg_{j \in i} G_j.$$

The X_i then form a projective system $(X_i)_{i \in I}$ in C, with epimorphic transition morphisms, and the corresponding functor

$$F(X) = \varinjlim_i \mathrm{Hom}(X_i, X) \qquad (*)$$

takes values in $\mathcal{U}$-sets, although the index set I plainly does not have cardinal in $\mathcal{U}$. More precisely, let us show that for every small full subcategory C_0 of C, there exists $i_0 \in I$ such that the restriction of F to C_0 is represented by X_{i_0}. This will prove both that F is $\mathcal{U}$-set-valued and that it commutes with small projective limits. To prove the assertion, it is enough to note that, for every $X \in \mathrm{Ob}\, C_0$, the cardinal of the set $J(X)$ of $j \in J$ for which there exists a nontrivial homomorphism from G_j to X is necessarily small, since such a morphism is necessarily a monomorphism, G_j being simple. Hence, if i_0 is the subset of J which is the union of the $J(X)$ for $X \in \mathrm{Ob}\, C_0$, then i_0 is small, that is, $i_0 \in I$, and it does the job. On the other hand, it is clear that F is not representable, since $\mathrm{card}\, I \notin \mathcal{U}$. From this and 8.12.7 one therefore concludes that *the category C of groups in $\mathcal{U}$ admits no small full cogenerating subcategory.* Since the object $\mathbb{Z}$ of C is nevertheless a generator, it follows from the proof of 7.12 that there exists a two-generator group G which does not embed into an injective object of the category C of groups in $\mathcal{U}$. It also seems plausible that C has no injective objects other than the unit groups.

8.13 Pro-representable functors and accessible functors

8.13.1

In this section we use some notions and results from the following paragraph, notably 9.11 and 9.13, in order to obtain a criterion of pro-representability which will be used incidentally in IV 9.16. In what follows, C denotes a $\mathcal{U}$-category satisfying condition L of 9.1(b). This condition is satisfied, for example, if small filtered inductive limits are representable in C.

Proposition 8.13.2. Let C be as above, and let

$$f : C \longrightarrow \mathbf{Set}$$

be a functor.

a) Suppose every object of C is accessible (9.3). If f is pro-representable, then f is accessible (9.2) and left exact.

b) Suppose finite projective limits are representable in C, and that C admits a cardinal filtration (9.12). If f is accessible and left exact, then f is pro-representable.

Proof.

a) The hypothesis on C means that the covariant representable functors from C to **Set** are accessible. Therefore the same is true of every small inductive limit of such functors (9.6(i)), and hence of every pro-representable functor.

b) By 8.3.3(iii), it remains to prove that $\mathrm{Ob}({}_F\backslash C)^{\mathrm{op}}$ contains a small cofinal subcategory. By hypothesis there exists a cardinal Π such that f is Π-accessible. Let (X, u), $u \in F(X)$, be an object of ${}_F\backslash C$. With the notation of 9.12(c), one then has $X = \varinjlim_i X_i$, with I filtered and large relative to Π, and with the X_i in $C' = \mathrm{Filt}^{\Pi}(C)$. Hence

$$F(X) \xleftarrow{\sim} \varinjlim_i F(X_i).$$

This shows that the small subcategory $({}_F\backslash C')^{\mathrm{op}}$ is cofinal in $({}_F\backslash C)^{\mathrm{op}}$, and completes the proof.

Corollary 8.13.3. Let C be a $\mathcal{U}$-category satisfying the following conditions:

a) finite projective limits are representable in C;

b) small filtered inductive limits are representable in C;

c) the functor Ker on the category of double arrows of C is accessible, for example it commutes with small filtered inductive limits;

d) every strict epimorphism of C is strictly universal (10.2);

e) there exists a small subcategory of C generating by strict epimorphisms (7.1).

Under these conditions, a functor $f : C \to \mathbf{Set}$ is pro-representable if and only if it is left exact and accessible (9.2).

Indeed, the hypotheses on C in 8.13.2(a) and (b) are satisfied by 9.11 and 9.13, respectively.

Corollary 8.13.4. Let $f : C \to C'$ be a functor between $\mathcal{U}$-categories satisfying conditions a) through e) of 8.13.3. For f to admit a pro-adjoint, it is necessary and sufficient that f be left exact and accessible.

Since the representable functors

$$h_{Y'} : X' \longmapsto \mathrm{Hom}(Y', X')\,, \qquad C' \to \mathbf{Set},$$

are left exact and accessible (9.11), if f has these same properties then so do their composites with f, and these composites are therefore pro-representable by 8.13.3; that is, f admits a pro-adjoint. Conversely, suppose that f admits a pro-adjoint, and let C'_1 be a small generating subcategory of C'. Choose a cardinal Π such that the objects $Y' \in \mathrm{Ob}\, C'_1$ are Π-accessible. To prove that f is Π-accessible, it is enough to prove this for its composites with the $h_{Y'}$, $Y' \in \mathrm{Ob}\, C'_1$. But by hypothesis these composites are pro-representable, hence accessible by 8.13.3, and therefore Π-accessible after increasing Π if necessary. $\square$

Continuation anchor. The next section is Section 9, “Accessible functors, cardinal filtrations, and construction of small generating subcategories.”

Exposé I. Presheaves

Section 9, beginning through Lemma 9.13.1

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 9 from the start through Lemma 9.13.1. The notation follows the previous batches: $\mathcal{U}$ denotes the fixed universe; $\mathcal{U}$-**Set** is written as $\mathcal{U}$-**Set**; and $\operatorname{card} \operatorname{Fl} C$ denotes the cardinal of the set of arrows of a category C.

9 Accessible functors, cardinal filtrations, and construction of small generating subcategories

The present section is more technical than the other sections of this exposé. In this seminar it will be used only in IV 9 and VI 4, which themselves are not used elsewhere in the Seminar. It is therefore advisable to omit the present section on a first reading.

9.0

All categories considered in this number are assumed to be $\mathcal{U}$-categories. Except for the small indexing categories $I, J, \ldots$ that we shall use, the developments below will mainly concern "large" categories $E, F, \ldots$ that are stable under small filtered inductive limits. Most often, however, it will be enough that a slightly weaker condition hold, namely condition L of Definition 9.1 below. All cardinals considered in this number are assumed to belong to $\mathcal{U}$.

Following a suggestion of P. Deligne, we shall study, for a functor $f : E \to F$ between large categories, a condition expressing commutation of f with certain types of filtered inductive limits. This condition is remarkably stable and is satisfied by the most important functors that occur naturally. The applications we have in view for this seminar are 9.13.3 and 9.13.4, used in VI 4, and above all 9.25, which gives, in a non-trivial case, the existence of a small generating family in a category of sections of a fibred category; that result will be used in IV 9.16.

Definition 9.1.

a) Let I be a preordered set and let π be a cardinal. We say that I is *large with respect to* π if I is filtered and if every subset of I of cardinal $\leq \pi$ has an upper bound in I.

b) Let E be a category. If π is a cardinal, we say that E satisfies condition L_π if, for every small ordered set I large with respect to π, E is stable under inductive limits of type I. We say that E satisfies condition L if there exists a cardinal $\pi \in \mathcal{U}$ such that E satisfies L_π.

9.1.1

When, in Definition 9.1(a), one has $\pi \geq 2$, the second stated condition already implies that I is filtered. If π is finite, saying that I is large with respect to π simply means that I is filtered. In what follows we shall be interested almost exclusively in the case where π is infinite. Notice that if π and π' are cardinals with $\pi' \geq \pi$, then I large with respect to π' plainly implies I large with respect to π.

9.1.2

As announced in 9.0, the conditions L_π and L should be regarded as technical variants of the stronger condition of stability under small filtered inductive limits. It is clear that if π and π' are cardinals with $\pi' \geq \pi$, then condition L_π implies condition $L_{\pi'}$.

Definition 9.2. Let $f : E \to F$ be a functor. If Π is a cardinal, we say that f is Π-*preaccessible* (respectively Π-*accessible*) if E satisfies L_Π (9.1) and if, for every ordered set $I \in \mathcal{U}$ large with respect to Π, and every inductive system $(X_i)_{i\in I}$ in E of type I, the canonical morphism

$$\varinjlim f(X_i) \longrightarrow f(\varinjlim X_i)$$

is a monomorphism (respectively an isomorphism). We say that f is *preaccessible* (respectively *accessible*), relative to the universe $\mathcal{U}$, if there exists a cardinal $\Pi \in \mathcal{U}$ such that f is Π-preaccessible (respectively Π-accessible).

The category of Π-accessible (respectively accessible) functors from E to F will be denoted by $\mathrm{Hom}(E, F)_\Pi$ (respectively $\mathrm{Hom}(E, F)_{\mathrm{acc}}$).

9.2.1

Plainly, a functor which commutes with small filtered inductive limits, for example a functor admitting a right adjoint, is Π-accessible for every cardinal $\Pi \geq 2$.

Definition 9.3. Let E be a category, let X be an object of E, and let

$$h^\circ_X : E \longrightarrow (\mathcal{U}\text{-}\mathbf{Set})$$

be the covariant functor it represents. Let $\Pi \in \mathcal{U}$ be a cardinal. We say that X is a Π-preaccessible (respectively Π-accessible) object of E if the functor h°_X is Π-preaccessible (respectively Π-accessible). We say that X is preaccessible (respectively accessible) if there exists a cardinal $\Pi \in \mathcal{U}$ such that X is a Π-preaccessible (respectively Π-accessible) object of E.

9.3.1

We denote by E_Π the strictly full subcategory of E consisting of the Π-accessible objects of E.

When Definition 9.3 is applied to a category of the form $\mathrm{Hom}(C, F)$, the terminology just introduced is, *a priori*, ambiguous with the analogous terminology introduced in 9.2, when one interprets the objects of $\mathrm{Hom}(C, F)$ as functors. There does not, however, seem to be any serious risk of confusion.

Definition 9.4. Let E be a category and let $\pi \in \mathcal{U}$ be a cardinal. We say that E is a π-preaccessible (respectively π-accessible) category if there exists in E a small full subcategory C which is generating (7.1) and whose objects are π-preaccessible (respectively π-accessible) (9.3). We say that E is preaccessible (respectively accessible) if there exists a cardinal $\pi \in \mathcal{U}$ such that E is π-preaccessible (respectively π-accessible).

For important examples, see 9.11.3 below.

Proposition 9.5. Let $f : E \to F$ be a functor between categories such that F is accessible and every object of E is accessible. If f admits a left adjoint, then f is accessible.

Indeed, by hypothesis, F admits a small conservative family of representable functors $F \to (\mathcal{U}\text{-}\mathbf{Set})$ which are accessible. Thus one is immediately reduced to showing that the composite of f with each of these functors is accessible. Since f admits a left adjoint, these composites are representable functors $E \to (\mathcal{U}\text{-}\mathbf{Set})$, hence are accessible by the hypothesis on E.

Proposition 9.6. Let E and F be two categories, let $\Pi \in \mathcal{U}$ be a cardinal, and consider the full subcategory $\mathrm{Hom}(E,F)_\Pi$ of $\mathrm{Hom}(E,F)$ formed by the Π-accessible functors (9.2) from E to F.

(i) This subcategory is stable under every type of inductive limit which is representable in F.

(ii) Suppose that F is Π-accessible (9.4), and let J be a category such that $\operatorname{card}\operatorname{Fl} J \leq \Pi$ and such that projective limits of type J are representable in F; then projective limits of type J are representable in $\mathrm{Hom}(E,F)$. The subcategory $\mathrm{Hom}(E,F)_\Pi$ is stable under projective limits of type J.

Corollary 9.7. Let E and F be two categories, with F accessible (9.4). Then the full subcategory $\mathrm{Hom}(E,F)_{\mathrm{acc}}$ of $\mathrm{Hom}(E,F)$ formed by accessible functors is stable under every inductive or projective limit, relative to a small indexing category J, which is representable in F and hence in $\mathrm{Hom}(E,F)$.

Proof. [Proof of Proposition 9.6] Assertion (i) follows trivially from the commutation of the functor $\varinjlim$ with arbitrary inductive limits. For (ii), let $(f_j)_{j\in J}$ be a projective system of Π-accessible functors $E \to F$, and let

$$f = \varprojlim_j f_j$$

be its projective limit, computed argument by argument. We prove that f is Π-accessible; in other words, for every ordered set $I \in \mathcal{U}$ large with respect to Π, and every inductive system $(X_i)_{i\in I}$ in E, the canonical morphism

$$\varinjlim_i((\varprojlim_j f_j)(X_i)) \longrightarrow (\varprojlim_j f_j)(\varinjlim_i X_i)$$

is an isomorphism. The argument-by-argument computation of the functor $\varprojlim_j f_j$ identifies this canonical morphism with

$$\varinjlim_i \varprojlim_j f_j(X_i) \longrightarrow \varprojlim_j \varinjlim_i f_j(X_i),$$

the canonical morphism attached to the bifunctor

$$(i,j) \longmapsto f_j(X_i) : I \times J \longrightarrow F.$$

Thus Proposition 9.6 follows from the following more general assertion.

Corollary 9.8. Let F be a Π-accessible category, let I be an ordered set large with respect to Π, let J be a small category such that $\operatorname{card}\operatorname{Fl} J \leq \Pi$ and such that projective limits of type J are representable in F, and let $h : I \times J \to F$ be a functor. Then the canonical morphism

$$\varinjlim_i \varprojlim_j h(i,j) \longrightarrow \varprojlim_j \varinjlim_i h(i,j) \tag{9.8.1}$$

is an isomorphism. In other words, the functor

$$\varinjlim_{i} : \mathrm{Hom}(I, F) \longrightarrow F \tag{9.8.2}$$

commutes with projective limits of type J, for every small category J such that $\operatorname{card} \mathrm{Fl}\, J \leq \Pi$ and such that projective limits of type J are representable in F. Equivalently, for every J as above, the functor

$$\varinjlim_{j} : \mathrm{Hom}(J, F) \longrightarrow F \tag{9.8.3}$$

is Π-accessible.

Since, by hypothesis, F admits a conservative family of covariant representable Π-accessible functors $g : F \to (\mathcal{U}\text{-}\mathbf{Set})$, one is immediately reduced to proving 9.8 in the case $F = \mathcal{U}\text{-}\mathbf{Set}$. We shall then prove the assertion in the form that (9.8.2) commutes with projective limits of type J, with $\operatorname{card} \mathrm{Fl}\, J \leq \Pi$. Since I is filtered, (9.8.2) is known to commute with finite projective limits (2.8). A standard argument (cf. 2.3) therefore reduces us to proving that it commutes with products indexed by a set J of cardinal $\leq \Pi$. This reduces to proving the bijectivity of (9.8.1) when J is discrete.

For injectivity, consider two elements a, b of the left-hand side. They come from $\prod_j h(i_0, j)$ for a suitable $i_0 \in I$, say from elements $\prod_j a(i_0, j)$ and $\prod_j b(i_0, j)$. Suppose that the elements $\prod_j h(\infty, j)$ of the right-hand side which they define are equal; that is, for every j there exists $i(j) \geq i_0$ such that $a(i_0, j)$ and $b(i_0, j)$ have the same image in $h(i, j)$. Since I is large with respect to $\operatorname{card} J$, there exists a common upper bound $i_1 \in I$ for all the $i(j)$. It follows that the two elements under consideration in $\prod_j h(i_0, j)$ have the same image in $\prod_j h(i_1, j)$, and therefore define the same element of the left-hand side of (9.8.1). This proves injectivity.

For surjectivity, let $\prod_j a(\infty, j)$ be an element of the right-hand side. For each $j \in J$, $a(\infty, j)$ comes from an element of $h(i(j), j)$ for a suitable $i(j) \in I$. As before, one can find a common upper bound $i \in I$ for all the $i(j)$. Then $\prod_j a(\infty, j)$ comes from an element $\prod_j a(i, j)$ of $\prod_j h(i, j)$, hence lies in the image of (9.8.1). This completes the proof.

Corollary 9.9. Let F be a Π-accessible (respectively accessible) category. Then, for every category J such that $\operatorname{card} \mathrm{Fl}\, J \leq \Pi$ (respectively every small category J), and for every functor $(X_j)_{j \in J} : J \to F$ whose inductive limit in F is representable, if the X_j are Π-accessible (respectively accessible), then so is $\varprojlim X_j$.

Indeed, the functor $F \to (\mathcal{U}\text{-}\mathbf{Set})$ represented by $\varinjlim X_j$ is the projective limit of the functors represented by the X_j, and one applies Proposition 9.6(ii) to the resulting projective system of functors.

Remark 9.10. In the statements 9.6, 9.7, 9.8, and 9.9 one may everywhere replace the words "Π-accessible", "accessible" by "Π-preaccessible", "preaccessible". The proof just given proves this variant of the preceding statements as well.

Proposition 9.11. Let E be a category satisfying condition L (9.1), and let C be a small full generating subcategory (7.1). Suppose that the following condition is satisfied:

a) E is stable under kernels of double arrows, and the functor Ker on the category of double arrows of E is accessible; for example, it commutes with small filtered inductive limits.

Then every object of E is preaccessible (9.3), and *a fortiori* E is preaccessible. Suppose moreover that C is generating by strict epimorphisms (7.1), and suppose that E satisfies the following conditions:

b) E is stable under fibre products.

c) Every small strict epimorphic family in E is universally strict epimorphic (10.3); or else small coproducts are representable in E and every strict epimorphism in E is a universal strict epimorphism.

Then every object of E is accessible, and *a fortiori* E is accessible.

The fact that, under (a), every object of E is preaccessible follows from the fact that, for every object X of E, the set of strict subobjects of X is small (7.4), and from the following lemma.

Lemma 9.11.1. Under the hypotheses of 9.11(a), let $X \in \operatorname{ob} E$ and let Π be a cardinal such that E satisfies L_Π (9.1), the functor Ker in 9.11(a) is Π-accessible, and the set of strict subobjects of X has cardinal bounded by Π. Then X is Π-preaccessible.

Indeed, let I be a set large with respect to Π, let $(Y_i)_{i\in I}$ be an inductive system in E with inductive limit Y, and let

$$(u_i, v_i : X \rightrightarrows Y_i)$$

be a double arrow such that the composite double arrow $u, v : X \rightrightarrows Y$ satisfies $u = v$. We prove that there exists $j \geq i$ in I such that $u_j = v_j$. For this, consider the inductive system of double arrows $(u_j, v_j : X \rightrightarrows Y_j)_{j\geq i}$, whose inductive limit is (u, v). By hypothesis,

$$X = \operatorname{Ker}(u, v) = \varinjlim_j \operatorname{Ker}(u_j, v_j) = \varinjlim_j X_j, \qquad X_j = \operatorname{Ker}(u_j, v_j).$$

The X_j are strict subobjects of X, so there exists a subset I' of I, consisting of indices $j \geq i$, with $\operatorname{card} I' \leq \Pi$, such that every X_j is equal to one of the $X_{i'}$ for $i' \in I'$. Since I is large with respect to Π, there is an upper bound j of I' in I. Then X_j contains all the $X_{j'}$ for $j' \geq i$, hence $X_j \to X$ is an epimorphism, since the family of maps $X_{j'} \to X$ is epimorphic. It is therefore an isomorphism, since it is a strict monomorphism. Thus $u_j = v_j$, proving 9.11.1.

The second assertion of 9.11 follows from the first and from the following lemma.

Lemma 9.11.2. Under the hypotheses of 9.11(a), (b), and (c), let X be an object of E, and let Π be an infinite cardinal such that E satisfies L_Π (9.1), such that $\operatorname{card} \operatorname{ob} C_{/X} \leq \Pi$, such that for two objects $X' \to X$ and $X'' \to X$ of $C_{/X}$ the fibre product $X' \times_X X''$ is Π-preaccessible, and finally such that X is Π-preaccessible. Then X is Π-accessible.

With the notation of the proof of 9.11.1, it remains only to prove that every morphism $u : X \to Y$ comes from a morphism $u_i : X \to Y_i$. Consider the family of morphisms $Y_i \to Y$, which is a strict epimorphic family. By (b) and (c), the family

$$Y_i \times_Y X \longrightarrow X$$

is also strictly epimorphic. On the other hand, for every i, since C is generating by strict epimorphisms, the family of maps

$$T \longrightarrow Y_i \times_Y X$$

with source in C is strictly epimorphic. In the second alternative in (c), one can find such a strict epimorphism with source a small coproduct of objects of C. By transitivity of universally strict epimorphic families (II 2.5), it follows that the family of maps $T_\alpha \xrightarrow{f_\alpha} X$ with source in C which factor through one of the $Y_i \times_Y X$ is strictly epimorphic. Let J be the set of indices of this family; its cardinal is bounded by $\operatorname{card} \operatorname{ob} C_{/X} \leq \Pi$.

For each $\alpha \in J$, choose an $i = i(\alpha) \in I$ and an X-morphism

$$T_\alpha \longrightarrow Y_i \times_Y X,$$

or equivalently a morphism $v_\alpha : T_\alpha \to Y_i$ such that $p_i v_\alpha = u f_\alpha$, where $p_i : Y_i \to Y$ is the canonical morphism. Since I is large with respect to Π, one may choose $i(\alpha)$ independently of α; call it i. For every pair of indices $\alpha, \beta \in J$, consider the two composites

$$\mathrm{pr}_1 \, v_\alpha, \mathrm{pr}_2 \, v_\beta : T_\alpha \times_X T_\beta \rightrightarrows Y_i.$$

Their composites with $Y_i \to Y$ are equal. Since $T_\alpha \times_X T_\beta$ is Π-preaccessible by hypothesis, there exists an index $i' = i(\alpha, \beta) \geq i$ such that the composites of the arrows under consideration with $Y_i \to Y_{i'}$ are equal. The set of pairs (α, β) has cardinal $\leq \Pi^2 = \Pi$, because Π is infinite; hence one may again choose i' independently of α and β. Plainly we may suppose $i' = i$. But then, since the family $(f_\alpha : T_\alpha \to X)$ is strictly epimorphic, one can find a morphism $u_i : X \to Y_i$ such that $u_i f_\alpha = v_\alpha$. We then have $p_i u_i = u$, because for every α,

$$(p_i u_i) f_\alpha = p_i (u_i f_\alpha) = p_i v_\alpha = u f_\alpha,$$

and the family of the f_α is epimorphic. This completes the proof of 9.11.2.

Remark 9.11.3. As a special case of 9.11, one sees that every object of E is accessible in each of the following two cases. First, E is an abelian $\mathcal{U}$-category with exact filtered inductive limits and admitting a small generating subcategory; here it is the second alternative in (c) which applies. Second, E is the category of sheaves of sets on a topological space $X \in \mathcal{U}$. More generally, it is enough that E be the category of sheaves of sets on a $\mathcal{U}$-site (II 2.1), or that E be a $\mathcal{U}$-topos (IV 1.1).

Definition 9.12. Let E be a category. A *cardinal filtration* of E is an increasing filtration $(\mathrm{Filt}^\Pi(E))_{\Pi \geq \Pi_0}$ of E by strictly full subcategories $\mathrm{Filt}^\Pi(E)$, indexed by the cardinals $\Pi \in \mathcal{U}$ with $\Pi \geq \Pi_0$, where Π_0 is a fixed infinite cardinal depending on the cardinal filtration under consideration, and satisfying the following conditions:

a) For every $\Pi \geq \Pi_0$, $\mathrm{Filt}^\Pi(E)$ is equivalent to a small category.

b) E satisfies L_{Π_0} (9.3), and for every $\Pi \geq \Pi_0$, $\mathrm{Filt}^\Pi(E)$ is stable in E under filtered inductive limits indexed by ordered sets I which are large with respect to Π_0 and such that $\operatorname{card} I \leq \Pi$.

c) For every $\Pi \geq \Pi_0$ and every $X \in \operatorname{ob} E$, one can find an isomorphism

$$X \simeq \varinjlim_I X_i,$$

where $(X_i)_{i \in I}$ is an inductive system in E indexed by an ordered set I large with respect to Π (9.1), and where the X_i belong to $\mathrm{Filt}^\Pi(E)$. Moreover, if $X \in \operatorname{ob} \mathrm{Filt}^\Pi(E)$ and $\Pi' \geq \Pi$, one can take I with $\operatorname{card} I \leq (\Pi')^\Pi$.

9.12.1

It follows from (b) and (c) that every $X \in \operatorname{ob} E$ belongs to some $\operatorname{Filt}^{\Pi}(E)$ for Π sufficiently large.

Example 9.12.2. Take $E = (\mathcal{U}\textbf{-Set})$, $\Pi_0 = \aleph_0$, and let $\operatorname{Filt}^{\Pi}(E)$ be the full subcategory of E consisting of the sets X such that $\operatorname{card} X \leq \Pi$. More generally:

Proposition 9.13. Let E be a $\mathcal{U}$-category. Suppose that E is stable under small filtered inductive limits, under sums of two objects, and under cokernels of double arrows; that E is stable under fibre products; and that epimorphic morphisms in E are universal strict epimorphisms. Let C be a small full subcategory of E which is generating by strict epimorphisms. Let Π_0 be an infinite cardinal $\geq \operatorname{card} \operatorname{Fl} C$. For every cardinal $\Pi \geq \Pi_0$, let $\operatorname{Filt}^{\Pi}(E)$ be the strictly full subcategory of E consisting of the objects X of E for which there exists a strict epimorphic family $(X_i \to X)_{i \in I}$ with target X, such that $\operatorname{card} I \leq \Pi$ and $X_i \in \operatorname{ob} C$ for every $i \in I$. Then $(\operatorname{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}$ is a cardinal filtration of E. Moreover, for every $X \in \operatorname{ob} \operatorname{Filt}^{\Pi}(E)$, the cardinal of the set of arrows of $C_{/X}$, and *a fortiori* of the set of objects of $C_{/X}$, is bounded by Π^{Π_0}.

The last assertion is precisely 7.6.

To prove that this is a cardinal filtration, it remains to prove conditions (a), (b), and (c) of 9.12. Condition (a) follows at once from 7.5.2. For (b), suppose that

$$X = \varinjlim_{I} X_i,$$

with $\operatorname{card} I \leq \Pi$ and $X_i \in \operatorname{ob} \operatorname{Filt}^{\Pi}(E)$. The family of canonical morphisms $X_i \to X$ is therefore strictly epimorphic. By the hypothesis on the X_i, for each $i \in I$ there is a strict epimorphic family $(X_{ij} \to X_i)_{j \in J_i}$ with $\operatorname{card} J_i \leq \Pi$. Passing to the sums $\coprod_i X_i$ and $\coprod_{i,j} X_{ij}$, one sees, by transitivity of universal strict epimorphisms (II 2.5), that the family of composites

$$X_{ij} \longrightarrow X_i \longrightarrow X$$

indexed by the disjoint union of the J_i, for $i \in I$, is strictly epimorphic. Since $\operatorname{card} J \leq \Pi^2 = \Pi$, the conclusion follows. Notice that we have used only the existence of $\varinjlim_I X_i$, and not the fact that I is large with respect to Π_0, nor even that it is filtered.

It remains to prove (c). First note that for every $X \in \operatorname{ob} E$, the family of arrows $X_i \to X$ with source in $\operatorname{ob} C$ is strictly epimorphic, because C is generating by strict epimorphisms, and is plainly small. Thus there is a cardinal $\Pi \geq \Pi_0$ such that $X \in \operatorname{ob} \operatorname{Filt}^{\Pi}(E)$. It remains to prove that, if $\Pi' \geq \Pi \geq \Pi_0$ and $X \in \operatorname{ob} \operatorname{Filt}^{\Pi'}(E)$, then there is an isomorphism

$$X \simeq \varinjlim_{I} X_i,$$

where I is large with respect to Π, $\operatorname{card} I \leq (\Pi')^{\Pi}$, and the X_i lie in $\operatorname{ob} \operatorname{Filt}^{\Pi}(E)$.

Since C is generating by strict epimorphisms, one has

$$X \simeq \varinjlim_{C_{/X}} T. \tag{$*$}$$

Furthermore, by successively adjoining to C sums and cokernels of double arrows in C, and then doing the same for the category so obtained, and so on, we are reduced to the case where C

is stable under sums of two objects in E and under cokernels of double arrows in E, without destroying the hypothesis $\Pi_0 \geq \operatorname{card} \operatorname{Fl} C$. We may also suppose that, if E contains a fixed initial object $\varnothing_E$, then $\varnothing_E \in \operatorname{ob} C$. This implies that the category $C_{/X}$ is stable under sums of two objects and under cokernels of double arrows. It is then filtered: it is non-empty, since otherwise the relation $(*)$ would show that X is an initial object, so that $C_{/X}$ would contain the arrow $\varnothing_E \to X$, a contradiction.

Let I be the set, ordered by inclusion, of full subcategories i of $C_{/X}$ which are filtered and such that $\operatorname{card} \operatorname{ob} i \leq \Pi$. For every $i \in I$, let $X_i \in \operatorname{ob} E$ be the inductive limit of the composite functor

$$i \longrightarrow C_{/X} \longrightarrow E,$$

which exists by hypothesis. Then one plainly has

$$\varinjlim_{I} X_i \simeq \varinjlim_{C_{/X}} T \simeq X.$$

On the other hand, we have already noted that

$$\operatorname{card} \operatorname{ob} C_{/X} \leq (\Pi')^{\Pi_0}.$$

It follows that I is large with respect to Π and that

$$\operatorname{card} I \leq ((\Pi')^{\Pi_0})^{\Pi} = (\Pi')^{\Pi\Pi_0} = (\Pi')^{\Pi}$$

by the following lemma, whose proof is left to the reader; in applying it one takes $J = C_{/X}$ and $c = (\Pi')^{\Pi_0}$.

Lemma 9.13.1. Let J be a filtered category, and let c and Π be two cardinals such that

$$c \geq \operatorname{Sup}(\operatorname{card} \operatorname{Fl} J, \Pi)$$

and such that $\operatorname{card} \operatorname{Hom}(j, j') \leq \Pi_0$ for every pair of objects j, j' of J. Let I be the set of full filtered subcategories i of J such that $\operatorname{card} \operatorname{ob} i \leq \Pi$. Then, ordered by inclusion, I is large with respect to Π, and one has $\operatorname{card} I \leq c^{\Pi}$.

This completes the proof of Proposition 9.13.

Exposé I. Presheaves

Section 9, Remark 9.13.2 through Corollary 9.26

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 9 from Remark 9.13.2 through Corollary 9.26. The terminology follows the previous batches: "large with respect to Π" translates *grand devant* Π; "accessible" is used in the sense of 9.2–9.3; and $\operatorname{card} \mathrm{Fl}\, C$ denotes the cardinal of the set of arrows of a category C. A few evident typographical slips in the source have been corrected where the formulas force the correction, notably in some subscripts in 9.14, 9.21, and 9.24.

9 Accessible functors, cardinal filtrations, and construction of small generating subcategories

9.13.2

Remark 9.13.2. A slight additional effort should make it possible, in Proposition 9.13, to replace the hypothesis that E is stable under small filtered inductive limits by the hypothesis that E satisfies L (9.1), provided that E is stable under small sums, or that in E every epimorphic *family* is universally epimorphic. In the proof of c), one must then choose Π_0 so that E satisfies L_{Π_0}, and restrict to those full subcategories i of $C_{/X}$ which are not only filtered, but whose preordered set $\operatorname{ob} i$ is large with respect to Π_0. Using Section 8 and the hypothesis that E satisfies L_{Π_0}, one then finds that the X_i exist; one should conclude by means of a suitable variant of Lemma 9.13.1, which the editor has not checked.

Proposition 9.14. Let $f : E \to F$ be an accessible functor (9.2) between two categories endowed with cardinal filtrations

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0} \qquad \text{and} \qquad (\mathrm{Filt}^{\Pi}(F))_{\Pi \geq \Pi'_0}.$$

Then there exists a cardinal $\Pi_1 \geq \operatorname{Sup}(\Pi_0, \Pi'_0)$ such that, for every cardinal $\Pi \geq \Pi_1$, one has

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi^{\Pi_1}}(F). \tag{9.14.1}$$

In particular, if $\Pi = 2^c$ with $c \geq \Pi_1$, whence

$$\Pi^{\Pi_1} = 2^{c\Pi_1} = 2^c = \Pi,$$

then

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi}(F). \tag{9.14.2}$$

Let us immediately record the following consequence.

Corollary 9.15. Let E be a category endowed with two cardinal filtrations

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0} \qquad \text{and} \qquad (\mathrm{Filt}'^{\Pi}(E))_{\Pi \geq \Pi'_0}.$$

Then there exists a cardinal $\Pi_1 \geq \operatorname{Sup}(\Pi_0, \Pi'_0)$ such that, for every cardinal $c \geq \Pi_1$, if $\Pi = 2^c$, then

$$\operatorname{Filt}^{\Pi}(E) = \operatorname{Filt'}^{\Pi}(E).$$

Proof. *Proof of Proposition 9.14.* Let c be a cardinal such that

$$c \geq \operatorname{Sup}(\Pi_0, \Pi'_0)$$

and such that f is c-admissible. Let also $\Pi_1 \geq c$ be such that

$$f(\operatorname{Filt}^c(E)) \subset \operatorname{Filt}^{\Pi_1}(F). \tag{$*$}$$

Such a Π_1 exists because $\operatorname{Filt}^c(E)$ is equivalent to a small category (9.12 a)); hence $f(\operatorname{Filt}^c(E))$ is also essentially small, and one may apply 9.12.1 to the objects of the latter category to find one $\operatorname{Filt}^{\Pi}(F)$ containing all of them, taking into account that the $\operatorname{Filt}^{\Pi}(F)$ are full subcategories.

Let $\Pi \geq \Pi_1$, and let $X \in \operatorname{ob}\operatorname{Filt}^{\Pi}(E)$. We prove that

$$f(X) \in \operatorname{Filt}^{\Pi^{\Pi_1}}(F).$$

Write

$$X = \varinjlim_{I} X_i,$$

where $X_i \in \operatorname{ob}\operatorname{Filt}^c(E)$, where I is large with respect to c, and where

$$\operatorname{card} I \leq \Pi^c \leq \Pi^{\Pi_1}$$

(9.12 c)). Since f is c-admissible and I is large with respect to c, one has

$$f(X) \simeq \varinjlim_{I} f(X_i).$$

Moreover $\operatorname{card} I \leq \Pi^{\Pi_1}$, and $f(X_i) \in \operatorname{ob}\operatorname{Filt}^{\Pi_1}(F) \subset \operatorname{ob}\operatorname{Filt}^{\Pi^{\Pi_1}}(F)$ by $(*)$. Therefore $f(X) \in \operatorname{Filt}^{\Pi^{\Pi_1}}(F)$ by 9.12 b). $\square$

9.15.1

The notion of a cardinal filtration (9.12) is of interest only when the objects of E are accessible. Let us point out that Proposition 9.11 implies that this condition is satisfied when, in addition to the hypotheses of Proposition 9.13, one assumes that the functor Ker on the category of double arrows of E is accessible. We also record the following.

Proposition 9.16. Let E be a category endowed with a cardinal filtration

$$(\operatorname{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}.$$

Suppose that the elements of $\operatorname{Filt}^{\Pi_0}(E)$ are accessible objects of E. Then there exists a cardinal $\Pi_1 \geq \Pi_0$ in $\mathcal{U}$ such that the objects of $\operatorname{Filt}^{\Pi_0}(E)$ are Π_1-accessible. If Π_1 is chosen in this way, then for every cardinal $\Pi \geq \Pi_1$ one has, with the notation of 9.3.1,

$$E_{\Pi} \subset \operatorname{Filt}^{\Pi}(E) \subset E_{\Pi^{\Pi_0}}. \tag{9.16.1}$$

The existence of Π_1 follows immediately from the fact that $\mathrm{Filt}^{\Pi_0}(E)$ is equivalent to a small category (9.12 a)). Let $\Pi \geq \Pi_1$. If X belongs to $\mathrm{Filt}^{\Pi}(E)$, write

$$X \simeq \varinjlim_I X_i$$

with $\operatorname{card} I \leq \Pi^{\Pi_0}$ and $X_i \in \operatorname{ob}\mathrm{Filt}^{\Pi_0}(E)$ for every i (9.12 c)). It then follows from Corollary 9.9 that X is Π^{Π_0}-accessible; this gives the second inclusion in (9.16.1).

Conversely suppose that X is Π-accessible, and write

$$X \simeq \varinjlim_I X_i,$$

where I is large with respect to Π and the X_i lie in $\operatorname{ob}\mathrm{Filt}^{\Pi}(E)$ (9.12 c)). By the hypothesis on X, the given isomorphism $X \xrightarrow{\sim} \varinjlim_I X_i$ factors through one of the X_i; hence X is isomorphic to a direct factor of this X_i. We conclude that $X \in \operatorname{ob}\mathrm{Filt}^{\Pi}(E)$, and hence obtain the first inclusion in (9.16.1), by the following lemma.

Lemma 9.16.2. Every object of E which is a direct factor of an object of $\mathrm{Filt}^{\Pi}(E)$ belongs to $\mathrm{Filt}^{\Pi}(E)$.

Indeed, if X is the image of an idempotent p on an object Y of E (i.e.an endomorphism p such that $p^2 = p$), and if I is a filtered ordered set, then X is the inductive limit of the filtered inductive system $(Y_i)_{i\in I}$ defined by $Y_i = Y$ for every i, and with transition map $p : Y_i \to Y_j$ whenever $i < j$. Taking I large with respect to Π_0 and with $\operatorname{card} I \leq \Pi$, we see that if $Y \in \mathrm{Filt}^{\Pi}(E)$, then the same is true of X by 9.12 b).

Corollary 9.17. Under the hypotheses of Proposition 9.16, for every cardinal $c \geq \Pi_1$, if $\Pi = 2^c$, then $\mathrm{Filt}^{\Pi}(E)$ is identical with the strictly full subcategory E_Π of E formed by the Π-accessible objects.

Corollary 9.18. Let E be a category satisfying condition L_Π (9.1), where Π is a cardinal, and let C be a full subcategory of E. Denote by $\mathrm{Ind}(C)_\Pi$ the full subcategory of the category $\mathrm{Ind}(C)$ of ind-objects of C (8.2) formed by ind-objects of the form $(X_i)_{i\in I}$, where I is an ordered set large with respect to Π. Consider the canonical functor

$$(X_i)_{i\in I} \longmapsto \varinjlim_I X_i : \mathrm{Ind}(C)_\Pi \longrightarrow E. \tag{9.18.1}$$

a) This functor is fully faithful if and only if every object X of C is a Π-accessible object (9.3) of E.

b) Assume the hypotheses of Corollary 9.17; in particular $\Pi = 2^c$, and take $C = \mathrm{Filt}^{\Pi}(E)$. Then the functor (9.18.1) is an equivalence of categories.

Assertion a) is an immediate generalization of 8.7.5 a), and is proved in the same way. Assertion b) follows from Corollary 9.17 and from condition 9.12 c) in the definition of cardinal filtrations, which implies that the functor under consideration is essentially surjective.

Corollary 9.19. Under the hypotheses of Corollary 9.17, let $C = \mathrm{Filt}^{\Pi}(E)$, let F be a $\mathcal{U}$-category, and consider the functor

$$\mathrm{Hom}(E, F)_\Pi \longrightarrow \mathrm{Hom}(C, F) \tag{9.19.1}$$

induced by "restriction to C", $f \mapsto f|_C$, where the source of (9.19.1) is the category of Π-accessible functors from E to F (9.2). This functor is fully faithful. If F satisfies condition L_Π (9.1), then the functor (9.19.1) is an equivalence of categories.

The first assertion is proved as in Proposition 7.8, using Corollary 9.18 b). For the second, one constructs a quasi-inverse to (9.19.1) by associating to a functor $f : C \to F$ the functor

$$(X_i)_{i \in I} \longmapsto \varinjlim_i f(X_i)$$

from $\mathrm{Ind}(E_\Pi)$ to F, and then using Corollary 9.18 b) to obtain a functor $\bar{f} : E \to F$. It remains only to show that this last functor is Π-accessible. This is proved as in the analogous assertion 8.7.3.

Corollary 9.20. Let E be a category admitting a cardinal filtration and such that every object of E is accessible (cf.Proposition 9.16), and let F be a category. Then:

a) The category $\mathrm{Hom}(E, F)_{\mathrm{acc}}$ of accessible functors from E to F (9.2) is a $\mathcal{U}$-category. (N.B. the given categories E, F are assumed to be $\mathcal{U}$-categories; see 9.0.)

b) Suppose that F is stable under small filtered inductive limits. For every full subcategory C of E equivalent to a small category, with inclusion functor $i : C \to E$, consider the corresponding functor
$$i_! : \mathrm{Hom}(C, F) \longrightarrow \mathrm{Hom}(E, F)$$
(5.1). A functor $f : E \to F$ is accessible if and only if there exists a small subcategory C of E such that f lies in the essential image of the preceding functor $i_!$.

Proof.

a) It is enough to prove that, for every cardinal Π such that E satisfies L_Π, the full subcategory $\mathrm{Hom}(E, F)_\Pi$ of $\mathrm{Hom}(E, F)_{\mathrm{acc}}$ is a $\mathcal{U}$-category. It plainly suffices to check this for cardinals of the form 2^c, with c sufficiently large. But then this follows from Corollary 9.19, since $C = \mathrm{Filt}^\Pi(E)$ is essentially small and therefore $\mathrm{Hom}(C, F)$ is plainly a $\mathcal{U}$-category.

b) By transitivity in the formation of the functors $i_!$, it suffices in the statement to restrict to subcategories C of the form $\mathrm{Filt}^\Pi(E)$, where Π is as in Corollary 9.17. One then sees easily that the composite functor

$$\mathrm{Hom}(\mathrm{Filt}^\Pi(E), F) \longrightarrow \mathrm{Hom}(E, F)_\Pi \longrightarrow \mathrm{Hom}(E, F),$$

where the first arrow is quasi-inverse to (9.19.1) and the second is the inclusion, is none other than the functor $i_!$, up to isomorphism. Thus b) follows from Corollary 9.19.

Exercise 9.20.1. This exercise uses the notions of site and topos, developed in Exposés II and IV. Let E be a $\mathcal{U}$-topos, let $\mathcal{V}$ be a universe such that $\mathcal{U} \in \mathcal{V}$, let C be a small full generating subcategory of the $\mathcal{U}$-topos E, and let $\Pi_0 \in \mathcal{U}$ be an infinite cardinal such that $\Pi_0 \geq \operatorname{card} \mathrm{Fl}\, C$. For every cardinal $\Pi \geq \Pi_0$, $\Pi \in \mathcal{U}$, let $\mathrm{Filt}^\Pi(E)$ be the strictly full subcategory of E formed by objects X for which there exists a strict epimorphic family $(X_i \to X)_{i \in I}$ with target X, such that $\operatorname{card} I \leq \Pi$ and $X_i \in \mathrm{Ob}\, C$ for every $i \in I$.

a) Show that $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}$ is a cardinal filtration of the $\mathcal{U}$-category E, and that one may choose Π_0 so that, for every $\Pi \geq \Pi_0$, one has inclusions

$$E_{\Pi} \subset \mathrm{Filt}^{\Pi}(E) \subset E_{\Pi^{\Pi_0}},$$

where for every cardinal c, E_c denotes the strictly full subcategory of E formed by the c-accessible objects.

b) Choose a cardinal $\Pi \geq \Pi_0$ such that $\Pi = \Pi^{\Pi_0}$, for example a cardinal of the form 2^c with $c > \Pi_0$, and put $C = \mathrm{Filt}^{\Pi}(E)$. Show that one has an equivalence of categories

$$\mathrm{Ind}(C)_{\Pi} \xrightarrow{\sim} E$$

(with the notation $\mathrm{Ind}(C)_{\Pi}$ of Corollary 9.18).

c) Keep the notation of b), let $\mathcal{V}$ be a universe such that $\mathcal{U} \subset \mathcal{V}$, and let $\widetilde{E}_{\mathcal{V}}$ be the $\mathcal{V}$-topos of $\mathcal{V}$-sheaves on the site E endowed with its canonical topology. Denote by $\mathrm{Ind}(C, \mathcal{V})_{\Pi}$ the category of $\mathcal{V}$-Ind-objects of C indexed by preordered sets of indices which are *large with respect to* Π. Show that one has an equivalence of categories

$$\mathrm{Ind}(C, \mathcal{V})_{\Pi} \xrightarrow{\sim} \widetilde{E}_{\mathcal{V}}.$$

d) Let $c \in \mathcal{V}$ be a cardinal, and let $\mathrm{Ind}(E, \mathcal{V})'_c$ be the full subcategory of $\mathrm{Ind}(E, \mathcal{V})$ formed by $\mathcal{V}$-ind-objects of E indexed by a preordered set which is large with respect to every cardinal $< c$. Taking $c = \mathrm{card}\, \mathcal{U}$, show that one has an equivalence of categories

$$\mathrm{Ind}(E, \mathcal{V})'_c \xrightarrow{\sim} \widetilde{E}_{\mathcal{V}}.$$

9.21

This section develops the technical preliminaries for the proof of Theorem 9.22 below, which is the main result of Section 9. Let

$$p : E \longrightarrow B \tag{9.21.1}$$

be a fibred functor, where B is a small category, and where the inverse image functors

$$f^* : E_{\beta} \longrightarrow E_{\alpha}$$

associated with arrows $f : \alpha \to \beta$ of B are accessible (9.2). In particular, the fibre categories E_{α} ($\alpha \in \mathrm{ob}\, B$) satisfy condition L (9.1). Since B is small, there is a cardinal $\Pi'_0 \in \mathcal{U}$ such that all the categories E_{α} satisfy $L_{\Pi'_0}$. Let $\Pi_0 \in \mathcal{U}$ be an infinite cardinal $> \Pi'_0$, so that

$$\Pi_0 > \Pi'_0, \qquad E_{\alpha} \text{ satisfies } L_{\Pi'_0} \text{ for all } \alpha \in \mathrm{ob}\, B. \tag{9.21.2}$$

Moreover, the hypotheses just made immediately imply the existence of a cardinal $c \in \mathcal{U}$ satisfying:

$$\begin{cases} \text{a)} & c \text{ is infinite,} \\ \text{b)} & c \geq \mathrm{card}\, \mathrm{Fl}\, B, \\ \text{c)} & \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_{\beta} \to E_{\alpha} \text{ is } c\text{-accessible.} \end{cases} \tag{9.21.3}$$

Suppose also that for each $\alpha \in \operatorname{ob} B$ one can find a full subcategory C_α of E_α satisfying:

$$\begin{cases} \text{a)} & \text{For every } X \in \operatorname{ob} C_\alpha,\ X \text{ is } c\text{-accessible in } E_\alpha. \\ \text{b)} & \text{For every } X \in \operatorname{ob} E_\alpha,\ \text{there is an isomorphism } X \simeq \varinjlim_I X_i \\ & \text{in } E_\alpha,\ \text{where the } X_i \text{ lie in } C_\alpha \text{ and } I \text{ is large with respect to } c. \end{cases} \tag{9.21.4}$$

Let Π be a cardinal such that

$$\Pi \geq \Pi_0, \qquad \text{and hence} \qquad \Pi > \Pi_0', \tag{9.21.5}$$

and for every $\alpha \in \operatorname{ob} B$ let

$$C_\alpha^\Pi \subset E_\alpha \tag{9.21.6}$$

be the full subcategory formed by objects which can be represented in the form $\varinjlim_I X_i$, where I is an ordered set large with respect to Π_0', $\operatorname{card} I \leq \Pi$, and the X_i lie in C_α. It is then clear, by 7.5.2, that C_α^Π is essentially small, i.e.equivalent to a small category.

Let

$$F = \operatorname{Hom}_B(B, E) \tag{9.21.7}$$

be the category of sections of E over B, and let

$$F^\Pi \subset F \tag{9.21.8}$$

be the strictly full subcategory of F formed by sections $X : \alpha \mapsto X(\alpha)$ such that, for every $\alpha \in \operatorname{ob} B$,

$$X(\alpha) \in C_\alpha^\Pi.$$

Since the C_α^Π are essentially small, it is clear that the same is true of F^Π. We shall show that this category is generating, and more precisely:

Lemma 9.21.9. Under the preceding hypotheses and with the preceding notation, the following hold.

(i) Every object of F is accessible (9.3). If d is a cardinal $\geq \Pi_0'$, and if $\alpha \mapsto X(\alpha)$ is an element of F such that, for every α, $X(\alpha)$ is d-accessible in E_α, then X is d-accessible in F.

(ii) Suppose that $\Pi \leq c$, or that Π_0' is finite (i.e.that the E_α are stable under small filtered inductive limits). Then every object X of F is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i lie in F^Π and I is large with respect to Π.

To prove (i), note first that by (9.21.4) and Corollary 9.9, every object of E_α is accessible. On the other hand, F satisfies condition $L_{\Pi_0'}$, by the following lemma.

Lemma 9.21.10. Let $p : E \to B$ be a fibred functor, let $F = \operatorname{Hom}_B(B, E)$, let I be a category, and let $i \mapsto X_i$ be a functor from I to F. In order that $\varinjlim_I X_i$ be representable in F, it is enough that, for every $\alpha \in \operatorname{ob} B$, $\varinjlim_I X_i(\alpha)$ be representable in the fibre category E_α. When this is the case, $\varinjlim_I X_i$ is computed "argument by argument". In particular, if the fibre categories satisfy condition $L_{\Pi_0'}$, where Π_0' is a fixed cardinal, then so does F.

We leave the easy proof of Lemma 9.21.10 to the reader, merely noting that it is convenient to use the following immediate result.

Corollary 9.21.10.1. With the hypotheses and notation of Lemma 9.21.10 for $p : E \to B$ and for I, for every $\alpha \in \operatorname{ob} B$, the inclusion functor $E_\alpha \to E$ commutes with inductive limits of type I.

Returning to the general hypotheses of Lemma 9.21.9, let X be an object of F. Since for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is accessible in E_α, there exists a cardinal $d \in \mathcal{U}$ such that for every α, $X(\alpha)$ is d-accessible in E_α. We may choose $d \geq \Pi'_0$, so that F satisfies L_d by Lemma 9.21.10. Using Lemma 9.21.10 again to compute inductive limits $\varinjlim_I Y_i$ in F, with I large with respect to d, we immediately see that X is d-accessible. This proves (i).

We now prove Lemma 9.21.9(ii) in several steps, namely 9.21.11 through 9.21.16.

Let

$$X : \alpha \longmapsto X(\alpha)$$

be an object of F. By (9.21.4 b)), for every $\alpha \in \operatorname{ob} B$ one can find an ordered set I_α large with respect to c, and an isomorphism

$$X(\alpha) = \varinjlim_{i \in I_\alpha} X(\alpha)_i,$$

where $i \mapsto X(\alpha)_i$ is an inductive system of type I_α in E_α, with $X(\alpha)_i$ lying in C_α. Consider the product ordered set

$$I = \prod_\alpha I_\alpha.$$

It is clear that it is large with respect to c, and that the preceding inductive systems give rise to inductive systems in the categories E,

$$i \longmapsto X(\alpha)_i \in \operatorname{ob} C_\alpha, \qquad i \in \operatorname{ob} I, \tag{9.21.11}$$

indexed by the same ordered set I, and to isomorphisms

$$X(\alpha) \xrightarrow{\sim} \varinjlim_I X(\alpha)_i \quad \text{in } E_\alpha, \tag{9.21.12}$$

for all $\alpha \in \operatorname{ob} B$. In what follows we fix data (9.21.11) and (9.21.12).

Lemma 9.21.13. Under the preceding hypotheses, one can find a map $\varphi : I \to I$ such that $\varphi(i) \geq i$ for every $i \in I$, and a map

$$(i, f) \longmapsto \lambda(i, f)$$

from $I \times \operatorname{Fl} B$ to $\operatorname{Fl} E$, satisfying the following conditions.

a) For $i \in I$ and $f : \alpha \to \beta$ in $\operatorname{Fl} B$, $\lambda(i, f)$ is an f-morphism from $X(\alpha)_i$ to $X(\beta)_{\varphi(i)}$, making commutative the diagram

$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \\
 & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"'] &
\end{tikzcd}$$

$$\alpha \xrightarrow[f]{} \beta.$$

Here the vertical arrows are the canonical morphisms deduced from (9.21.12).

b) For every $i \in I$ and every $f : \alpha \to \beta$ in $\mathrm{Fl}\, B$, one has commutativity in

$$\begin{tikzcd}
 & & X(\beta)_{\varphi^2(i)} \\
X(\alpha)_{\varphi(i)} \arrow[urr, "\lambda(\varphi(i),f)"] & & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[u] \arrow[urr, "\lambda(i,f)"'] & & \\
\alpha \arrow[rr, "f"'] & & \beta.
\end{tikzcd}$$

c) For every pair of consecutive arrows

$$\alpha \xrightarrow{f} \beta \xrightarrow{g} \gamma$$

of B, one has commutativity in the following diagram:

$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \arrow[r, "X(g)"] & X(\gamma) \\
 & & X(\gamma)_{\varphi^2(i)} \arrow[u] \\
 & X(\beta)_{\varphi(i)} \arrow[uu] \arrow[ur, "\lambda(\varphi(i),g)"'] & X(\gamma)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uuu] \arrow[ur, "\lambda(i,f)"] \arrow[urr, "\lambda(i,gf)"'] & & \\
\alpha \arrow[r, "f"'] & \beta \arrow[r, "g"'] & \gamma.
\end{tikzcd}$$

Again, the vertical arrows are deduced from (9.21.12).

Notice also that the commutativity of the two upper trapezia in b) is already contained in a); thus b) is really asserting the commutativity of the lower triangle of the diagram considered.

We prove Lemma 9.21.13. Let $i \in I$, and let $f : \alpha \to \beta$ be an arrow of B. Consider the composite

$$X(\alpha)_i \longrightarrow X(\alpha) \xrightarrow{X(f)} X(\beta) \simeq \varinjlim_j X(\beta)_j.$$

It defines a morphism in E_α

$$X(\alpha)_i \longrightarrow f^*(X(\beta)) \simeq f^*\left(\varinjlim_j X(\beta)_j\right) \simeq \varinjlim_j f^*(X(\beta)_j),$$

where the last equality follows from the fact that f^* is c-accessible (9.21.3 c)) and that I is large with respect to c. By (9.21.4 a)), since $X(\alpha)_i$ lies in C_α, the morphism under consideration factors through one of the $f^*(X(\beta)_j)$; here j *a priori* depends on i and on $f \in \mathrm{Fl}\, B$. But by (9.21.3 b)), and since I is large with respect to c, we may choose j independently of f; write

$j = \varphi(i)$. Since I is filtered, we may suppose $\varphi(i) \geq i$. We thus obtain, for $i \in I$ and $f \in \mathrm{Fl}\, B$, a morphism

$$X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)}),$$

or equivalently an f-morphism

$$\lambda(i, f) : X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)}$$

which, by construction, makes the diagram in a) commute.

Next fix i, f, g and consider the lower triangle of the diagram in 9.21.13 c). It is not clear that this triangle commutes, but the two composites

$$X(\alpha)_i \rightrightarrows X(\gamma)_{\varphi^2(i)}$$

become equal after composition with $X(\gamma)_{\varphi^2(i)} \to X(\gamma)$, as follows from the commutativity of the two upper trapezia and of the outer trapezium; these are respectively the diagrams $D(f)$, $D(g)$, and $D(gf)$. Since $X(\alpha)_i$ is c-accessible (9.21.4 a)) and

$$X(\gamma) \simeq \varinjlim_{j \in I} X(\gamma)_j$$

with I large with respect to c, one can find an element $j \geq \varphi^2(i)$ of I such that the two arrows just considered become equal after composition with

$$X(\gamma)_{\varphi^2(i)} \longrightarrow X(\gamma)_j.$$

A priori j depends on i, f, g. But for fixed i, the set of possible pairs f, g has cardinal $\leq c^2 = c$, by (9.21.3 a) and b)); since I is large with respect to c, we may choose j independently of f and g. Write $j = \varphi'(i) \geq \varphi^2(i) \geq \varphi(i)$. Define, for every $i \in I$ and every $f : \alpha \to \beta$, $\lambda'(i, f)$ to be the composite f-morphism

$$X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)} \longrightarrow X(\beta)_{\varphi'(i)},$$

where the second arrow is the transition morphism. It is then immediate from the construction that (φ', λ') satisfies conditions 9.21.13 a) and c). Proceeding in the same way for condition b), one sees that φ' can be chosen so that this condition is also satisfied. This completes the proof of Lemma 9.21.13.

9.21.14

Let

$$J \subset I$$

be a subset of I satisfying the following conditions:

a) J is filtered;

b) for every $j \in J$, one has $\varphi(j) \in J$;

c) for every $\alpha \in \mathrm{ob}\, B$, the object
$$X_J(\alpha) = \varinjlim_{i \in J} X(\alpha)_i$$
is representable in E_α.

Conditions a) and c) are satisfied in particular if J is large with respect to Π_0' (9.21.2). It then follows from Corollary 9.21.10.1 that, for every $\alpha \in \operatorname{ob} B$, $X_J(\alpha)$ is the inductive limit $\varinjlim_{i\in J} X(\alpha)_i$ in E. Now for an arrow $f : \alpha \to \beta$ of B, the arrows $\lambda(i, f)$, with i varying in J, define by 9.21.13 b) a morphism of ind-objects from $(X(\alpha)_i)_{i\in J}$ to $(X(\beta)_i)_{i\in J}$. Hence they induce a homomorphism

$$X_J(f) : X_J(\alpha) \longrightarrow X_J(\beta)$$

on inductive limits, which is visibly an f-homomorphism. Using 9.21.13 c), one finds the transitivity relations

$$X_J(gf) = X_J(g)X_J(f),$$

so that a section $X_J \in \operatorname{ob} F$ of E over B has been defined. Finally, 9.21.13 a) shows that the canonical homomorphisms

$$X_J(\alpha) \longrightarrow X(\alpha), \qquad \alpha \in \operatorname{ob} B,$$

are functorial in α; thus we have a canonical homomorphism

$$u_J : X_J \longrightarrow X.$$

Moreover, if

$$J' \supset J$$

is another subset of I satisfying conditions a), b), c) above, one obtains a canonical homomorphism

$$u_{J',J} : X_J \longrightarrow X_{J'},$$

so that the X_J form an inductive system in F, parametrized by the set, ordered by inclusion, of subsets J of I satisfying the conditions under consideration. Finally, the homomorphisms u_J above define a homomorphism from this inductive system to the constant inductive system defined by X.

9.21.15

Let K be a set of subsets J of I satisfying conditions a), b), c) of 9.21.14, and suppose that K is filtered and has union I. Then it is clear that

$$\varinjlim_{J\in K} X_J \xrightarrow{\sim} X,$$

using Corollary 9.21.10.1 to reduce the verification to an isomorphism argument by argument.

For example, take K to be the set of all subsets J of I which are large with respect to Π_0', stable under φ, and such that $\operatorname{card} J \leq \Pi$. Then, by the definition (9.21.6) of C_α^Π, one has

$$X_J(\alpha) \in \operatorname{ob} C_\alpha^\Pi$$

for every $\alpha \in \operatorname{ob} B$, and hence $X_J \in \operatorname{ob} F^\Pi$. Therefore Lemma 9.21.9(ii) will be proved if we show that K is large with respect to Π (hence filtered) and has union I. For this it plainly suffices to prove that every subset S of I with $\operatorname{card} S \leq \Pi$ is contained in some $J \in K$. This follows from the preliminary hypothesis stated in Lemma 9.21.9(ii), and from the following lemma.

Lemma 9.21.16. Let I be an ordered set, let Π be an infinite cardinal, and let $\varphi : I \to I$ be a map.

a) Suppose that I is filtered. Then for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a filtered subset $J \supset S$ of I such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.

b) Let Π'_0 be a cardinal $< \Pi$, and suppose that I is large with respect to Π. Then for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a subset $J \supset S$ of I which is large with respect to Π'_0, such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.

We prove b); the proof of a) is analogous and simpler. Let P be the set of subsets of I of cardinal $\leq \Pi$, and let $T \mapsto i_T : P \to I$ be a map such that, for $T \in P$, i_T is an upper bound for T in I. The existence of this map is simply the assertion that I is large with respect to Π. Let A be a well-ordered set such that $\operatorname{card} A = \Pi$ and such that, for every $a \in A$, the set of $a' \leq a$ has cardinal $< \Pi$. In particular, since $\Pi'_0 < \Pi$, it follows that A is large with respect to Π'_0.

Define, by transfinite recursion, maps

$$S \longmapsto S_a : P \to P \qquad (a \in A)$$

as follows:

$$S_0 = S, \qquad S_{a+1} = S_a \cup \varphi(S_a) \cup \{i_{S_a}\}, \qquad S_a = \bigcup_{a'<a} S_{a'} \quad \text{if } a \text{ is a limit ordinal.}$$

Then put, for $S \in P$,

$$J(S) = \bigcup_{a \in A} S_a.$$

It is clear that $J(S) \supset S$, that $J(S)$ is stable under φ, that $\operatorname{card} J(S) \leq \Pi$ (since $\operatorname{card} A = \Pi$), and finally that $J(S)$ is large with respect to Π'_0 (using that A is large with respect to Π'_0). This proves Lemma 9.21.16 and completes the proof of Lemma 9.21.9.

We also note the following variant of Lemma 9.21.9.

Lemma 9.21.17. The notation is that of Lemma 9.21.9. Denote by F' the full subcategory

$$F' = \operatorname{Hom}^{\mathrm{cart}}_B(B, E)$$

of $F = \operatorname{Hom}_B(B, E)$ formed by the cartesian sections of E over B. Then:

(i) Every object of F' is accessible. More precisely, let $\alpha \mapsto X(\alpha)$ be an element of F', and let d be a cardinal such that $d \geq \Pi'_0$, $d \geq c$, and such that for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is a d-accessible object of E_α. Then X is a d-accessible object of F'.

(ii) Suppose that $\Pi \leq c$ and that the functors $f^* : E_\beta \to E_\alpha$ are Π'_0-accessible, or else that the categories E_α are stable under small filtered inductive limits and that the functors $f^* : E_\beta \to E_\alpha$ commute with these inductive limits. Then every object X of F' is isomorphic to an object of the form $\varinjlim_I X_i$, where for every $i \in I$, X_i is an object of the full subcategory

$$F'^{\Pi} = F' \cap F^{\Pi}$$

of F', and where I is large with respect to Π.

The proof being entirely analogous to that of Lemma 9.21.9, we merely indicate where modifications of that proof are necessary. First note the following.

Lemma 9.21.18. The statement of Lemma 9.21.10 remains valid when one replaces $F = \mathrm{Hom}_B(B, E)$ by $F' = \mathrm{Hom}_B^{\mathrm{cart}}(B, E)$, provided that one assumes that the inverse image functors

$$f^* : E_\beta \longrightarrow E_\alpha$$

commute with inductive limits of type I.

Thus, under the hypotheses of Lemma 9.21.17, if d is a cardinal such that $d \geq c$ and $d \geq \Pi_0'$, it follows from what precedes and from hypothesis (9.21.3 c)) that for every ordered set I large with respect to d, inductive limits of type I are representable in F' and are computed argument by argument. Lemma 9.21.17(i) follows by an immediate argument.

To prove (ii), one must supplement Lemma 9.21.13.

Lemma 9.21.19. Under the hypotheses of Lemma 9.21.13, suppose that $X \in \mathrm{ob}\, F'$, i.e.that X is a cartesian section of E over B. Then the conclusion can be strengthened as follows: there exists a function μ which assigns, to every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, a morphism

$$\mu(i, f) : f^* X(\beta)_i \longrightarrow X(\alpha)_{\varphi(i)}$$

in E_α, in such a way that:

d) For every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, the two following diagrams commute:

$$\begin{tikzcd}
 & f^*X(\beta)_{\varphi^2(i)} \\
X(\alpha)_{\varphi(i)} \arrow[ur, "\lambda(\varphi(i),f)"] & \\
 & f^*X(\beta)_i \arrow[uu] \arrow[ul, "\mu(i,f)"']
\end{tikzcd}
\qquad
\begin{tikzcd}
X(\alpha)_{\varphi^2(i)} & \\
 & f^*X(\beta)_{\varphi(i)} \arrow[ul, "\mu(\varphi(i),f)"'] \\
X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"'] &
\end{tikzcd}$$

where the unlabeled arrows are the evident transition maps.

To prove this, one proceeds as in the proof of 9.21.13, by considering the composite morphism

$$f^*(X(\beta)_i) \longrightarrow f^*(X(\beta)) \xrightarrow{X(f)^{-1}} X(\alpha),$$

where, as already in the notation for condition d), an f-morphism $R \to S$ of E is identified with the corresponding morphism $R \to f^*(S)$ in E_α. Since

$$X(\alpha) = \varinjlim_j X(\alpha)_j$$

and I is large with respect to c, the preceding morphism can be factored through one of the $X(\alpha)_j$. Here j *a priori* depends on i and on f, but can be chosen independently of f; write it $\psi(i)$. Enlarging the function φ of Lemma 9.21.13 if necessary, we may suppose that $\psi = \varphi$. Proceeding as for conditions b) and c) of Lemma 9.21.13, we see that, after enlarging φ again if necessary, the two diagrams of Lemma 9.21.19 d) may be assumed commutative.

9.21.20

With φ chosen as in Lemma 9.21.19, resume the argument of 9.21.14. One must, however, assume that J satisfies, besides conditions a), b), c) stated there, the following condition:

d) For every arrow $f : \alpha \to \beta$ in B, the functor $f^* : E_\beta \to E_\alpha$ commutes with inductive limits of type J.

I claim that, under this additional condition, the section X_J of E over B is cartesian; that is, for every arrow $f : \alpha \to \beta$ of B, the morphism

$$X_J(\alpha) \longrightarrow f^* X_J(\beta) \tag{$*$}$$

is an isomorphism. Indeed, by condition d) above, the target of this morphism is identified with

$$\varinjlim_{i \in J} f^*(X(\beta)_i),$$

and the morphism itself is obtained by passing to $\varinjlim_J$ from the morphism of ind-objects in E_α

$$(X(\alpha)_i)_{i\in J} \longrightarrow (f^* X(\beta)_i)_{i \in J},$$

deduced from the morphisms

$$\lambda(i, f) : X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)})$$

for $i \in J$. But the conditions made explicit in Lemma 9.21.19 ensure that this morphism of ind-objects is in fact an isomorphism, with inverse obtained from the morphism deduced from the system of the $\mu(i, f)$, $i \in J$. This proves that $(*)$ is an isomorphism, hence that $X_J \in \operatorname{ob} F'$.

9.21.21

We now assume that the functors $f^* : E_\beta \to E_\alpha$ are Π'_0-*accessible*. Then the conditions on J considered in 9.21.20 are satisfied if J is large with respect to Π'_0, stable under φ, and has $\operatorname{card} J \leq \Pi$. The proof of Lemma 9.21.17 is then completed as in 9.21.15.

Theorem 9.22. Let $p : E \to B$ be a fibred functor, where B is a category essentially equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ of B, the inverse image functor

$$f^* : E_\beta \longrightarrow E_\alpha$$

is accessible (9.2). Suppose further that, for every $\alpha \in \operatorname{ob} B$, the fibre category E_α admits a cardinal filtration (9.12), and that every object of E_α is accessible (9.3). Then each of the categories

$$F = \operatorname{Hom}_B(B, E), \qquad F' = \operatorname{Hom}_B^{\mathrm{cart}}(B, E)$$

admits a small full subcategory which is generating by strict epimorphisms (7.1), and all of its objects are accessible.

It is immediate that the statement is not essentially changed when B is replaced by a full subcategory B_0 such that the inclusion functor $B_0 \to B$ is an equivalence, and E is replaced by $E_0 = E \times_B B_0$. Thus we may suppose that B is a small category.

For every $\alpha \in \operatorname{ob} B$, let

$$(\mathrm{Filt}^{\Pi}(E_\alpha))_{\Pi \geq \Pi_\alpha}$$

be a cardinal filtration of E_α. Let $c_0 \in \mathcal{U}$ be a cardinal satisfying:

$$\begin{cases} c_0 \geq \mathrm{Sup}_{\alpha \in \operatorname{ob} B} \Pi_\alpha, \qquad c_0 \geq \operatorname{card} \mathrm{Fl}\, B; \\ \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_\beta \to E_\alpha \text{ is } c_0\text{-accessible;} \\ \text{for every } \alpha \in \operatorname{ob} B, \text{ the objects of } \mathrm{Filt}^{\Pi_\alpha}(E_\alpha) \text{ are } c_0\text{-accessible.} \end{cases} \tag{9.22.1}$$

Put

$$c = 2^{c_0},$$

so that $c > c_0$, and for every $\alpha \in \operatorname{ob} B$ put

$$C_\alpha = \mathrm{Filt}^c(E_\alpha).$$

I claim that the preliminary hypotheses of Lemmas 9.21.9 and 9.21.17 are satisfied, taking $\Pi = \Pi_0 = c$. This is clear for (9.21.2) and (9.21.3). Condition (9.21.4 a)) follows from Corollary 9.17, and (9.21.4 b)) follows from condition 9.12 c) for a cardinal filtration. Notice also that condition 9.12 b) in the definition of a cardinal filtration implies that, for every $\alpha \in \operatorname{ob} B$, one has

$$C_\alpha^\Pi = C_\alpha = \mathrm{Filt}^c(E_\alpha),$$

where C_α^Π is defined in (9.21.6). Theorem 9.22 therefore follows from Lemmas 9.21.9 and 9.21.17. More precisely, we have proved:

Corollary 9.23. Under the hypotheses of Theorem 9.22, if $c_0 \in \mathcal{U}$ is a cardinal satisfying (9.22.1), and if $c = 2^{c_0}$, then the subcategory F^c of F (respectively F'^c of F') formed by objects X such that $X(\alpha) \in \operatorname{ob} \mathrm{Filt}^c(E_\alpha)$ for every $\alpha \in \operatorname{ob} B$ is essentially small and generating by strict epimorphisms. More precisely, every object X of F (respectively of F') is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i lie in F^c (respectively in F'^c), and where I is an ordered set large with respect to c.

Under slightly stronger hypotheses in Theorem 9.22, one can also make Theorem 9.22 and Corollary 9.23 considerably more precise.

Corollary 9.24. Under the hypotheses of Theorem 9.22, suppose that each of the categories E_α ($\alpha \in \operatorname{ob} B$) is stable under small filtered inductive limits, and that for every $\Pi \geq \Pi_\alpha$, $\mathrm{Filt}^\Pi(E_\alpha)$ is stable under filtered inductive limits indexed by filtered ordered sets I such that $\operatorname{card} I \leq \Pi$; this is a slight strengthening of condition 9.12 b) for cardinal filtrations. In the case where F' is considered, suppose moreover that, for every arrow $f : \alpha \to \beta$ of B, the functor

$$f^* : E_\beta \to E_\alpha$$

commutes with small filtered inductive limits. Finally let $c_0 \in \mathcal{U}$ be a cardinal satisfying (9.22.1), and for every cardinal $\Pi \geq c = 2^{c_0}$ consider the strictly full subcategory F^Π (respectively F'^Π) of F (respectively of F') formed by the X such that

$$X(\alpha) \in \mathrm{Filt}^\Pi(E_\alpha)$$

for every $\alpha \in \operatorname{ob} B$. Then the subcategories under consideration define a cardinal filtration (9.12) of F (respectively of F').

One must verify conditions a), b), c) of 9.12. Conditions a) and b) follow from the analogous conditions for the given cardinal filtrations of the E_α, and from Lemma 9.21.10 (respectively Lemma 9.21.18, taking into account the second condition in (9.22.1)). It remains to prove c). The first part of c) follows at once from Lemma 9.21.9 (respectively Lemma 9.21.17), by taking $\Pi'_0 = 0$ and $\Pi_0 = c$.

It remains to prove that if $\Pi' \geq \Pi \geq c$ are two cardinals, then for every X in $F^{\Pi'}$ (respectively $F'^{\Pi'}$) one has

$$X \simeq \varinjlim_{i \in K} X_i,$$

where the X_i lie in F^{Π} (respectively in F'^{Π}), where K is large with respect to Π, and where $\operatorname{card} K \leq \Pi'^{\Pi}$. To this end, resume the proofs of Lemma 9.21.9(ii) and Lemma 9.21.17(ii). We may suppose that each I_α has cardinal $\leq \Pi'$ by condition 9.12 c) on the cardinal filtration of E_α. Hence

$$\operatorname{card} I \leq (\Pi'^{\Pi})^{\operatorname{card} \operatorname{ob} B} \leq (\Pi'^{\Pi})^c = \Pi'^{\Pi c} = \Pi'^{\Pi}.$$

The indexing set K used in 9.21.15, respectively 9.21.21, is the set of subsets J of I which are filtered (i.e.large with respect to $\Pi'_0 = 0$), stable under φ, and such that $\operatorname{card} J \leq \Pi$. It is then immediate that

$$\operatorname{card} K \leq (\operatorname{card} I)^{\Pi} \leq (\Pi'^{\Pi})^{\Pi} = \Pi'^{\Pi^2} = \Pi'^{\Pi},$$

which completes the proof of Corollary 9.24.

To finish, it is appropriate to give a statement freed from the somewhat technical hypotheses of Theorem 9.22, by replacing them, for the E_α, by the conjunction of the hypotheses occurring in Proposition 9.11, which ensure accessibility of the objects of the E_α, and in Proposition 9.13, which ensure the existence of a cardinal filtration in E_α.

Corollary 9.25. Let $p : E \to B$ be a fibred functor, where B is a category equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ of B, the inverse image functor

$$f^* : E_\beta \to E_\alpha$$

is accessible (9.2). Suppose further that, for every $\alpha \in \operatorname{ob} B$, the fibre category E_α satisfies the following conditions:

a) E_α admits a small full subcategory generating by strict epimorphisms (7.1).

b) E_α is stable under fibre products, under kernels of double arrows, under sums of two objects, under cokernels of double arrows, and finally under small filtered inductive limits.[1]

c) The functor $\operatorname{Ker}(u, v)$ on the category of double arrows of E_α is accessible; for example, it commutes with small filtered inductive limits.

d) Every strict epimorphism of E_α (10.2) is a universal strict epimorphism.

Under these hypotheses, each of the categories

$$F = \operatorname{Hom}_B(B, E), \qquad F' = \operatorname{Hom}^{\mathrm{cart}}_B(B, E)$$

admits a small subcategory generating by strict epimorphisms, and all of its objects are accessible (9.3).

[1] This last condition is probably unnecessary; cf.9.13.2.

Moreover, Corollary 9.24 gives:

Corollary 9.26. Under the hypotheses of 9.25, and in the case where one considers $F' = \mathrm{Hom}_B^{\mathrm{cart}}(B, E)$, suppose that the functors

$$f^* : E_\beta \to E_\alpha$$

commute with small filtered inductive limits. Then F (respectively F') admits a cardinal filtration, which can be made explicit by the procedure of Corollary 9.24.

Continuation anchor. The next part of Exposé I is Section 10, the glossary of terms used in the preceding sections.

Exposé I. Presheaves

Section 10 and Appendix II: Glossary; Universes

A. Grothendieck and J.-L. Verdier; Appendix attributed to N. Bourbaki

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 10 and the appended note "Universes." The range corresponds to source lines 7424–8921 of the Orgogozo–Laszlo TeX snapshot. The translation keeps the period terminology where it is still mathematically useful: *strict epimorphism*, *effective equivalence relation*, *direct factor*, and *base-changeable* for French *quarrable*. A few evident typographical slips in the source have been silently normalized where the formulas force the correction.

10 Glossary

For the convenience of the reader, we collect here the definitions of several terms used in the preceding sections. Throughout this section, C denotes a category.

10.1 Cartesian, cocartesian

A diagram in C

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ C \arrow[r] & D \end{tikzcd}$$

is called *cartesian* if it is commutative and if the canonical morphism from A to the fiber product $C \times_D B$ is an isomorphism. It is called *cocartesian* if the corresponding diagram in the opposite category C° is cartesian.

10.2 Epimorphism, strict epimorphism

See 10.3.

10.3 Epimorphic family, strictly epimorphic family

A family

$$f_i : A_i \longrightarrow B, \qquad i \in I,$$

of arrows with common target in C is called an *epimorphic family* if, for every object T of C, the map induced by the f_i,

$$\operatorname{Hom}_C(B, T) \longrightarrow \prod_i \operatorname{Hom}_C(A_i, T), \tag{10.3.1}$$

is injective.

The family is called *strictly epimorphic* if the image of (10.3.1) consists of those families (g_i) such that, for every object R of C, every pair of indices $i, j \in I$, and every pair of arrows $u : R \to A_i$, $v : R \to A_j$ satisfying $f_i u = f_j v$, one also has $g_i u = g_j v$. When the fiber products

$A_i \times_B A_j$ are representable for $i, j \in I$, this is equivalent to saying that the image of (10.3.1) consists of those (g_i) such that, for all i, j,

$$g_i \, \mathrm{pr}_1 = g_j \, \mathrm{pr}_2,$$

where $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections from $A_i \times_B A_j$. The family $(f_i)_{i \in I}$ is called an *effective epimorphic family* if this condition is satisfied, that is, if it is strictly epimorphic and the fiber products $A_i \times_B A_j$ are representable.

The family $(f_i)_{i \in I}$ is called *universally epimorphic* (respectively, *universally effectively epimorphic*) if the morphisms f_i are base-changeable (10.7) and if, for every arrow $B' \to B$, the family of arrows obtained from (f_i) by base change along $B' \to B$ is epimorphic (respectively, effective epimorphic).

The notion of a *universally strictly epimorphic* family will be defined in II 2.5–2.6. Note that when the morphisms f_i are base-changeable, for example if fiber products are representable in C, this notion agrees with that of a universally effectively epimorphic family, using II 2.4. In general, between the six variants of epimorphicity considered here, one has the following logical implications:

universally effective epimorphic ⇒ effective epimorphic ⇒ strictly epimorphic ⇒ epimorphic,
universally effective epimorphic ⇒ universally strictly epimorphic ⇒ strictly epimorphic,
universally strictly epimorphic ⇒ universally epimorphic ⇒ epimorphic.

A morphism $f : A \to B$ of C is called an *epimorphism* (respectively, a strict epimorphism, effective epimorphism, universal epimorphism, universally effective epimorphism, universally strict epimorphism) if the one-element family consisting of f is epimorphic (respectively, ...).

10.4 Monomorphic family, strictly monomorphic family

A family of arrows $f_i : A_i \to B$ of C is called *monomorphic* (respectively, strictly monomorphic, ...) if, as a family of arrows in the opposite category C°, it is epimorphic (respectively, strictly epimorphic, ...). An arrow $f : A \to B$ of C is called a *monomorphism* (respectively, a strict monomorphism, ...) if the one-element family consisting of f is monomorphic (respectively, strictly monomorphic, ...), that is, if f, viewed as an arrow of C°, is an epimorphism (respectively, a strict epimorphism, ...).

10.5 Monomorphism, strict monomorphism

See 10.4.

10.6 Projector, image of a projector, direct factor of an object

Let $f : X \to X$ be an endomorphism of an object of C. We say that f is a *projector* if $f^2 = f$. Then the pair (f, id_X) admits a representable cokernel X' if and only if it admits a representable kernel X''; and when this is so, there is a unique isomorphism $u : X' \to X''$ in C such that $iup = f$, where $p : X \to X'$ and $i : X'' \to X$ are the canonical morphisms. The representability of the kernel, or of the cokernel, is also equivalent to the existence of morphisms i, p and of an isomorphism u such that $iup = f$ and $pi = u^{-1}$. In particular, this property is preserved by any functor.

One then usually identifies X' and X'', and calls it the *image of the projector* f. We say that the projector f *admits an image* if $\mathrm{Ker}(f, \mathrm{id}_X)$ is representable, equivalently if $\mathrm{Coker}(f, \mathrm{id}_X)$ is representable. A subobject (respectively, a quotient) Y of X is called a *direct factor* of X if one can find a projector f in X which factors through Y, and which admits Y as image as a subobject (respectively, as a quotient) of X. By abuse of language, one sometimes says that an object of C is a *direct factor of* X if it is isomorphic to the image of a projector in X.

10.7 Base-changeable

An arrow $f : A \to B$ of C is called *base-changeable* (French: *quarrable*) if, for every arrow $B' \to B$ of C, the fiber product $A \times_B B'$ is representable in C.

10.8 Quotient, strict quotient

Let A be an object of C. Two epimorphisms $f : A \to B$, $f' : A \to B'$ with source A are called *equivalent* if there exists an isomorphism $u : B \xrightarrow{\sim} B'$ such that $uf = f'$. This gives an equivalence relation on the set of epimorphisms with source A; its classes are called the *quotients*, or *quotient objects*, of A. For every quotient of A one usually chooses an element of this class, say $f : A \to B$, and one often speaks, by abuse of language, of the quotient B of A, or of the quotient $f : A \to B$ of A. We say that B is a *strict quotient* (respectively, an effective quotient, universal quotient, ...) if the morphism $f : A \to B$ is a strict epimorphism (respectively, an effective epimorphism, universal epimorphism, ...) in the sense of 10.3.

10.9 Equivalence relations

Let F and G be two presheaves of sets on C. A diagram $p_1, p_2 : F \rightrightarrows G$ is called an *equivalence relation on* G if, for every object A of C, the map

$$(p_1(A), p_2(A)) : F(A) \longrightarrow G(A) \times G(A)$$

induces a bijection from $F(A)$ onto the graph of an equivalence relation on the set $G(A)$.

A diagram $p_1, p_2 : B \rightrightarrows C$ in C is called an equivalence relation on C if the corresponding diagram of presheaves is an equivalence relation. When finite products and fiber products are representable in C, a functor $\Phi : C \to C'$ commuting with finite products and fiber products carries equivalence relations on C to equivalence relations on $\Phi(C)$.

Let $\Pi : C \to D$ be a morphism of C such that the fiber product $C \times_D C$ is representable. Then the diagram

$$\mathrm{pr}_1, \mathrm{pr}_2 : C \times_D C \rightrightarrows C$$

is an equivalence relation on C.

10.10 Effective equivalence relation, universally effective equivalence relation

An equivalence relation $p_1, p_2 : R \rightrightarrows A$ on an object A of C is called an *effective equivalence relation* if there exists a morphism $\pi : A \to B$ such that the square

$$\begin{tikzcd} R \arrow[r, "p_2"] \arrow[d, "p_1"'] & A \arrow[d, "\pi"] \\ A \arrow[r] & B \end{tikzcd}$$

is cartesian and cocartesian. The morphism π is the cokernel of the pair (p_1, p_2), and is therefore determined up to unique isomorphism. The morphism π is then an effective epimorphism; if it is a universally effective epimorphism in the sense of 10.3, one says that the *equivalence relation* is *universally effective*.

10.11 Subobject, strict subobject

These are the notions dual to those of quotient, strict quotient, and so on; see 10.8.

References

[1] B. Mitchell, *Theory of Categories*, Academic Press, 1965.

II. Appendix: Universes (by N. Bourbaki*)

*We reproduce here, with his permission, secret papers of N. Bourbaki. The references in this text are to his learned work.

1 Definition and first properties of universes

Definition 1. A set U is called a *universe* if it satisfies the following conditions:

(U.I) if $x \in U$ and $y \in x$, then $y \in U$;

(U.II) if $x, y \in U$, then $\{x, y\} \in U$;

(U.III) if $x \in U$, then $\mathcal{P}(x) \in U$;

(U.IV) if $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then the union $\bigcup_{\alpha \in I} x_\alpha$ belongs to U;

(U.?) if $x, y \in U$, then the pair $(x, y) \in U$.

N.B. Since, I believe, it has been decided for future editions to define the ordered pair à la Kuratowski by

$$(x, y) = \{\{x, y\}, \{x\}\},$$

condition (U.?) is useless, because it follows from (U.II).

Examples.

1) The empty set is a universe, denoted U_0.

2) Consider the nonempty finite words formed with the four symbols "{", "}", ",", and "∅" (cf. Alg. I). Define, by induction on the length n of such a word, the notion of a *meaningful* word:

   (a) for $n = 1$, only the word $\emptyset$ is meaningful;

   (b) for a word A of length n to be meaningful, it is necessary and sufficient that there exist p distinct meaningful words $A_1, \ldots, A_p$, with $p \geq 1$ and of lengths $< n$, such that

$$A = \{A_1, A_2, \ldots, A_p\}.$$

For example, $\emptyset$, $\{\emptyset\}$, and

$$\{\{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$$

are meaningful words. It is clear, by induction on the length, that every meaningful word denotes a term of set theory; for example, if $A = \{A_1, A_2, \ldots, A_p\}$, then A denotes the set whose elements are $A_1, A_2, \ldots, A_p$. These terms are of course *finite sets.* Let U_1 be the set of the sets thus obtained. One easily checks that U_1 satisfies conditions (U.I), (U.II), (U.III), and (U.IV) of Definition 1, though not that idiotic (U.?), which is not serious if one is willing to de-cannulate the pair. Thus U_1 is a *universe.* Notice that the elements of U_1 are finite and that U_1 is countable.

In the statements that follow, U denotes a universe.

Proposition 1. If $x \in U$ and $y \subset x$, then $y \in U$.

Indeed $\mathcal{P}(x) \in U$ by (U.III), whence $y \in \mathcal{P}(x)$ and $y \in U$ by (U.I).

Corollary. If $x \in U$, every quotient set y of x is an element of U.

Indeed y is a subset of $\mathcal{P}(x)$. Hence $y \in U$ by (U.III) and Proposition 1.

Proposition 2. If $x \in U$, then $\{x\} \in U$.

This follows from (U.II), applied with $y = x$.

Proposition 3. Every pair, triple, and quadruple of elements of U is an element of U.

For pairs this is true by the preceding N.B. or by (U.?). The case of triples follows since $(x, y, z) = ((x, y), z)$, and the case of quadruples then follows since $(x, y, z, t) = ((x, y, z), t)$.

Proposition 4. If $X, Y \in U$, then $X \times Y \in U$.

Indeed, for $x \in X$, the set $\{x\} \times Y$ is the union of the family $\{(x, y)\}_{y \in Y}$; since $\{(x, y)\} \in U$ by (U.I), (U.II), and Proposition 3, we have $\{x\} \times Y \in U$ by (U.IV). Finally, $X \times Y$ is the union of the family $(\{x\} \times Y)_{x \in X}$, so it is again an element of U by (U.IV).

Corollary 1. If $X, Y, Z, \ldots$ are elements of U, then all sets in the scale built on $X, Y, Z, \ldots$ are elements of U (cf. Chapter IV, §).

This follows from successive applications of Proposition 4 and (U.III).

Corollary 2. If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then the sum set $\sum_{\alpha \in I} x_\alpha$ is an element of U.

Indeed this sum set is a subset of the product $(\bigcup_{\alpha \in I} x_\alpha) \times I$, whose two factors are elements of U, by (U.IV) and the hypothesis. One then applies Propositions 4 and 1.

Proposition 5. If X and Y are elements of U, then every correspondence between X and Y, in particular every map from X to Y, is an element of U.

Indeed such a correspondence C is a triple (X, Y, Γ), where Γ is a subset of $X \times Y$, the graph of C (Chapter II, §). We have $\Gamma \in U$ by Propositions 4 and 1. Hence $C \in U$ by Proposition 3.

Proposition 6. If $X, Y \in U$, then every set Z of correspondences between X and Y, in particular every set of maps from X to Y, is an element of U.

Indeed Z is a subset of $\{X\} \times \{Y\} \times \mathcal{P}(X \times Y)$. This product is an element of U, by Proposition 2 and the corollary to Proposition 4. Therefore $Z \in U$ by Proposition 1.

Corollary. If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then $\prod_{\alpha \in I} x_\alpha \in U$.

Indeed this product is a set of maps from I to $\bigcup_{\alpha \in I} x_\alpha$, and this union is an element of U by (U.IV).

Proposition 7. If X is a subset of U whose cardinal is at most that of an element of U, then X is an element of U.

Let I be an element of U such that $\operatorname{card}(X) \leq \operatorname{card}(I)$. There is a surjection $i \mapsto x_i$ from a subset I' of I onto X. Then X is the union of the family $(\{x_i\})_{i \in I'}$. This union is an element of U by Proposition 1, Proposition 2, and (U.IV).

Corollary. If U is nonempty, every finite subset of U is an element of U, and U has elements of arbitrarily large finite cardinality.

On the other hand, if $U = \emptyset$, then $\emptyset$ is a finite subset of $\emptyset$ but not an element of $\emptyset$.

Indeed, if U is nonempty, Proposition 2 shows that a one-element set x_0 belongs to U. By induction on n, put $x_{n+1} = \mathcal{P}(x_n)$. Then $x_n \in U$ by (U.III), and $\operatorname{card}(x_{n+1}) = 2^{\operatorname{card}(x_n)}$, so U has elements of arbitrarily large finite cardinality.

Remark. It follows from Proposition 2 and the corollary to Proposition 7 that every nonempty universe U contains the universe U_1 of Example 2; indeed $\emptyset \in U$ by Proposition 1. Thus U_1 is the intersection of all nonempty universes. More generally:

Proposition 8. If $(U_\lambda)_{\lambda\in L}$ is a nonempty family of universes, then $U = \bigcap_{\lambda\in L} U_\lambda$ is a universe.

This follows immediately from Definition 1.

2 Universes and species of structures

Let $\mathcal{E}$ be a species of structure. To fix ideas and lighten the exposition, suppose that every structure of species $\mathcal{E}$ is defined on one underlying set. Let (X, S) be a structure of species $\mathcal{E}$, where X is the underlying set and S the structure, and let U be a universe. If $X \in U$, then all the constituent objects of the structure S on X are elements of U, by Corollary 1 to Proposition 4 of Section 1 and by (U.I); hence the structure (X, S) is an element of U.

Now suppose that $\mathcal{E}$ is a species of structure with *morphisms*. If X and X' are elements of U endowed with structures of species $\mathcal{E}$, then the set of morphisms from X to X' is again an element of U, by Proposition 6 of Section 1.

Consider the category $(U\text{-}\mathcal{E})$ defined in Section 1, no. 2, Example c). Since the objects and arrows of $(U\text{-}\mathcal{E})$ are elements of U, $(U\text{-}\mathcal{E})$ is a pair of subsets of U, endowed with a quadruple of maps. It is not, in general, an element of U. The notation $X \in (U\text{-}\mathcal{E})$ will mean that X is an element of U endowed with a structure of species $\mathcal{E}$.

The category $(U\text{-}\mathcal{E})$ is stable under many operations:

a) If $X \in (U\text{-}\mathcal{E})$, and if X' is a subset of X admitting an *induced structure*, then $X' \in (U\text{-}\mathcal{E})$. This follows from Proposition 1 of Section 1.

b) If $X \in (U\text{-}\mathcal{E})$, and if X'' is a quotient set of X admitting a *quotient structure*, then $X'' \in (U\text{-}\mathcal{E})$, by the corollary to Proposition 1 of Section 1.

c) If $X, Y \in (U\text{-}\mathcal{E})$, and if $X \times Y$ admits a *product structure*, then $X \times Y \in (U\text{-}\mathcal{E})$, by Proposition 4 of Section 1. More generally, if $(X_i)_{i\in I}$ is a family of elements of $(U\text{-}\mathcal{E})$, if one has $I \in U$, and if $\prod_{i\in I} X_i$ admits a *product structure*, then $\prod_{i\in I} X_i \in (U\text{-}\mathcal{E})$, by the corollary to Proposition 6. The analogous assertions hold for *sum structures*, by Corollary 2 to Proposition 4.

d) Let I be a preordered set, and let $(X_i, f_{ij})_{i,j\in I}$ be a *projective* (respectively, *inductive*) system of sets endowed with structures of species $\mathcal{E}$ and morphisms. Let L be the limit of this system, in the sense of Chapter III. If $X_i \in U$ for all i, if $I \in U$, and if L admits a projective (respectively, inductive) limit structure, then $L \in (U\text{-}\mathcal{E})$: indeed L is a subset of $\prod_{i\in I} X_i$ (respectively, a quotient set of $\sum_{i\in I} X_i$), and is therefore an element of U by Section 1.

Here are a few more particular examples:

* 1) Let $X, Y \in (U\text{-}\mathbf{Top})$ be two *locally compact* spaces. Then the set $\mathcal{C}(X, Y)$ of continuous maps from X to Y, endowed with the compact-open topology (General Topology, X), is an element of $(U\text{-}\mathbf{Top})$. Indeed $\mathcal{C}(X, Y) \in U$ by Proposition 6 of Section 1.

* 2) Let $X, Y \in (U\text{-}\mathbf{Ab})$. Then the group $\mathrm{Hom}_{\mathbb{Z}}(X, Y)$ is an element of $(U\text{-}\mathbf{Ab})$ by Proposition 6 of Section 1. If, moreover, $\mathbb{Z} \in U$, then the tensor product $X \otimes_{\mathbb{Z}} Y$ is an element of $(U\text{-}\mathbf{Ab})$. Indeed, this tensor product is a quotient of the free abelian group $\mathbb{Z}^{(X\times Y)}$, whose underlying set is a subset of $\mathbb{Z}^{X\times Y}$, and the latter is an element of U by Propositions 4 and 6 of Section 1.

* 3) Let $X \in (U\text{-}\mathbf{Unif})$ be a uniform space. Then its *completion* $\widehat{X}$ is an element of $(U\text{-}\mathbf{Unif})$. Indeed $\widehat{X}$ is a set of equivalence classes of Cauchy filters on X; but a filter on X is an element of $\mathcal{P}(\mathcal{P}(X))$, and hence an equivalence class of filters is an element of $\mathcal{P}(\mathcal{P}(\mathcal{P}(X)))$. Thus $\widehat{X}$ is an element of $\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathcal{P}(X))))$, and therefore an element of U by (U.III) and (U.I).

N.B. Moral: Bourbaki must take care to "canonify" his constructions. For example, in constructions in which one adjoins an element ∞, such as the Alexandroff compactification or the projective field, it will be useful to take $\infty = \emptyset$, because $\emptyset$ is an element of every nonempty universe, and to form the sum set of $\{\emptyset\}$ and the given set. Of course one should also "canonify" the two-element set used to construct sum sets; $\{\emptyset, \{\emptyset\}\}$ seems to me a good candidate, since it belongs to every nonempty universe.

3 Universes and categories

Let U be a universe and C a category. We write $C \in U$ if $\mathrm{Ob}(C) \in U$ and $\mathrm{Fl}(C) \in U$. This notation is justified by the fact that C is the sextuple consisting of $\mathrm{Ob}(C)$, $\mathrm{Fl}(C)$, and the four structural maps. Thus if $\mathrm{Ob}(C) \in U$ and $\mathrm{Fl}(C) \in U$, these four maps are elements of U, by Corollary 1 to Proposition 4 of Section 1, and so is the sextuple C, by Proposition 3 of Section 1, *cum grano salis*. This is moreover a special case of Section 2 if one considers the species of structure "category". With the notation of Section 2, the relation $C \in U$ is also written $C \in (U\text{-}\mathbf{Cat})$.

Notice that a category such as $(U\text{-}\mathbf{Set})$, $(U\text{-}\mathbf{Ord})$, $(U\text{-}\mathbf{Top})$, or $(U\text{-}\mathbf{Gr})$ is not, in general, an element of U.

Let C, D be two categories and U a universe such that $C, D \in U$. If C' is a subcategory of C, then $C' \in U$. The product category $C \times D$, the sum category of C and D, and the opposite categories C° and D° are also elements of U, by Section 2. The functor category $E = \mathrm{Hom}(C, D)$ is also an element of U: indeed, $\mathrm{Ob}(E)$ is a set of pairs of maps $\mathrm{Ob}(C) \to \mathrm{Ob}(D)$, $\mathrm{Fl}(C) \to \mathrm{Fl}(D)$, whence $\mathrm{Ob}(E) \in U$ by Section 1. As for $\mathrm{Fl}(E)$, it is a set of functorial morphisms, that is, of maps $\mathrm{Ob}(C) \to \mathrm{Fl}(D)$; hence $\mathrm{Fl}(E) \in U$ by Proposition 6 of Section 1.

Proposition 9. Let $\mathcal{E}$ be a species of structure with morphisms, and let U', U be two universes such that $U' \subset U$. Then $(U'\text{-}\mathcal{E})$ is a full subcategory of $(U\text{-}\mathcal{E})$.

Indeed, if x and y are objects of $(U'\text{-}\mathcal{E})$, then by definition they are objects of $(U\text{-}\mathcal{E})$. As for $\mathrm{Hom}(x, y)$, it is the same set in the two categories; see Section 1, no. 2, Example c).

N.B. The hypothesis that U and U' are universes is useless, so Proposition 9 would advantageously belong back in Section 1, no. 4. But the Tribe asked that it be placed here.

Remark. It may happen that the axioms of the species of structure $\mathcal{E}$ imply that the cardinals of the sets endowed with structures of species $\mathcal{E}$ are *bounded* by a fixed cardinal, say $\mathfrak{c}$; for example, the species of finite groups, finitely generated groups, finitely generated modules over a fixed ring A, or algebras of finite type over A. Suppose then that there exists an element z of U' such that $\mathfrak{c} \leq \operatorname{card}(z)$. Then, for every set x endowed with a structure of species $\mathcal{E}$, there exists an element x' of U' equipotent to x, for instance a subset of z. Endow x' with the structure transported from that of x; one obtains an element of $(U'\text{-}\mathcal{E})$. It follows that the inclusion functor from $(U'\text{-}\mathcal{E})$ into $(U\text{-}\mathcal{E})$ is then essentially surjective (Section 4, no. 2, Definition 2), and is therefore an equivalence of categories (Section 4, no. 3, Theorem 1).

4 The axiom of universes

The very agreeable stability results of Section 2 are of interest only if they can be applied to something other than the two small universes U_0 and U_1 described in Section 1. We will therefore add to the axioms of set theory the following axiom:

(A.6) *For every set x, there exists a universe U such that $x \in U$.*

This axiom implies that, if $(x_\alpha)_{\alpha \in I}$ is a *family* of sets, then there exists a universe U such that $x_\alpha \in U$ for every $\alpha \in I$. Indeed, it suffices to apply (A.6) to $x = \bigcup_{\alpha \in I} x_\alpha$, and then to apply Proposition 1 of Section 1.

In particular, given a category C, there exists a universe U such that $C \in U$ in the sense of Section 3: apply the preceding paragraph to the family $(\operatorname{Ob}(C), \operatorname{Fl}(C))$. This applies to categories of the form $(V\text{-}\mathcal{E})$, where $\mathcal{E}$ is a species of structure with morphisms and V is a set, for example a universe; in general $V \neq U$.

For example, if V is a universe, then $\operatorname{Ob}(V\text{-}\mathbf{Set}) = V$ and $\operatorname{Fl}(V\text{-}\mathbf{Set}) \subset V$, by Proposition 5 of Section 1. The universes U such that $(V\text{-}\mathbf{Set}) \in U$ are therefore those such that $V \in U$. But the relation $V \in V$ is impossible for a universe: indeed, for every subset A of V, one would have $A \in V$, by Proposition 1 of Section 1, whence $\operatorname{card}(\mathcal{P}(V)) \leq \operatorname{card}(V)$, which is impossible.

5 Universes and strongly inaccessible cardinals

Let U be a universe. By condition (U.I) of Definition 1, every element x of U is a subset of U; hence $\operatorname{card}(x) \leq \operatorname{card}(U)$. Since the cardinals of subsets of U form a well-ordered set, the cardinal

$$\mathfrak{c}(U) = \sup_{x \in U} \operatorname{card}(x) \tag{1}$$

exists. Notice that, for every cardinal $\mathfrak{c} < \mathfrak{c}(U)$, there exists an element x of U such that $\operatorname{card}(x) = \mathfrak{c}$. Indeed, by definition there exists $y \in U$ such that $\mathfrak{c} \leq \operatorname{card}(y) \leq \mathfrak{c}(U)$, and one takes for x a suitable subset of y. Conversely, if $x \in U$, then $\operatorname{card}(x) < \mathfrak{c}(U)$: indeed $\mathcal{P}(x) \in U$ by (U.III), whence $2^{\operatorname{card}(x)} \leq \mathfrak{c}(U)$. Thus the cardinal $\mathfrak{c}(U)$ has the following properties:

1) If $\mathfrak{c}$ is a cardinal $< \mathfrak{c}(U)$, then $2^{\mathfrak{c}} < \mathfrak{c}(U)$. Indeed, if x is an element of U of cardinal $\mathfrak{c}$, then $\mathcal{P}(x) \in U$ by (U.III), whence $2^{\mathfrak{c}} < \mathfrak{c}(U)$.

2) If $(\mathfrak{c}_\lambda)_{\lambda \in I}$ is a family of cardinals such that $\mathfrak{c}_\lambda < \mathfrak{c}(U)$ for every $\lambda \in I$, and if $\operatorname{card}(I) < \mathfrak{c}(U)$, then the sum cardinal $\sum_{\lambda \in I} \mathfrak{c}_\lambda$ is $< \mathfrak{c}(U)$. Indeed, let $x_\lambda \in U$ be such that $\operatorname{card}(x_\lambda) = \mathfrak{c}_\lambda$. Replacing I, if necessary, by an equipotent index set, we may suppose that $I \in U$. Then the sum set of the x_λ is an element of U, by Corollary 2 to Proposition 4 of Section 1, proving the assertion.

We make the following definition.

Definition 2. A cardinal $\mathfrak{d}$ is called *strongly inaccessible* if:

(FI.1) if $\mathfrak{c}$ is a cardinal such that $\mathfrak{c} < \mathfrak{d}$, then $2^{\mathfrak{c}} < \mathfrak{d}$;

(FI.2) if $(\mathfrak{c}_\lambda)_{\lambda \in I}$ is a family of cardinals such that $\mathfrak{c}_\lambda < \mathfrak{d}$ for all $\lambda \in I$, and if $\operatorname{card}(I) < \mathfrak{d}$, then $\sum_{\lambda \in I} \mathfrak{c}_\lambda < \mathfrak{d}$.

Examples. The cardinal 0 and the countably infinite cardinal are strongly inaccessible. No nonzero finite cardinal is strongly inaccessible. Thus we have just proved that the universe axiom (A.6) implies:

(A'.6) *Every cardinal is strictly dominated by a strongly inaccessible cardinal.*

Conversely:

Theorem 1. The assertion (A'.6) implies the axiom of universes (A.6).

Indeed, let A be a set. We must construct a universe U of which A is an element. Define by recursion a sequence $(A_n)_{n \geq 0}$ of sets by

(2) $A_0 = A$, and A_{n+1} is the union of the elements of A_n, that is, the set of elements of elements of A_n.

Put $B = \bigcup_{n=0}^{\infty} A_n$. By (A'.6), choose a strongly inaccessible cardinal $\mathfrak{c}$ such that $\operatorname{card}(B) < \mathfrak{c}$.

There exists a well-ordered set I such that $\operatorname{card}(I) = \mathfrak{c}$. Replacing I by its least segment of cardinal $\mathfrak{c}$, we may suppose that every proper segment of I has cardinal $< \mathfrak{c}$. We denote by $\mathcal{E}$ the least element of I. It follows from the hypothesis on segments that I has no largest element, and hence every element of I has a successor; for $\alpha \in I$, we denote the successor of α by $s(\alpha)$.

Now define, by transfinite recursion, a family $(B_\alpha)_{\alpha \in I}$ of sets by

$$\begin{cases} B_{\mathcal{E}} = B, \\ B_{s(\alpha)} = B_\alpha \cup \mathcal{P}(B_\alpha), \\ B_\alpha = \bigcup\limits_{\beta < \alpha} B_\beta \quad \text{if } \alpha \text{ has no predecessor.} \end{cases} \tag{3}$$

Put $U = \bigcup_{\alpha \in I} B_\alpha$. We will show that U is the required universe.

First $A \in U$, because $A = A_0$ is a subset of $B = B_{\mathcal{E}}$, and hence an element of $\mathcal{P}(B_{\mathcal{E}}) \subset B_{s(\mathcal{E})}$.

Next note that, for every $\alpha \in I$,

$$\operatorname{card}(B_\alpha) < \mathfrak{c}. \tag{4}$$

This is proved by transfinite induction on α. It is true for $\alpha = \mathcal{E}$ by hypothesis. From (4) one obtains

$$\operatorname{card}(B_{s(\alpha)}) \leq \operatorname{card}(B_\alpha) + 2^{\operatorname{card}(B_\alpha)} < \mathfrak{c} + \mathfrak{c} = \mathfrak{c},$$

using (FI.1) and the fact that $\mathfrak{c}$ is infinite. Finally, if α has no predecessor, the set I' of $\beta < \alpha$ has cardinal $< \mathfrak{c}$ by construction. Thus, if $\operatorname{card}(B_\beta) < \mathfrak{c}$ for all $\beta < \alpha$, then

$$\operatorname{card}(B_\alpha) = \operatorname{card}\Big(\bigcup_{\beta \in I'} B_\beta\Big) \leq \sum_{\beta \in I'} \operatorname{card}(B_\beta) < \mathfrak{c}$$

by (FI.2). This proves (4).

We now show that

$$\operatorname{card}(x) < \mathfrak{c} \quad \text{for every } x \in U. \tag{5}$$

It is enough to show, for every $\alpha \in I$, that $\operatorname{card}(x) < \mathfrak{c}$ for all $x \in B_\alpha$. Again proceed by transfinite induction on α. For $\alpha = \mathcal{E}$, if $x \in B_{\mathcal{E}} = B$, then $x \in A_n$ for some $n \geq 0$; hence $x \subset A_{n+1}$, so $x \subset B$ and $\operatorname{card}(x) \leq \operatorname{card}(B) < \mathfrak{c}$. The case where α has no predecessor is evident. Finally, if $x \in B_\alpha$ and $\alpha = s(\beta)$, then either $x \in B_\beta$, and the assertion follows by induction, or $x \in \mathcal{P}(B_\beta)$, whence $x \subset B_\beta$, and the assertion follows from (4).

We can now prove that U is indeed a universe:

(U.I) Let $x \in U$ and $y \in x$. We must show that $y \in U$. In other words, we must prove, for every $\alpha \in I$, the relation

$$x \in B_\alpha \text{ and } y \in x \Longrightarrow y \in B_\alpha.$$

Proceed once more by transfinite induction on α. It is true for $\alpha = \mathcal{E}$, since $x \in B_{\mathcal{E}} = B$ implies $x \in A_n$ for some n, and so, if $y \in x$, then $y \in A_{n+1}$, whence $y \in B = B_{\mathcal{E}}$. The passage to an element with no predecessor is evident. Finally pass from α to $s(\alpha)$. If $x \in B_{s(\alpha)}$ and $y \in x$, then either $x \in B_\alpha$, so $y \in B_\alpha \subset B_{s(\alpha)}$ by induction, or $x \in \mathcal{P}(B_\alpha)$, and again $y \in B_\alpha \subset B_{s(\alpha)}$.

(U.II) Let $x, y \in U$. Since the family $(B_\alpha)_{\alpha \in I}$ is increasing by (3), there exists $\alpha \in I$ such that $x, y \in B_\alpha$. Then $\{x, y\} \in \mathcal{P}(B_\alpha) \subset B_{s(\alpha)} \subset U$.

(U.III) Let $x \in U$. There exists $\alpha \in I$ such that $x \in B_\alpha$. It suffices to show, for every $\alpha \in I$, that

$$x \in B_\alpha \Longrightarrow \mathcal{P}(x) \in B_{s(s(\alpha))}.$$

Again proceed by transfinite induction on α. If $\alpha = \mathcal{E}$, then $x \in A_n$ for some n, whence every subset y of x is contained in $A_{n+1} \subset B = B_{\mathcal{E}}$. Thus $y \in \mathcal{P}(B_{\mathcal{E}}) \subset B_{s(\mathcal{E})}$ for every $y \in \mathcal{P}(x)$, and hence $\mathcal{P}(x) \subset B_{s(\mathcal{E})}$ and $\mathcal{P}(x) \in \mathcal{P}(B_{s(\mathcal{E})}) \subset B_{s(s(\mathcal{E}))}$. The passage to an element α with no predecessor is immediate, because $\beta < \alpha$ then implies $s(s(\beta)) < \alpha$. Finally pass from α to $s(\alpha)$. Let $x \in B_{s(\alpha)}$. If $x \in B_\alpha$, then $\mathcal{P}(x) \in B_{s(s(\alpha))} \subset B_{s(s(s(\alpha)))}$ by induction. If $x \in \mathcal{P}(B_\alpha)$, then $x \subset B_\alpha$, so $\mathcal{P}(x) \subset \mathcal{P}(B_\alpha)$ and $\mathcal{P}(x) \in \mathcal{P}(\mathcal{P}(B_\alpha)) \in B_{s(s(s(\alpha)))}$.

(U.IV) Let $(x_\lambda)_{\lambda \in K}$ be a family of elements of U such that $K \in U$. We must show that $x = \bigcup_{\lambda \in K} x_\lambda$ is an element of U. For every $\lambda \in K$, choose $\alpha(\lambda) \in I$ such that $x_\lambda \in B_{\alpha(\lambda)}$. We show that the set $\alpha(K)$ of the $\alpha(\lambda)$ is bounded in I. If it were not, then

$$I = \bigcup_{\lambda \in K} [\mathcal{E}, \alpha(\lambda)],$$

contradicting (FI.2), since $\operatorname{card}(K) < \mathfrak{c}$ by (5) and $\operatorname{card}([\mathcal{E}, \alpha(\lambda)]) < \mathfrak{c}$ for every $\lambda \in K$. Let β be an upper bound for $\alpha(K)$. Then $x_\lambda \in B_\beta$ for every $\lambda \in K$, because the family $(B_\alpha)_{\alpha\in I}$ is increasing. By what was proved in the verification of (U.I), each x_λ is a subset of B_β. Hence $x = \bigcup_{\lambda\in K} x_\lambda \subset B_\beta$. By (3), it follows that $x \in B_{s(\beta)}$, whence $x \in U$. □

Definition 3. Let x, y be two sets and let n be an integer ≥ 0. We say that y is a *component of order* n of x if there exists a sequence $(x_j)_{j=0,\dots,n}$ such that $x_0 = x$, $x_n = y$, and $x_{j+1} \in x_j$ for $j = 0, \dots, n-1$.

Thus x is the only component of order 0 of x. The components of order 1, respectively 2, of x are the elements of x, respectively the elements of elements of x. We say that y is a *component* of x if there exists $n \geq 0$ such that y is a component of order n of x. The relation "y is a component of x" is a preorder relation. By the selection-union scheme (Chapter II, ...), the relation "y is a component of x" is collectivizing with respect to y, so the components of x form a set.

Definition 4. Let $\mathfrak{c}$ be a cardinal. A set x is said to be of *type* $\mathfrak{c}$ (respectively, of *strict type* $\mathfrak{c}$, of *finite type*) if all components of x have cardinal $\leq \mathfrak{c}$ (respectively, $< \mathfrak{c}$, finite).

Examples. The elements of the universe U_1 of Section 1, Example 2, are all of finite type. If $\mathfrak{c}$ is a cardinal and x is of type $\mathfrak{c}$ (respectively, strict type $\mathfrak{c}$, finite type), then every component of x and every subset of x are of type $\mathfrak{c}$ (respectively, strict type $\mathfrak{c}$, finite type). Similarly, $\mathcal{P}(x)$ is of type $2^{\mathfrak{c}}$ (respectively, of strict type $2^{\mathfrak{c}}$, of finite type). If x is of type $\mathfrak{c}$ and $\mathfrak{c} \leq \mathfrak{c}'$, then x is of type $\mathfrak{c}'$.

Lemma. Let $\mathfrak{c}$ be a noncountable strongly inaccessible cardinal, and let X be a set of strict type $\mathfrak{c}$. Then there exists a cardinal $\mathfrak{d} < \mathfrak{c}$ such that X is of type $\mathfrak{d}$.

Indeed, for every integer n, let $\mathfrak{d}_n$ be the cardinal of the set X_n of components of order n of X. We have $\mathfrak{d}_0 = 1$, $\mathfrak{d}_1 = \operatorname{card}(X) < \mathfrak{c}$, and, since $X_n = \bigcup_{y\in X_{n-1}} y$, it follows by induction on n and by use of (FI.2) that $\mathfrak{d}_n < \mathfrak{c}$ for every n. Then the $\mathfrak{d}_n$ are bounded above by the cardinal $\mathfrak{d} = \sum_{j\geq 0} \mathfrak{d}_j$, which is $< \mathfrak{c}$ by (FI.2) and the noncountability of $\mathfrak{c}$. If Y is a component of order n of X, then $\operatorname{card}(Y) \leq \operatorname{card}(X_{n+1}) \leq \mathfrak{d}$. □

Proposition 10. Let U be a universe and let $\mathfrak{c}$ be a strongly inaccessible cardinal. Then the set U' of those $x \in U$ which are of strict type $\mathfrak{c}$ is a universe.

Indeed, if $x, y \in U'$ and $z \in x$, then obviously $z \in U'$ and $\{x, y\} \in U'$. If $x \in U'$, then $\operatorname{card}(x) < \mathfrak{c}$, whence $\operatorname{card}(\mathcal{P}(x)) < \mathfrak{c}$ by (FI.1). Since a component of $\mathcal{P}(x)$ is either $\mathcal{P}(x)$, or a subset of x, or a component of x, one has $\mathcal{P}(x) \in U'$. Finally, if $(x_\alpha)_{\alpha\in I}$ is a family of elements of U' such that $I \in U'$, then $\operatorname{card}(\bigcup_\alpha x_\alpha) < \mathfrak{c}$ by (FI.2); since every component of order > 0 of $\bigcup_\alpha x_\alpha$ is a component of some x_α, we do indeed have $\bigcup_\alpha x_\alpha \in U'$. □

Remark on the cardinal $\mathfrak{c}(U)$. Let U be a universe and let $\mathfrak{c}(U)$ be the cardinal defined by (1), that is,

$$\mathfrak{c}(U) = \sup_{x\in U} \operatorname{card}(x).$$

Obviously $\mathfrak{c}(U) \leq \operatorname{card}(U)$, because, recall, $x \in U$ implies $x \subset U$. But the equality $\mathfrak{c}(U) = \operatorname{card}(U)$ is not always true. For example, let $(x_\alpha)_{\alpha\in I}$ be a noncountable family of symbols with the axioms

$$\text{“}x_\alpha = \{x_\alpha\} \text{ for every } \alpha \in I\text{''} \quad \text{and} \quad \text{“}x_\alpha \neq x_\beta \text{ if } \alpha \neq \beta\text{''}.$$

In other words, x_α is a one-element set, namely itself. As in Example 2 of Section 1, form the "meaningful words" made from the symbols $\emptyset$, x_α, braces, and so on. Each denotes a *finite* set. Let U be the set of these. One easily checks that the conditions of Definition 1 are satisfied, so U is a universe. Since meaningful words are finite sequences of elements of a set of cardinal $\mathrm{card}(I) + 4 = \mathrm{card}(I)$, we have $\mathrm{card}(U) = \mathrm{card}(I)$; in fact $\mathrm{card}(U) \geq \mathrm{card}(I)$ would suffice, and this is evident. On the other hand, $\mathfrak{c}(U)$ is the countable cardinal, whence $\mathfrak{c}(U) < \mathrm{card}(U)$.

The strict inequality $\mathfrak{c}(U) < \mathrm{card}(U)$ is due to the fact that here we have introduced sets x_α such that $x_\alpha \in x_\alpha$. If one forbids horrors of this kind, one obtains $\mathrm{card}(U) = \mathfrak{c}(U)$ for every universe U, as well as other very pretty results "which cannot serve any purpose." This is what we shall do in the next section.

6 Artinian sets and artinian universes

Definition 5. A set x is called *artinian* if there exists no infinite sequence $(x_n)_{n\geq 0}$ such that $x_0 = x$ and $x_{n+1} \in x_n$ for every $n \geq 0$.

Examples. The sets $\emptyset$, $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}$, and more generally the elements of the universe U_1 of Section 1, are artinian.

In pictorial terms, if x is artinian and one takes an element x_1 of x, then an element x_2 of x_1, and so on, the process must stop, and one arrives at a component x_n of x which is *empty.* In other words, artinian sets are "built out of $\emptyset$." This will be made precise later (cf. (*) in the proof of Theorem 2).

Artinian sets evidently enjoy the following properties:

(AR.I) If x is artinian, every subset of x and every component of x is artinian. For x to be artinian, it is necessary and sufficient that every element of x be artinian.

(AR.II) If x and y are artinian, then $\{x, y\}$ is artinian.

(AR.III) If x is artinian, then $\mathcal{P}(x)$ is artinian.

(AR.IV) Every union of artinian sets is an artinian set.

These properties immediately show:

Proposition 11. If U is a universe, the set of artinian elements of U is a universe, necessarily artinian.

Corollary. If x is an artinian set, there exists an artinian universe U such that $x \in U$.

Indeed, by axiom (A.6), x is an element of a universe V; take for U the set of artinian elements of V.

The following proposition is still less useful than the rest of the section:

Proposition 12. Let A be an artinian set. Then:

a) for every $x \in A$, one has $x \notin x$;

b) if $x, y \in A$, one cannot have both $x \in y$ and $y \in x$;

c) for every $x \in A$, the relation "y is a component of z" between components y, z of x is an order relation;

d) for every nonempty element x of A, there exists $y \in x$ such that $x \cap y = \emptyset$.

Indeed, the negation of a), respectively of b), entails the existence of an infinite sequence $(x_n)_{n\geq 0}$ contradicting Definition 5, namely $(x, x, x, \ldots)$, respectively $(x, y, x, y, x, y, \ldots)$. The negation of c) means that there exist $x \in A$, distinct components y, z of x, and membership chains

$$y \in y_1 \in \cdots \in y_q \in z, \qquad z \in z_1 \in \cdots \in z_r \in y;$$

then, as in b), one obtains an infinite sequence $(x_n)_{n\geq 0}$ contradicting Definition 5. Finally, if d) is false, there exists a nonempty element x of A such that $x \cap y \neq \emptyset$ for every $y \in x$. Put $x_0 = x$, and choose $x_1 \in x$. Since $x \cap x_1 \neq \emptyset$, choose $x_2 \in x \cap x_1$, and so on. More formally, define by recursion an infinite sequence $(x_n)_{n\geq 1}$ of elements of x by choosing $x_1 \in x$ and $x_{n+1} \in x_n \cap x$ for $n \geq 1$. Then the sequence $(x_n)_{n\geq 0}$ contradicts Definition 5.

Remark. Let B be a set. For B to be artinian, it is necessary and sufficient that every set A of sets of components of B satisfy condition d) of Proposition 12. Necessity follows from (AR.I) and Proposition 12. Conversely, if B is not artinian, there exists an infinite sequence $(x_n)_{n\geq 0}$ with $x_0 = B$ and $x_{n+1} \in x_n$ for every $n \geq 0$. Take for A the singleton consisting of the set X of the x_n. Thus A contains a nonempty element X such that, for every element y of X, one has $y \cap X \neq \emptyset$: indeed $y = x_n$ for some n, and $x_{n+1} \in y \cap X$.

Theorem 2. Let $\mathfrak{c}$ be an infinite cardinal. Then:

a) The relation "x is an artinian set of type $\mathfrak{c}$" (respectively, of strict type $\mathfrak{c}$) is collectivizing with respect to x. The set $A_{\mathfrak{c}}$ of artinian sets of type $\mathfrak{c}$ has cardinal $2^{\mathfrak{c}}$.

b) If $\mathfrak{c}$ is strongly inaccessible, the set $U_{\mathfrak{c}}$ of artinian sets of strict type $\mathfrak{c}$ is a universe of cardinal $\mathfrak{c}$; the cardinal $\mathfrak{c}(U_{\mathfrak{c}}) = \sup_{x\in U_{\mathfrak{c}}} \operatorname{card}(x)$ is $\mathfrak{c}$.

c) If a universe U has an element of cardinal $\mathfrak{c}$, then every artinian set of type $\mathfrak{c}$ belongs to U, that is, $A_{\mathfrak{c}} \subset U$.

c') If a universe U is nonempty, then every artinian set of finite type is an element of U.

Before proving Theorem 2, let us derive a few illuminating corollaries.

Corollary 1. If a universe U is artinian, then $\operatorname{card}(U)$ is strongly inaccessible, and U is the set of artinian sets of strict type $\operatorname{card}(U)$.

Indeed put $\mathfrak{c}(U) = \sup_{x\in U} \operatorname{card}(x)$. This is a strongly inaccessible cardinal, by the beginning of Section 5. First suppose it is noncountable. Then, for every infinite cardinal $\mathfrak{c} < \mathfrak{c}(U)$, every artinian set of type $\mathfrak{c}$ is an element of U by c); hence every artinian set of strict type $\mathfrak{c}(U)$ is an element of U, by the lemma of Section 5. This last assertion remains true if $\mathfrak{c}(U)$ is countable, by c'). Conversely, by (U.I), every set of U is of strict type $\mathfrak{c}(U)$. Thus U is the universe $U_{\mathfrak{c}(U)}$ of b), whence $\mathfrak{c}(U) = \operatorname{card}(U)$ by b).

It follows from Corollary 1 that an artinian universe is uniquely determined by its cardinal, which is strongly inaccessible. Thus one has a "one-to-one correspondence" between artinian universes and strongly inaccessible cardinals. In particular:

Corollary 2. The inclusion relation $U \subset U'$ between artinian universes is a well-ordering relation.

Indeed, the relation $\mathfrak{c} \leq \mathfrak{c}'$ between cardinals is a well-ordering relation (Chapter III), and, with the notation of Theorem 2 b), the relations $\mathfrak{c} \leq \mathfrak{c}'$ and $U_{\mathfrak{c}} \subset U_{\mathfrak{c}'}$ are equivalent.

Note that Theorem 2 b) gives a second proof of Theorem 1.

We now prove Theorem 2. Given a set A, call a *chain* of A any finite sequence $(x_j)_{j=0,\ldots,n}$ such that $x_0 = A$ and $x_{j+1} \in x_j$ for $j = 0, \ldots, n-1$. The chains of A form a set, by the selection-union scheme; denote it by $G(A)$. Given a chain $X = (x_j)_{j=0,\ldots,n}$, the chains of the form $(x_i)_{i=0,\ldots,q}$, with $q \leq n$, are said to be smaller than X. This gives an ordered-set structure on $G(A)$. For A to be artinian, it is necessary and sufficient that $G(A)$ be a noetherian ordered set, in the sense of the equivalent conditions of Chapter III, §6, no. 5.

We shall show that:

> (*) *If A is artinian, then A is uniquely determined by the isomorphism class of the ordered set $G(A)$.*

Indeed, given an ordered set G and an element $g \in G$, write $S(g)$ for the set of $g' \in G$ such that $g < g'$ and such that $g \leq h \leq g'$ implies $h = g$ or $h = g'$. In other words, $S(g)$ is the set of immediate successors of g. Consider the map θ which sends a chain $X = (x_0, \ldots, x_n)$ of A to the set x_n. Then, for every $X \in G(A)$,

$$\theta(X) = \{\theta(X') \mid X' \in S(X)\}.$$

Since A is the image under θ of the least element $X_0 = (A)$ of $G(A)$, it will be enough to show that θ is uniquely determined by the isomorphism class of the ordered set $G(A)$. This follows from the following lemma.

Lemma 1. Let G be a noetherian ordered set, and let φ be a map on G such that, for every $g \in G$,

$$\varphi(g) = \{\varphi(g') \mid g' \in S(g)\}. \tag{1}$$

Then φ is uniquely determined. Moreover, if U is a universe containing an element equipotent to G, then φ takes its values in U.

The hypothesis implies that $\varphi(g) = \emptyset$ if g is a maximal element of G. Indeed, let φ and φ' be two maps satisfying (1). If $\varphi \neq \varphi'$, the set of $g \in G$ such that $\varphi(g) \neq \varphi'(g)$ is nonempty, hence has a maximal element h, since G is noetherian. Then $\varphi(g) = \varphi'(g)$ for every $g > h$, in particular for every successor $g \in S(h)$, whence $\varphi(h) = \varphi'(h)$ by (1), a contradiction. Thus $\varphi = \varphi'$.

Now let U be a universe containing an element x equipotent to G. We show that φ takes its values in U. Otherwise, let h be maximal among the $g \in G$ such that $\varphi(g) \notin U$. We have $\varphi(g') \in U$ for every $g' \in S(h)$, so

$$\varphi(h) = \{\varphi(g') \mid g' \in S(h)\}$$

is a subset of U. Since its cardinal is less than or equal to $\mathrm{card}(G)$, hence to the cardinal of an element of U, one has $\varphi(h) \in U$ by Proposition 7 of Section 1. This contradiction shows that φ takes its values in U.

We now prove Theorem 2 a). It is enough to treat the non-respective assertion, since the other follows at once. Let $\mathfrak{c}$ be an infinite cardinal. If A is a set of type $\mathfrak{c}$, then

$$\operatorname{card}(G(A)) \leq \mathfrak{c}. \tag{2}$$

Indeed, if A_n denotes the set of components of order n of A, then $\operatorname{card}(A_1) \leq \mathfrak{c}$ and $\operatorname{card}(A_{n+1}) \leq \mathfrak{c} \cdot \operatorname{card}(A_n)$. Hence $\operatorname{card}(A_n) \leq \mathfrak{c}^n$ by induction; but $\mathfrak{c}^n = \mathfrak{c}$, because $\mathfrak{c}$ is infinite (Chapter III). Therefore

$$\operatorname{card}\Big(\bigcup_{n=0}^{\infty} A_n\Big) \leq \operatorname{card}(\mathbb{N}) \cdot \mathfrak{c} = \mathfrak{c}.$$

Since $G(A)$ is a set of finite sequences of elements of $\bigcup_n A_n$, inequality (2) follows.

Now let E be a set of cardinal $\mathfrak{c}$. Giving an order structure on a subset E' of E is equivalent to giving the subset of $E \times E$ consisting of the pairs (x, y) such that $x \in E'$, $y \in E'$, and $x \leq y$. Therefore, by (2), the isomorphism classes of ordered sets $G(A)$, where A is of type $\mathfrak{c}$, form a set $\mathfrak{F}_\mathfrak{c}$, and

$$\operatorname{card}(\mathfrak{F}_\mathfrak{c}) \leq \operatorname{card}(\mathcal{P}(E \times E)) = 2^\mathfrak{c}.$$

Let $\mathfrak{F}'_\mathfrak{c}$ be the subset of $\mathfrak{F}_\mathfrak{c}$ formed by the classes of noetherian ordered sets having a least element. If, to each $G \in \mathfrak{F}'_\mathfrak{c}$, we associate the value of the function φ of Lemma 1 at the least element of G, we obtain a map θ on $\mathfrak{F}'_\mathfrak{c}$ whose image contains *all* artinian sets of type $\mathfrak{c}$. These therefore form a set $A_\mathfrak{c}$, and

$$\operatorname{card}(A_\mathfrak{c}) \leq \operatorname{card}(\mathfrak{F}'_\mathfrak{c}) \leq \operatorname{card}(\mathfrak{F}_\mathfrak{c}) \leq 2^\mathfrak{c}.$$

It remains to see that $\operatorname{card}(A_\mathfrak{c}) = 2^\mathfrak{c}$. For this it is enough to see that there exists an artinian set B of type $\mathfrak{c}$ and cardinal $\mathfrak{c}$, since the subsets of B will then be elements of $A_\mathfrak{c}$. This existence follows from the following lemma.

Lemma 2. For every cardinal $\mathfrak{c}$, there exists an artinian set B of type $\mathfrak{c}$ and cardinal $\mathfrak{c}$.

Proceed by transfinite induction on $\mathfrak{c}$. For $\mathfrak{c} = 0$ there is no choice: take $B = \emptyset$. If $\mathfrak{c}$ has a predecessor $\mathfrak{c}'$, let B' be an artinian set of type $\mathfrak{c}'$ and of cardinal $\mathfrak{c}'$. Then $\mathfrak{c} \leq 2^{\mathfrak{c}'}$, so there exists a subset B of $\mathcal{P}(B')$ of cardinal $\mathfrak{c}$. The elements of B are subsets of B', and hence have cardinals $\leq \mathfrak{c}' \leq \mathfrak{c}$; the components of higher order of B are components of B', and hence also have cardinals $\leq \mathfrak{c}' \leq \mathfrak{c}$. Finally, if $\mathfrak{c}$ has no predecessor, choose, for every cardinal $\mathfrak{c}_\lambda < \mathfrak{c}$, an artinian set B_λ of type $\mathfrak{c}_\lambda$ and cardinal $\mathfrak{c}_\lambda$. Then $B = \bigcup_\lambda B_\lambda$ has the required property. This proves Lemma 2 and completes the proof of part a).

We turn to b). Let $\mathfrak{c}$ be a strongly inaccessible cardinal. We already know, by a), that the artinian sets of strict type $\mathfrak{c}$ form a set $U_\mathfrak{c}$. The fact that $U_\mathfrak{c}$ is a universe follows immediately from properties (AR.I)–(AR.IV) of artinian sets, at the beginning of this section, and from cardinal estimates analogous to those of Proposition 10 of Section 5. The relation

$$\sup_{x \in U_\mathfrak{c}} \operatorname{card}(x) = \mathfrak{c}$$

follows from Lemma 2 applied to cardinals $< \mathfrak{c}$. Finally, to prove $\operatorname{card}(U_\mathfrak{c}) = \mathfrak{c}$, first suppose that $\mathfrak{c}$ is noncountable. By the lemma of Section 5, $U_\mathfrak{c}$ is the union

$$\bigcup_{\mathfrak{d} < \mathfrak{c}} A_\mathfrak{d},$$

where $A_{\mathfrak{d}}$ denotes the set of artinian sets of type $\mathfrak{d}$. But $\operatorname{card}(A_{\mathfrak{d}}) = 2^{\mathfrak{d}} < \mathfrak{c}$ by a); on the other hand, the set of cardinals $\mathfrak{d} < \mathfrak{c}$ has cardinal $\leq \mathfrak{c}$ (Chapter III). Hence

$$\operatorname{card}(U_{\mathfrak{c}}) \leq \mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c},$$

and also $\operatorname{card}(U_{\mathfrak{c}}) \geq \mathfrak{c}$, since $\operatorname{card}(U_{\mathfrak{c}}) \geq \operatorname{card}(A_{\mathfrak{d}}) = 2^{\mathfrak{d}}$ for every $\mathfrak{d} < \mathfrak{c}$.

The case $\mathfrak{c} = 0$ is trivial. It remains to treat the case where $\mathfrak{c}$ is the countably infinite cardinal. In this case $U_{\mathfrak{c}}$ is the set of artinian sets of finite type, that is, sets which are finite, as are all their components; and one uses a pretty combinatorial result.

Lemma 3 (D. König?). Consider two infinite sequences $(E_n)_{n\geq 1}$, $(f_n)_{n\geq 1}$ of finite sets E_n and maps $f_n : E_{n+1} \to E_n$. If there exists no infinite sequence $(x_n)_{n\geq 1}$ such that $x_n \in E_n$ and $f_n(x_{n+1}) = x_n$ for every n, then E_n is empty for all sufficiently large n.

In other words, if all sequences (x_n) such that $f_n(x_{n+1}) = x_n$ are finite, then their lengths are bounded. This can be expressed in terms of projective limits: a projective limit of nonempty finite sets is nonempty; see General Topology, Chapter I, 2nd ed., §9, no. 6, Proposition 8, 2°.

Argue by contradiction. If there exist nonempty E_n for arbitrarily large n, then no E_n is empty, because $E_n = \emptyset$ implies $E_{n+1} = \emptyset$, given the existence of $f_n : E_{n+1} \to E_n$. Call a finite sequence $(x_j)_{1\leq j\leq n}$ *coherent* if $f_j(x_{j+1}) = x_j$ for $j = 1, \ldots, n-1$. We prove by induction on n the existence of a coherent sequence $(a_1, \ldots, a_n)$, with $a_i \in E_i$, which, for every $q \geq n$, can be extended to a coherent sequence $(a_1, \ldots, a_n, x_{n+1}, \ldots, x_q)$ of length q. This is evident for $n = 0$, since no E_q is empty. Passing from n to $n+1$, if, for every $x \in f_n^{-1}(\{a_n\}) \subset E_{n+1}$, all coherent sequences extending $(a_1, \ldots, a_n, x)$ had lengths bounded by an integer $q(x)$, then all coherent sequences extending $(a_1, \ldots, a_n)$ would have bounded lengths, by $\sup_x q(x)$, since E_{n+1} is finite. Hence there exists $a_{n+1} \in f_n^{-1}(\{a_n\})$ such that the coherent sequence $(a_1, \ldots, a_n, a_{n+1})$ has extensions of arbitrary length. This gives an infinite sequence $(a_n)_{n\geq 1}$, contrary to the hypothesis.

It follows from Lemma 3 that if A is an artinian set of finite type, then the ordered set $G(A)$ of its chains is *finite*. Indeed, take E_n to be the set of chains

$$x_n \in x_{n-1} \in \cdots \in x_1 \in A$$

with $n+1$ terms. This is finite because the components of A of order $\leq n+1$ are finite in number and are themselves finite. Take f_n to be the map which deletes the last term:

$$(x_{n+1} \in x_n \in \cdots \in x_1 \in A) \longmapsto (x_n \in x_{n-1} \in \cdots \in x_1 \in A).$$

The set of isomorphism classes of finite ordered sets is countable: indeed, giving an order structure on a finite subset of $\mathbb{N}$ is equivalent to giving its graph, a finite subset of $\mathbb{N} \times \mathbb{N}$, and the set of finite subsets of a countable set is countable (Chapter III). It follows from (*) and Lemma 1 that the set $\mathfrak{a}$ of artinian sets of finite type is countable. It is infinite by Lemma 2, or more simply by using the sequence $(z_n)_{n\geq 0}$ defined by $z_0 = \emptyset$, $z_{n+1} = \{z_n\}$. This completes the proof of b).

Remark. We have just proved that, if A is an artinian set of finite type, then it has only finitely many components. Thus there exists an integer n such that A is of type n.

We now prove c). Let $\mathfrak{c}$ be an infinite cardinal, let U be a universe having an element of cardinal $\mathfrak{c}$, and let A be an artinian set of type $\mathfrak{c}$. The ordered set $G(A)$ has cardinal $\leq \mathfrak{c}$, by formula (2). The second assertion of Lemma 1, applied to $G(A)$, shows that the map

$$(x_0, \ldots, x_n) \rightsquigarrow x_n$$

from $G(A)$ takes its values in U. In other words, all components of A are elements of U. This proves c). The proof of c') is analogous: if A is an artinian set of finite type, then we have just seen that $G(A)$ is finite; since U is nonempty, it contains an element equipotent to $G(A)$, by the corollary to Proposition 7 of Section 1. $\square$

7 Vague metamathematical remarks

a) The axiom "*every set is artinian*" is *inoffensive*. Indeed, if one has a model M of set theory, the set M' of artinian elements of M is also a model (cf. Proposition 11).

b) The universe axiom (A.6), and the equivalent axiom (A'.6) on strongly inaccessible cardinals, is *independent* of the rest of set theory. Indeed, let $\mathfrak{c}$ be the first noncountable strongly inaccessible cardinal. The universe $U_{\mathfrak{c}}$ of artinian sets of strict type $\mathfrak{c}$ (Theorem 2 b)) is a model of set theory without (A.6): one calls the elements of $U_{\mathfrak{c}}$ "sets," the "membership relation" is the restriction to $U_{\mathfrak{c}}$ of the ordinary one, and so on. The "universes" of the model are therefore the ordinary universes which are elements of $U_{\mathfrak{c}}$. But we have seen that the only universes which are elements of $U_{\mathfrak{c}}$ are the two pequeños $U_0 = \emptyset$ and U_1. Thus U_1 is a "set" which is an element of no "universe." Hence we have a model of set theory in which (A.6) is false.

c) Bourbaki was too cautious in merely "presuming" that the *axiom of infinity* (A.5) is independent of the preceding axioms and schemes. It is indeed independent, because the countable universe U_1 of artinian sets of finite type is a model in which (A.5) is false and in which the preceding axioms and schemes are true.

d) It would be very interesting to prove that the universe axiom (A.6) is inoffensive. This seems difficult, and is even unprovable, says Paul Cohen.

The adjective "vague" in the title means that, in constructing models, one did not take the trouble to cannulate the symbol τ so that it does not spill out. The key to this, if one considers a model M, is to replace the ordinary

$$\tau_x(R(x))$$

by

$$\tau_x(R(x) \text{ and } x \in M).$$

One still has to check that this really transforms ordinary quantifiers into the quantifiers formerly called "typical":

$$(\forall x \in M) \qquad (\exists x \in M).$$

Exercises.

1) Let $n \geq 1$ be an integer. Show that the set of artinian sets of type n is infinite. The sets $z_0 = \emptyset$, $z_1 = \{\emptyset\}$, and $z_{q+1} = \{z_q\}$ are of type 1.

2) Call the *height* of a set A the least upper bound, finite or infinite, of the integers n such that there exists a sequence

$$x_n \in x_{n-1} \in \cdots \in x_0 = A.$$

Show that the sets of height $\leq n$ are finite and form a finite set whose cardinal p_n is calculated from $p_0 = 1$, $p_{n+1} = 2^{p_n}$. Proceed by induction on n, noting that the elements of a set A of height $\leq n$ are sets of height $\leq n-1$.

3) Let G be a noetherian ordered set having a least element g_0, and such that, for every $h \in G$, the set of $g \leq h$ is finite and totally ordered.

   a) Show that every element $h \neq g_0$ of G has a unique predecessor.

   b) Let φ be the map on G defined in Lemma 1, that is, $\varphi(g)$ is the set of the $\varphi(g')$ where g' runs over the successors of g. Put $A = \varphi(g_0)$. Show that A is an artinian set. For $g \in G$, let
$$g_0 < g_1 < \cdots < g_n = g$$
be the sequence of elements $\leq g$, and put
$$f(g) = (\varphi(g_0), \varphi(g_1), \dots, \varphi(g_n)).$$
Show that f is an increasing surjective map from G onto the ordered set $G(A)$ of the text. Show that, if f is *injective*, it is an isomorphism from G onto $G(A)$.

   c) For $g \in G$, let M_g be the set of majorants of g. Suppose that, for every pair of distinct elements g, g' having the same predecessor p, the ordered sets M_g and $M_{g'}$ are not isomorphic. Show that the map f of b) is then an *isomorphism* from G onto $G(A)$. Hint: if $f(g) = f(g')$ with $g \neq g'$, and if
$$g_0 < g_1 < \cdots < g_n = g, \qquad g'_0 < g'_1 < \cdots < g'_{n'} = g'$$
are the sequences of elements $\leq g$ and $\leq g'$, show that $n = n'$, and that one may suppose that g and g' have the same predecessor p. Then consider the set of $p \in G$ such that there exist two distinct successors g, g' of p with $f(g) = f(g')$; take a maximal element q of this set and two distinct successors h, h' of q such that $f(h) = f(h')$. Observe that the restrictions of f to M_h and to $M_{h'}$ are injective, and hence, by b), are isomorphisms from M_h to $G(\varphi(h))$ and from $M_{h'}$ to $G(\varphi(h'))$. Deduce from the non-isomorphism hypothesis on M_h and $M_{h'}$ that $\varphi(h) \neq \varphi(h')$, contradicting $f(h) = f(h')$.

N.B. Exercise 3 gives very precise information on how artinian sets are "made." We already knew that such a set A is determined by the isomorphism class of the ordered set $G(A)$ (Lemma 1). We now know how to *characterize* the ordered sets G isomorphic to some $G(A)$:

α) G is noetherian and has a least element;

β) for every $g \in G$, the set of $h \leq g$ is totally ordered and finite, whence the existence and uniqueness of the predecessor of g;

γ) if g, g' are distinct elements having the same predecessor, then the set M_g of majorants of g and the set $M_{g'}$ of majorants of g' are not isomorphic.

4) Let $(A_n)_{n \geq 0}$ be the sequence of finite ordinals defined by $A_0 = \emptyset$, $A_{n+1} = A_n \cup \{A_n\}$. Show that the ordered set $G(A_n)$ has 2^n elements, and that the number of its elements of height q, in the sense of Exercise 2, is $\binom{n}{q}$.

References

[1] B. Mitchell, *Theory of Categories*, Academic Press, 1965.